PART 1

The Digital SAT Math

Introduction to the Digital SAT Math

The College Board has made the SAT digital. The digitalized SAT will be administered in schools or test centers with a proctor present starting in **spring 2024** if you plan to take the SAT in the U.S. The digital SAT still scores on a 1600 point-scale. (Reading and Writing section score: 200-800 and Math section score: 200-800)

The Digital SAT Math test formats (Adaptive Test)

Math section has two adaptive test design.

- **Module 1**: 22 questions (including 2 unscored questions- research purposes only)
  Students are given a broad mix of easy, medium, and hard questions.
  -Time: 35 min, the use of calculator is permitted.

- **Module 2**: 22 questions (including 2 unscored questions- research purposes only)
  Students are given a targeted mix of questions of varying difficulties based on their performance in module 1.
  -Time: 35 min, the use of calculator is permitted.

Important! – After completing the first-stage module on math section, **the test application routes students to one of two possible second-stage modules of questions**. One whose questions are, on average, of lower difficulty than those in the first module and the other whose questions are, on average, of higher difficulty than those in the first module.
And you will get the score based on the route and your performance.

The Digital SAT Math test formats (Nonadaptive Test)

Math section has two nonadaptive test design.

- **Module 1**: 27 questions (including 2 unscored questions- research purposes only)
  Students are given a broad mix of easy, medium, and hard questions.
  -Time: 43 min, the use of calculator is permitted.

- **Module 2**: 27 questions (including 2 unscored questions- research purposes only)
  Same difficulty level as module 1.
  -Time: 43 min, the use of calculator is permitted.

- Any student who wants to take the digital SAT on a paper braille test in an approved accommodation will need to be approved by College Board.

Math section Content domains.

- **Algebra** (13-15 questions, about 35%)
    - linear equations (functions) with one or two variables, System of linear equations, and linear inequalities one or two variables. Meaning of numbers or variables in the context.

- **Advanced Math** (13-15 questions, about 35%)
    - Equivalent equations, non-linear equations in one variable and systems of equations in two variables, non-linear functions. Solving rational equations and graphs of rational and polynomial functions.

- **Problem-solving and Data Analysis** (5-7 questions, about 15%)
- Ratios, rates, proportional relationship, units. Percentages, one or two variable data: distributions and measures of center and spread, Scatterplots. Probability and conditional probability, Inference from sample statistics and margin of error.

- **Geometry and Trigonometry** (5-7 questions, about 15%)
- Area and volume. Lines, angles, and triangles, right triangles, trigonometry, Circles.

How to calculate your scores in this book

- Use the answer keys on the end of each test to find the number of questions in module 1 and module 2 that you answered correctly.

- Add the number of correct answers you got on both module 1 and module 2. That will be your raw scores. Remember, there are no guess penalties. Even though the real College Board digital SAT counts only 20 questions only for scoring (2 unscored questions for research purposes), **we count all 22 questions to get your score in this book.**

- Look for the expected Math score (200-800) in the Score Conversion table provided below the answer keys using your calculated raw scores.

Test-taking Strategies

- **Know the test basics**: 2 modules. 22 questions each module. 35 minutes. Remembering the importance of doing well in the first stage module in order to route the higher difficulty level in module 2. No guessing penalties.

- **Get familiar with Digital Testing Tools**: Cross out answer choices in multiple choice questions. Display or hide a countdown timer. Access the Math section's reference sheet during testing. Flag questions within a given test module to return to. Access a display informing you of how many questions in each module you've either flagged or left unanswered and allowing you to jump to any questions within a module. Use of the built-in Desmos Graphing Calculator or your own calculator on the entire math sections.

- **Know the two test questions formats**: About 75% of questions are multiple choice and the rest are SPR format (Students-Produced Response)

- **Get Familiar with the Various Question Types**: Study the SAT Math point notes summarized in the beginning of this book first and do all Practice tests available on both College Board web site (see below) and 15 full-length practice tests provided in this book.

- 4 free official full-length digital College Board SAT tests are located at https://satsuite.collegeboard.org/digital/digital-practice-preparation/practice-tests/linear

THE MATH BREAKDOWN

No Need to Know
Here are a few things you won't need to know to answer Digital SAT Math questions: calculus, logarithms, and matrices. Essentially, the Digital SAT tests a whole lot of algebra and some arithmetic, statistics, and geometry.

The Math section of the Digital SAT is split into two modules. Each module contains 22 questions, of which 16 or 17 are multiple-choice questions and the rest are student-produced response questions (SPR), meaning that you fill in your own answer instead of choosing from four answers. Questions on the second module are, on average, easier or harder based on your performance on the first module. Each module has two "pre-test" questions that do not count towards your score, but they are not identified, so treat every question as if it counts.

The Math section is further broken down by question type and content area, as described below. Unlike the Verbal section, the question types and content areas do not go in any predictable order. Everything is mixed together, so you could see a trig question, then a word problem about averages, then a question about the vertex of a parabola. We'll cover these topics and many more in the next several chapters, so you'll be ready for all of it.

Question Type Breakdown	
70% Problem Solving	15–16 questions per module
30% Word Problems	6–7 questions per module
Content Breakdown	
35% Algebra	7–8 questions per module
35% Advanced Math	7–8 questions per module
15% Problem-Solving & Data Analysis	3–4 questions per module
15% Geometry and Trigonometry	3–4 questions per module

Fill-In Questions

Approximately 25% of the questions on the Math section of the Digital SAT are what College Board calls Student-Produced Response questions, or SPRs. These are the only questions on the test that are not multiple-choice. Instead of selecting the correct answer from among several choices, you will have to find the answer on your own and type it into a box. We call these fill-ins because you fill in your answer. The fill-in questions cover the same math topics as the multiple-choice questions do, and they fit within the order of difficulty of the module. The fill-in format has special characteristics, and we'll tell you more about them in Chapter 26.

You Don't Have to Finish

We've all been taught in school that when you take a test, you have to finish it. If you answered only two-thirds of the questions on a high school math test, you probably wouldn't get a very good grade. But as we've already seen, the Digital SAT is not at all like the tests you take in school. Most students don't know about the difference, so they make the mistake of trying to work all of the questions on both Math modules of the Digital SAT.

Quick Note

Remember, this is not a math test in school! It is not scored on the same scale your math teacher uses. You don't need to get all the questions right to get an above-average score.

Because they have only a limited amount of time to answer all the questions, most students rush through the questions they think are easy in order to get to the harder ones as soon as possible. At first, it seems reasonable to save more time for the more challenging questions, but think about it this way. When students rush through a Math section, they're actually spending too little time on the easier questions (which they have a good chance of getting right), just so they can spend more time on the harder questions (which they have less chance of getting right). Does this make sense? Of course not.

Here is the secret: on the Math section, you don't have to answer every question in each module. In fact, unless you are aiming for a top score, you should intentionally skip some harder questions. Most students can raise their Math scores by concentrating on correctly answering all of the questions that they think are easy or medium difficulty. In other words…

Slow Down!

Most students do considerably better on the Math section when they slow down and spend less time worrying about the more complex questions (and more time working carefully on the more straightforward ones). Haste causes careless errors, and careless errors can ruin your score. In most cases, you can actually raise your score by answering fewer questions. That doesn't sound like a bad idea, does it? If you're shooting for an 800, you'll have to answer every question correctly. But if your target is 550, you should ignore the hardest questions in each module and use your limited time wisely.

POOD and the Math Section

The questions in both modules of the Digital SAT Math section are arranged in a loose order of difficulty. The earlier questions are generally easier and the last few are harder, but the level of difficulty may jump around a little. Furthermore, the second module might start with a question that feels much easier than the last question of the first module. Assessing the difficulty of a question is also complicated by the fact that in College Board's view, "hard" on the Digital SAT means that a higher percentage of students tend to get it wrong, often due to careless errors or lack of time.

The two experimental questions in each module can also alter the order of difficulty. If you encounter a question that seems surprisingly easy or surprisingly hard based on the questions before and after, use your POOD to decide whether to do it, mark it for later, or enter a guess.

Because difficulty levels can go up and down a bit, don't worry too much about how hard the test-writers think a question is. Focus instead on the questions that are easiest for you, and do your best to get those right—no matter where they appear—before moving on to the tougher ones. So which will be the easy ones for you? It is *personal* order of difficulty, but here are some things to consider:

- **Math knowledge:** Do you know the topic cold? Do you see exactly how to start solving it? Then the question is worth attempting, but read and work carefully!
- **SAT knowledge:** Is there a Princeton Review technique from this book that would be perfect for this question? Now is the time to put your skills to use.
- **Self-knowledge:** Do your eyes glaze over halfway through a word problem? Do you think, "More like trigoNOmetry" when you see a trig question? Then come back to that question later or just pick a random answer to select.
- **Take the first bite:** A great way to decide whether a question deserves your time is to think about Bite-Sized Pieces. If you know immediately how to start a question, there's a good chance you'll be able to finish it and get it right.

> Don't forget: Fill in answers for questions you decide to skip, use the Mark for Review tool to mark questions to come back to later, and enter an answer for every question.

USING THE ONLINE TOOLS AND SCRATCH PAPER

Online Tools

Several of the built-in features of the Digital SAT will be useful on the Math section.

- Mark for Review tool to mark questions to come back to later
- Built-in calculator, which can be accessed at any time
- Reference sheet with common geometry formulas, which can be accessed at any time
- The Annotate tool is NOT available on the Math section, so you will not be able to underline or highlight parts of the question.

Scratch Paper

The proctor at the test center will hand out three sheets of scratch paper, and you can use your own pen or pencil. Plan ahead about how to use the scratch paper in combination with what's on the screen.

Use the Tools Effectively!

Online Tools

Eliminate wrong answers
Work steps on the calculator
Look up geometry formulas

Scratch Paper

Rewrite key parts of the question
Write out every calculation
Redraw geometric figures and label them
Rewrite answer choices as needed

Here's a question from your first practice test with an image of the testing app screen on the left and scratch paper on the right. On a question like this, use your scratch paper to rewrite parts of the question and translate them into math, and then use the Answer Eliminator tool to cross out answers that don't match that piece. Always include the question number next to your work on the scratch paper to keep yourself organized.

16

Stella had 211 invitations to send for an event. She has already sent 43 invitations and will send them all if she sends 24 each day for the next d days. Which of the following equations represents this situation?

(A) $\cancel{24d - 43 = 211}$ **Undo**

(B) $24d + 43 = 211$ (B)

(C) $\cancel{43d - 24 = 211}$ **Undo**

(D) $\cancel{43d + 24 = 211}$ **Undo**

16)

24 a day for d days
$24d$

add to 43 sent,
not subtract

Calculators

Calculators are permitted on every Math question on the Digital SAT. In addition, the testing app includes a built-in calculator with many, many features. Practice with the built-in calculator or the one you're planning to bring with you. We'll tell you more about calculators as we go along.

The Princeton Review Approach

We're going to give you the tools you need in order to handle the easier questions on the Digital SAT Math section, along with several great techniques to help you crack some of the more difficult ones. But you must concentrate first on getting the easier questions correct. Don't worry about the questions you find difficult on the Math section until you've learned to work carefully and accurately on the easier questions.

When it does come time to look at some of the harder questions, use Process of Elimination to help you avoid trap answers and to narrow down your choices if you have to guess. You will learn to use POE to improve your odds of finding the answer by getting rid of answer choices that can't possibly be correct.

Generally speaking, each chapter in this section begins with the basics and then gradually moves into more advanced principles and techniques. If you find yourself getting lost toward the end of the chapter, don't worry. Concentrate your efforts on principles that are easier to understand but that you still need to master.

THE BIG PICTURE AND IMPORTANT STRATEGIES

The Reading and Writing section of this book describes various ways to eliminate wrong answers. That idea comes into play on the Digital SAT Math section, as well. There are also ways to break down math questions and avoid trap answers. This chapter provides an overview of the strategies you should know in order to maximize your Math score.

BALLPARKING STRATEGY

One way to eliminate answers on the Math section is by looking for ones that are the wrong size, or that are not "in the ballpark." We call this strategy **Ballparking.** Although you can use your calculator on the following question, you can also eliminate one answer without doing any calculations.

1 Mark for Review

In a garden, the corn on the north edge of the garden is 30% shorter than that on the south edge. If the corn on the south edge of the garden is 50 inches tall, how tall is the corn on the north edge of the garden, in inches?

Ⓐ 30

Ⓑ 33

Ⓒ 35

Ⓓ 65

Here's How to Crack It

The question asks for the height of the corn on the north edge of the garden and states that the corn there is shorter than the corn on the south edge, which is 50 inches tall. You are asked to find the height of the corn on the north edge, so the correct answer must be less than 50. Eliminate (D), which is too high. Often, one or more of the bad answers on these questions is the result you would get if you applied the percentage to the wrong value. To find the right answer, take 30% of 50 by multiplying 0.3 by 50 to get 15; then subtract that from 50. The corn on the north edge is 35 inches tall. The correct answer is (C).

READ THE FINAL QUESTION STRATEGY

It's a bad idea to assume you know what a question is going to ask you to do. Make sure to always read the final question *before* starting to work on the question. Write key words or the entire final question on the scratch paper. Then, try to ballpark before you solve.

2 Mark for Review

If $16x - 2 = 30$, what is the value of $8x - 4$?

Ⓐ 12

Ⓑ 15

Ⓒ 16

Ⓓ 28

Here's How to Crack It

The question asks for the value of an expression, but don't just dive in and solve for the variable. First, see if you can eliminate answers by Ballparking, which can also work on algebra questions. To go from $16x$ to $8x$, you would just divide by 2. Dividing 30 by 2 gives you 15, so 28 is way too big. Eliminate it. The correct answer is not likely to be 15, either, because that ignores the -2 and the -4 in the question.

To solve this one, add 2 to each side of the equation to get $16x = 32$. Divide both sides by 2, which gives you $8x = 16$. But don't stop there! The final question asks for $8x - 4$, so (C) is a trap answer. You have to take the last step and subtract 4 from both sides to find that $8x - 4 = 12$. The correct answer is (A).

RTFQ:
Read
The
Final
Question

Get started faster and avoid trap answers by reading and rewriting the actual question being asked.

BITE-SIZED PIECES STRATEGY

When dealing with complicated math questions, take it one little piece at a time. We call this strategy **Bite-Sized Pieces**. If you try to do more than one step at a time, especially if you do it in your head, you are likely to make mistakes or fall for trap answers. After each step, take a look at the answer choices and determine whether you can eliminate any.

Try the following question.

3 Mark for Review

A paper airplane is thrown horizontally from the top of a hill. It travels in a straight line, and as it moves forward, it also descends. The plane moves horizontally at 9 feet per second while descending one foot for every 3 feet traveled horizontally. After 5 seconds of travel, how many feet has the plane descended from the height at which it was thrown?

Ⓐ 3

Ⓑ 10

Ⓒ 15

Ⓓ 20

Bite-Sized Pieces: Do one small, manageable piece at a time and keep writing things down.

Here's How to Crack It

The question asks how many feet the plane has descended after 5 seconds. There are a few things going on here. The plane is traveling horizontally, and it is also descending. Start by figuring out how far it travels horizontally. It moves in that direction at 9 feet per second for 5 seconds, so it moves horizontally $9 \times 5 = 45$ feet. It descends 1 foot for every 3 traveled horizontally. If it goes 45 feet horizontally, it will descend more than 3 feet, so eliminate (A). Now figure out how many "3 feet" are in 45 feet—for each one of them, the plane will descend 1 foot. Since $45 \div 3 = 15$, the plane descends 15 feet. The correct answer is (C).

You may also have noticed that all the numbers in the question are odd. This makes it unlikely that the answer would be 10 or 20, which are even. If you see things like that, use them as opportunities to eliminate.

Here's another example.

4 Mark for Review

$$(5ck^2 + 5c^2 - 2c^2k) - (ck^2 + 2c^2k + 5c^2)$$

Which of the following is equivalent to the expression above?

(A) $4ck^2$

(B) $4ck^2 - 4c^2k$

(C) $5c^2k^4 - 10c^4k$

(D) $8c^2k^3 + 7c^2k - 5c^2$

Here's How to Crack It

The question asks for an expression that is equivalent to the difference of two polynomials. In math class, your teacher would want you to combine all like terms and show your work, but this isn't math class. Start with one tiny piece of this intimidating question. The first set of parentheses starts with a term containing ck^2. Check the second set of parentheses for the same combination of variables and exponents. The first term there matches, so the first step to take is $5ck^2 - ck^2 = 4ck^2$. There are no other terms with ck^2, so the correct answer must contain $4ck^2$. Eliminate (C) and (D). Now you have a fifty-fifty chance of getting it right, so you can guess and go, *or* you can do one more step to determine whether the answer is (A) or (B). The difference between the two answers is the $-4c^2k$ term, so focus on the terms in the expression that contain c^2k. In the first set of parentheses, you have $-2c^2k$, and then you subtract the $2c^2k$ term in the second set of parentheses to get $-2c^2k - 2c^2k = -4c^2k$. The correct answer is (B).

WORD PROBLEMS

The two strategies we just showed you—RTFQ and Bite-Sized Pieces—are a large part of the approach to word problems on the Digital SAT. The test-writers will try to make things difficult to understand by making a story out of the math. To make sure you have the best shot at reaching the correct answer quickly and accurately, follow this basic approach.

Word Problem Basic Approach

1. **Read the Final Question (RTFQ)**—Understand the actual question being asked. Write down key words.
2. **Let the answers point the way**—Use the answer type to help determine how to start working on the question.
3. **Work in bite-sized pieces**—Find one piece to start with, then work piece-by-piece until the final question has been answered.
4. **Use POE**—Check to see whether any answers can be eliminated after each bite-sized piece.

This approach can be helpful on questions that are "just" math, but they are vital on word problems. Here are some details about the word problem basic approach.

RTFQ—The final question will start with something like *Which of the following, What is,* or *How many.* Find the final question (it's usually at the end) and write down key words. If the question asks for the value of a variable or the measure of an angle, write down which variable or which angle. If it asks for a specific part of a graph or a word problem, write down which part. Terms and units, such as *median, positive, minutes,* or *miles,* also go on the scratch paper.

Let the answers point the way—On multiple-choice questions, the answer type often gives a clue about how to approach the question. Do the answers have numbers? variables? equations? graphs? a bunch of words? Use that information to get started.

Work in bite-sized pieces—Rather than trying to plan the entire question up front, just get started. Work the question one bite-sized piece at a time, reading more along the way and making notes on the scratch paper. The final question and the answer types usually reveal the best approach. If it's not obvious, either mark the question to come back to later or enter a guess.

Use POE—On some questions, it's possible to eliminate answers along the way while working in bite-sized pieces. If the question asks about an equation representing a situation, for example, an answer that gets any piece of the equation wrong can be eliminated. Eliminate answers that don't work when you plug them in, answers that are clearly too big, too small, or have the wrong sign, and answers that contradict information given in the question.

THE CALCULATOR

You are allowed to use a calculator on every question in the Math section, although it doesn't help on every question. You have two calculator options:

- Use the built-in Desmos calculator within the testing app.
- Bring your own approved scientific or graphing calculator.

Whether you use the built-in calculator or your own, practice, practice, practice! A calculator can make some questions much easier to answer and will save you time on other questions. To practice with the built-in calculator, download the Bluebook app or just use the free version on the Desmos website.

Calculator Guide

Head to your Student Tools to read our Guide for the built-in calculator. There you will find basic information about opening and using the calculator, details about how to use the keypads and advanced functions, and instructions for getting the most out of the graphing calculator. This Guide also includes a number of questions from your first practice test with detailed instructions and screenshots to show you how to solve them using the built-in calculator.

Even if you are planning to use your own calculator, this Guide might help you think of ways to use it that you hadn't considered before.

Call on the Calculator
Practice using the built-in calculator in your Student Tools or your own throughout your test preparation. Use it for example questions in this book, use it while taking practice tests, and use it any time you're doing practice questions for the Digital SAT.

Personal Calculator

If you have a good calculator that you are familiar with and like using, you may take it with you and use it on the test. Make sure that your calculator is either a scientific or a graphing calculator and can perform the order of operations correctly. To test your calculator, try the following problem, typing it in exactly as written without hitting the ENTER or "=" key until the end: $3 + 4 \times 6 =$. The calculator should give you 27. If it gives you 42, it's not a good calculator to use.

If you do decide to use your own graphing calculator, keep in mind that it *cannot* have a QWERTY-style keyboard (like the TI-95). Most of the graphing calculators have typing capabilities, but because they don't have typewriter-style keyboards, they are perfectly legal. To see the full College Board calculator policy, visit satsuite.collegeboard.org/digital/what-to-bring-do/calculator-policy.

Also, you *cannot* use the calculator on your phone. In fact, on test day, you will have to turn your phone off and put it away.

The only danger in using a calculator on the Digital SAT is that you may be tempted to use it in situations in which it won't help you. Some students believe that their calculator will solve many difficulties they have with math. This type of thinking may even occasionally cause students to miss a question they might have otherwise answered correctly on their own.

Remember that your calculator is only as smart as you are. But if you practice and use a little caution, you will find that your calculator will help you a great deal.

What a Calculator Is Good at Doing

Here's a list of some of the things a calculator is good for on the Digital SAT:

- arithmetic
- decimals
- fractions
- square roots
- percentages
- graphs (if it is a graphing calculator)

We'll discuss the calculator's role in most of these areas in the next few chapters.

Calculator Arithmetic

Adding, subtracting, multiplying, and dividing integers and decimals is easy on a calculator. But, you need to be careful when you key in the numbers. A calculator will give you an incorrect answer to an arithmetic calculation if you press the wrong keys.

Calculators Don't Think for You
A calculator crunches numbers and often saves you a great deal of time and effort, but it is not a substitute for your problem-solving skills.

The main thing to remember about a calculator is that it can't help you find the answer to a question you don't understand. If you wouldn't know how to solve a particular problem using pencil and paper, you won't know how to solve it using a calculator either. Your calculator will help you, but it won't take the place of a solid understanding of basic Digital SAT mathematics.

Use Your Paper First

When you decide to use a calculator to help answer a question, the first step should be to set up the problem or equation on paper; this will keep you from getting lost or confused. This is especially important when solving the problem involves a number of separate steps. The basic idea is to use the scratch paper to make a plan, and then use your calculator to execute it.

Write Things Down
You have scratch paper, so make the most of it. Keep track of your progress through each question by writing down each step.

Working on paper first will also give you a record of what you have done if you change your mind, run into trouble, or lose your place. If you suddenly find that you need to try a different approach to a question, you may not have to go all the way back to the beginning. This will also make it easier for you to check your work, if you have time to do so.

Don't use the memory function on your calculator (if it has one). Because you can use your scratch paper, you don't need to juggle numbers within the calculator itself. Instead of storing the result of a calculation in the calculator, write it on your scratch paper, clear your calculator, and move to the next step of the question. A calculator's memory is fleeting; scratch paper is forever.

Order of Operations

In the next chapter, we will discuss the proper order of operations for solving equations that require several operations to be performed. Be sure you understand this information, because it applies to calculators as much as it does to pencil-and-paper computations. You may remember PEMDAS from school. PEMDAS is the order of operations. You'll learn more about it and see how questions on the Digital SAT require you to know the order of operations. You must always perform calculations in the proper order.

Fractions

Most scientific calculators have buttons that will automatically simplify fractions or convert fractions from decimals. (For instance, on the TI-81, TI-83, and TI-84, hitting "Math" and then selecting the first option, "Answer → Fraction," will give you the last answer calculated as a fraction in the lowest terms.) Find out if your calculator has this function! If it does, you can use it to simplify messy fractions. This function is also very useful when you get an answer as a decimal, but the answer choices given are all fractions.

The Digital SAT Calculator Guide has details about working with fractions, along with other types of math and graphs. Pull up the Guide in your Student Tools and become an expert!

Batteries

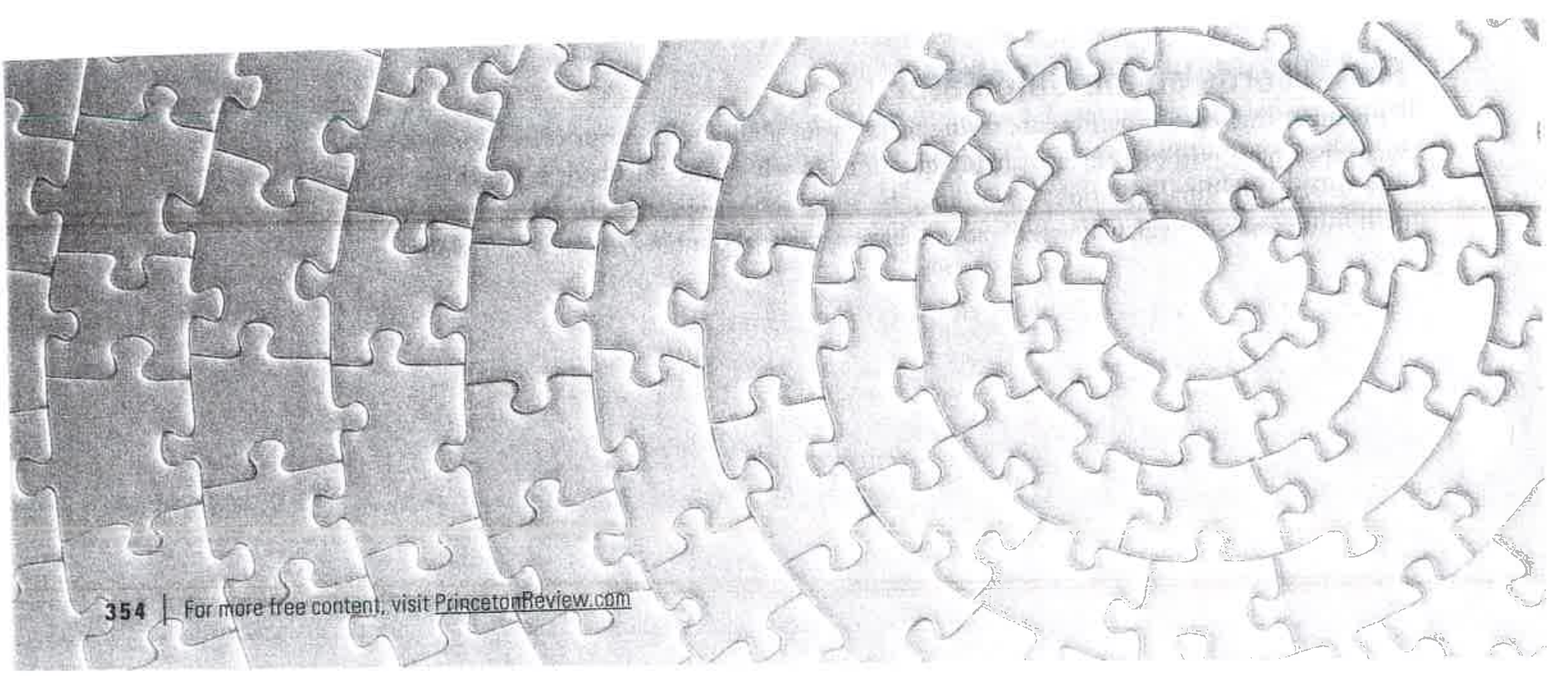
Change the batteries on your calculator a week before the Digital SAT so that you know your calculator won't run out of power halfway through the test. You can also take extra batteries with you, just in case. Although it isn't very likely that the batteries will run out on your calculator on the day of the test, it could happen—so you want to be prepared.

Final Words on the Calculator

Remember that the test-writers are trying to test your ability to use your calculator wisely. As such, they have purposely created many questions in which the calculator is worthless. So be sure to read the final question—there may be some serious surprises in there. Practice your math skills *and* your calculator skills so you have options about how you solve questions.

Summary

- Look for ways to eliminate answer choices that are too big or too small. Ballparking can help you find the right answer without extensive calculations, avoid trap answers, and improve your chances of getting the question right even if you have to guess.
- Use the built-in tools as much as possible. The most useful ones on the Math section are the Calculator, Reference Sheet, and Answer Eliminator tool.
- Use your scratch paper constantly: number the work for each question, write down key words from the final question, redraw geometric figures, and write down every step of math. Even if you use a calculator, it's worth setting up the math on your scratch paper to stay organized and avoid mistakes.
- Utilize the Word Problem Basic Approach: Read the Final Question (RTFQ), Let the Answers Point the Way, Work in Bite-Sized Pieces, and use Process of Elimination (POE).
- Practice with the calculator you plan to use for the test: either the built-in Desmos calculator or your personal scientific or graphing calculator.
- If you are going to use the built-in calculator, read the Digital SAT Calculator Guide in your Student Tools to maximize its effectiveness.
- If you are going to use your own calculator, make sure it is on the approved list and has fresh batteries.
- Set up the question on the scratch paper before entering anything into a calculator. By doing so, you will eliminate the possibility of getting lost or confused.
- A calculator can't help you find the answer to a question you don't understand. Be sure to use your calculator as a tool, not a crutch.
- Whether you are using a calculator or not, you must always perform calculations in the proper order (PEMDAS).



CHAPTER 17: MATH INTRODUCTION AND GENERAL STRATEGIES

There are often several ways to solve a math problem. You don't have to solve it the way the SAT® wants you to. Since our purpose is to get you to the correct answer in the quickest amount of time, we'll walk you through the strategies, techniques, tips, etc., that our students find extremely helpful. A number of these strategies may seem unfamiliar. However, they have been proven to work and, with practice, they can work for you too.

The College Board tells us that the questions on the Math section fall into four content domains:

1. Algebra
2. Advanced Math
3. Problem-Solving and Data Analysis
4. Geometry and Trigonometry

We've identified the specific categories most commonly tested within these headings, with each question category representing its own chapter, with the specific strategy for tackling each one.

MATH GENERAL STRATEGIES

1. **NEVER leave a question blank.** There is no penalty for guessing.

2. **Complete easy questions first.** Questions follow a loose order of difficulty, meaning they tend to get harder as you go. Spend your time wisely on the questions likely to award you points. Pick and choose which questions you can complete when time is running out.

3. **Abide by a 5-second rule (or something close to it).** If after 5 or so seconds you have no idea what to do, skip that question and return to it if time allows.

5. **BALLPARK:** You may be able to get an answer in the ballpark, even if you don't know how to fully solve a problem.

6. **BITE-SIZE:** Break the question down into manageable Bite-sized steps when you get to a stopping point after each step.

7. **Use your CALCULATOR wisely.** Don't trust it unquestioningly; you should always be mentally checking to see if what it's spitting out makes sense.

8. **Know when to walk away!** Beware time-sucker problems. Take a guess and move on.

9. For EVERY math question, ask if the question would be easier if I **Plugged In** for the Variable or Plugged in the Answer.

MATH

QUESTION CATEGORY	STRATEGY
Plug In for the Variable	Find the Target Value
Plug in the Answer	Start with B and Label Columns
Basic Algebra	Isolate the variable
Slope Intercept Form of a Line	Write out the Formulas
Proportions	Write out Top and Bottom Units
Functions	Think of f(x) as y
Systems of Equations	Substitute, Stack, or Graph
Inequalities	Flip the Sign
Quadratics	Know the Basic Equations
Graphs	Select Quickest Option
Percents	Translate Text to Math
Exponents/Radicals	MADSPM
Mean/Median/Mode/Range	Use Average Pie for Mean
Probability	Success/Total
Represents Situation	Bite-size and POE
Geometry	Draw/Label/Complete Formulas/Carve
Trigonometry	SOHCAHTOA
Statistics and Survey Design	Large and Random Sample Sizes are Best

WHAT IS A FILL-IN?

Both Math modules on the Digital SAT have several questions without answer choices. The exact breakdown per module varies, but you can expect to see about 11 total questions in this format. The test-writers call these Student-Produced Response questions, but we're going to keep things simple and call them fill-ins.

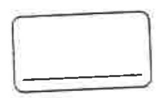
Different Format, Same Content

Fill-in questions test the same math topics as multiple-choice questions:

- Algebra
- Advanced Math
- Problem Solving and Data Analysis
- Geometry and Trig

Despite the different format, many of the techniques that you've learned so far still apply to fill-in questions. You can't use Process of Elimination or PITA, of course, but you can still use Plugging In, Bite-Sized Pieces, and other great techniques. Your calculator will still help you out on many of these questions, as well.

The only difficulty with fill-ins is getting used to the way in which you are asked to answer the question. For each fill-in question, you will have a box like this:



To enter your answer, click inside the box and start typing. The numbers you enter will automatically appear left to right, and the testing app will show a preview of your answer so you can make sure it looks right.

THE INSTRUCTIONS

The fill-in instructions will appear on the left side of the screen for every fill-in question. There are buttons in the middle of the screen that look like this:

You can use the buttons to make the instructions on the left and the question on the right take up more or less of the screen. The instructions and examples in the testing app look like those on the next page.

Student-produced response questions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	31/2 3 1/2
$\frac{2}{3}$	2/3 .6666 .6667 0.666 0.667	0.66 .66 0.67 .67
$-\frac{1}{3}$	−1/3 −.3333 −0.333	−.33 −0.33

Fill the Space
Know what you can and can't enter into the fill-in box:

- 5 characters for a positive answer
- 6 characters for a negative answer
- Don't enter extra zeros if the answer is short
- Do enter as much of a long decimal as will fit
- Don't enter a fraction that doesn't fit
- Do enter reduced fractions that fit

You don't want to have to spend time rereading the instructions every time, so make sure you know them well. Here is all the information you need to know about entering a fill-in answer:

1. There is space to enter 5 characters if the answer is positive and 6 characters—including the negative sign—if the answer is negative.
2. You do not need to type the comma for numbers longer than three digits, such as 4,200. In fact, the testing app will not allow it.
3. The testing app also will not allow symbols such as %, $, or π. Square roots, units, and variables cannot be entered.
4. You can enter your answer as either a fraction or a decimal. For example, .5, 0.5, and 1/2 are all acceptable answers. Use the forward slash for fractions.
5. If your answer is a fraction, it must fit within the space. Do not try to enter something like $\frac{200}{500}$ as a fraction: either reduce it or convert it to a decimal. Entering 20/50, 2/5, .4, and 0.4 would all count as the correct answer.
6. Fractions do not need to be in the lowest reduced form. As long as it fits, it's fine.
7. You cannot fill in mixed numbers. Convert all mixed numbers to improper fractions or decimals. If your answer is $2\frac{1}{2}$, you must convert it to 5/2 or 2.5. If you enter 21/2, the testing app will read your answer as $\frac{21}{2}$.
8. If your answer is a decimal that will not fit in the space provided, either enter as many digits as will fit or round the last digit. The fraction $-\frac{2}{3}$ can be entered in decimal form as –0.666, –0.667, –.6666, or –.6667.
9. Some questions will have more than one right answer. Any correct answer you enter will count as correct; do not try to enter multiple answers.

FILL-INS: A TEST DRIVE

To get a feel for this format, let's work through two examples. As you will see, fill-in questions are just regular Digital SAT Math questions.

1 Mark for Review

If $a + 2 = 6$ and $b + 3 = 21$, what is the value of $\frac{b}{a}$?

Here's How to Crack It

The question asks for the value of $\frac{b}{a}$. You need to solve the first equation for a and the second equation for b. Start with the first equation, and solve for a. By subtracting 2 from both sides of the equation, you should see that $a = 4$.

Now move to the second equation, and solve for b. By subtracting 3 from both sides of the second equation, you should see that $b = 18$.

The question asked you to find the value of $\frac{b}{a}$. That's easy. The value of b is 18, and the value of a is 4. Therefore, the value of $\frac{b}{a}$ is $\frac{18}{4}$.

That's an odd-looking fraction. How in the world do you fill it in? Ask yourself this question: "Does $\frac{18}{4}$ fit?" Yes! Fill in $\frac{18}{4}$.

18/4

Your math teacher wouldn't like it, but the scoring computer will. You shouldn't waste time reducing $\frac{18}{4}$ to a prettier fraction or converting it to a decimal. Spend that time on another question instead. The fewer steps you take, the less likely you will be to make a careless mistake.

2 Mark for Review

The radius of circle O is 212 times the radius of circle P. If the area of circle O is t times the area of circle P, what is the value of t?

Here's How to Crack It

The question asks for the relationship between the areas of two geometric figures. It doesn't matter that this is a fill-in instead of a multiple-choice question: you still start with the Geometry Basic Approach. Draw two circles on your scratch paper. Next, label the figure. Mark the center of one circle as O and the center of the other as P, and draw in the radius of each circle. The question doesn't give you any numbers for the radius or area of circle P, so plug in. Make the radius 2 and label that on the figure. Finally, write down formulas. The area of a circle is given by $A = \pi r^2$. Plug in $r = 2$ to get $A = \pi(2)^2$, or $A = 4\pi$ for circle P.

The question states that *the radius of circle O is 212 times the radius of circle P*, so multiply 2 by 212 to get $r = (2)(212) = 424$. Label the radius of circle O as 424. Plug $r = 424$ into the area formula to get $A = \pi(424)^2$, or $A = 179{,}776\pi$. To solve for t, divide the area of circle O by the area of circle P to get $t = \dfrac{179{,}776\pi}{4\pi}$, or $t = 44{,}944$. This is a big number! However, it's still only 5 characters long, so it will fit in the fill-in box. The fill-in box doesn't accept commas, so don't worry about that. The correct answer is 44944.

44944

MORE POOD

The fill-in questions are mixed in with the multiple-choice questions, and both math modules have an approximate order of difficulty. More important than the question order is your Personal Order of Difficulty (POOD), a strategy that encourages you to focus on the questions you know how to answer first. Don't spend too much time on a question you are unsure about, no matter which format it is.

Keep in mind, of course, that many of the math techniques that you've learned are still very effective on fill-in questions. The Geometry Basic Approach and Plugging In both worked well on the previous question. If you're able to plug in or take an educated guess, go ahead and fill in that answer. As always, there's no penalty for getting it wrong.

Here's another fill-in question that you can answer by using a technique you've learned before.

---

3 Mark for Review

Town A has 2,200 residents. The mean age of the residents of Town A is 34. Town B has 3,680 residents with a mean age of 40. What is the mean age of the residents of Town A and Town B combined?

Here's How to Crack It

The question asks for a mean, or average. Work the question in bite-sized pieces and start with Town A. For averages, use the formula $T = AN$, in which T is the *Total*, A is the *Average*, and N is the *Number of things*. The question states that *Town A has 2,200 residents*, so that is the *Number of things*. The question also states that the *mean age of the residents of Town A is 34*, so that is the *Average*. Plug these numbers into the average formula to get $T = (34)(2{,}200)$, or $T = 74{,}800$. Do the same thing for Town B: the *Number of things* is 3,680 residents, and the *Average* is the mean age of 40, so the formula becomes $T = (40)(3{,}680)$, or $T = 147{,}200$.

Next, add the two totals to get $74{,}800 + 147{,}200 = 222{,}000$. This is the *Total* for the two towns combined. The *Number of things* for the two towns combined is $2{,}200 + 3{,}680 = 5{,}880$ residents. Use the average formula one more time to get $222{,}000 = (A)(5{,}880)$. Divide both sides of the equation by 5,880 to get $37.7551020408 = A$.

There clearly isn't room to enter this answer in the fill-in box, so either cut it off or round when you run out of room. You can enter 37.75 or 37.76 and get the question right. Don't round too much, though: if you enter 37.8, you'll get the question wrong. Enter the full five characters to get credit for a positive answer. The correct answer is 37.75 or 37.76.

37.75 or 37.76

Careless Mistakes

On fill-in questions, you obviously can't use POE to get rid of bad answer choices, and Plugging In the Answers won't work either. In order to earn points on fill-in questions, you're going to have to find the answer yourself, as well as be extremely careful when you enter your answers in the fill-in box. If you need to, double-check your work to make sure you have solved correctly. If you suspect that the question is a difficult one and you get an answer too easily, you may have made a careless mistake or fallen into a trap.

Try the example below with this in mind.

4 Mark for Review

A teacher is grading two assignments that each had to be a specific length: research papers and short stories. Each research paper has 5 more pages than each short story. How many pages are in a research paper if 7 research papers and 5 short stories have a total of 275 pages?

Here's How to Crack It

The question asks for the number of pages in a research paper given other information about two assignments. Use another skill from earlier in this book and translate English to math in bite-sized pieces. The question states that *each research paper has 5 more pages than each short story*. Let r represent the number of pages in a research paper. The word *has* translates to =. The phrase *5 more than* translates to 5 +. Finally, let s represent the number of pages in a short story. The sentence, therefore, translates to $r = 5 + s$. Do the same thing with the information that *7 research papers and 5 short stories have a total of 275 pages*. Use r and s again for the number of pages in a research paper and a short story, respectively. Translate ***and*** as + and ***have a total of*** as =, and the sentence translates to $7r + 5s = 275$. You now have two equations with the same two variables:

$$r = 5 + s$$

$$7r + 5s = 275$$

Substitute $5 + s$ for r in the second equation to get

$$7(5 + s) + 5s = 275$$

Distribute the 7.

$$35 + 7s + 5s = 275$$

Combine like terms on the left side, then subtract 35 from both sides.

$$12s = 240$$

Isolate s.

$$s = 20$$

It's tempting to fill in 20 and call it a day, but always read the final question! The question asks for the number of pages in a research paper, not in a short story. Plug 20 for s into the first equation to solve for r.

$$r = 5 + 20 = 25$$

Thus, $r = 25$, so fill in that value. The correct answer is 25.

$$\boxed{25}$$

MULTIPLE CORRECT ANSWERS

As you've already seen, some fill-in questions will have more than one possible correct answer. It won't matter which correct answer you enter as long as it really is correct. This happens frequently when the answer is a fraction or a decimal. It can also happen when there is more than one solution to an equation.

Let's look at one of those.

5 Mark for Review

What is one possible solution to the equation $|a + 3| = 7$?

Here's How to Crack It

The question asks for a possible solution to an equation with an absolute value. With an absolute value, the value inside the absolute value bars can be either positive or negative. Set $a + 3$ equal to both 7 and –7, and solve both equations. When $a + 3 = 7$, subtract 3 from both sides of the equation to get $a = 4$. When $a + 3 = -7$, subtract 3 from both sides of the equation to get $a = -10$. Enter either 4 or –10 and you'll get the question right.

In this case, that was more work than you needed to do. The question asked for *one possible solution*, so you could have stopped after finding one value. However, questions about absolute value might ask for a specific solution—either the positive solution or the negative solution—so always read the final question (RTFQ) to make sure you don't enter a value that isn't correct.

4 **or** –10

Fill-In Drill

Answers and explanations can be found starting on page 591.

1 Mark for Review

If $a^b = 4$, and $3b = 2$, what is the value of a?

2 Mark for Review

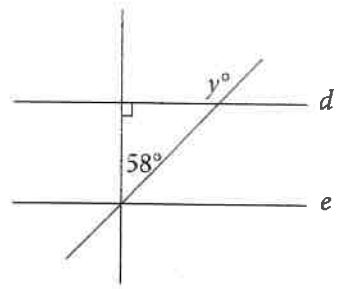
If $4x + 2y = 24$ and $\frac{7y}{2x} = 7$, what is the value of x?

3 Mark for Review

$$n = 12(2)^{\frac{t}{3}}$$

The number of mice in a certain colony is shown by the formula above, such that n is the number of mice and t is the time, in months, since the start of the colony. If 2 <u>years</u> have passed since the start of the colony, how many mice does the colony contain now?

4 Mark for Review



In the figure above, if d is parallel to e, what is the value of y?

5 Mark for Review

The function g is defined by $g(x) = -(x - 3)(x + 11)$. For what value of x is the value of $g(x)$ at its maximum?

6 Mark for Review

If line m is defined by the equation $-3x = -2y - 12$ and line n is parallel to line m, what is the slope of line n?

7 Mark for Review

If Alexandra pays \$56.65 for a table, and this amount includes a tax of 3% on the price of the table, what is the amount, in dollars, that she pays in tax?

8 Mark for Review

In triangle ABC, where angle A is a right angle, $\sin(C)$ is $\frac{13}{85}$. What is the value of $\tan(B)$?

9 Mark for Review

The kinetic energy (KE) of a ball in motion is given by the equation $KE = \frac{1}{2}mv^2$, where m is the mass of the ball in kilograms (kg) and v is the velocity in meters per second $\left(\frac{\text{m}}{\text{s}}\right)$. A ball with a mass of 5 kg and a kinetic energy of 18.225 kg$\left(\frac{\text{m}^2}{\text{s}^2}\right)$ is to be rolled along the ground. What is the velocity of the ball in meters per second, assuming there is no friction?

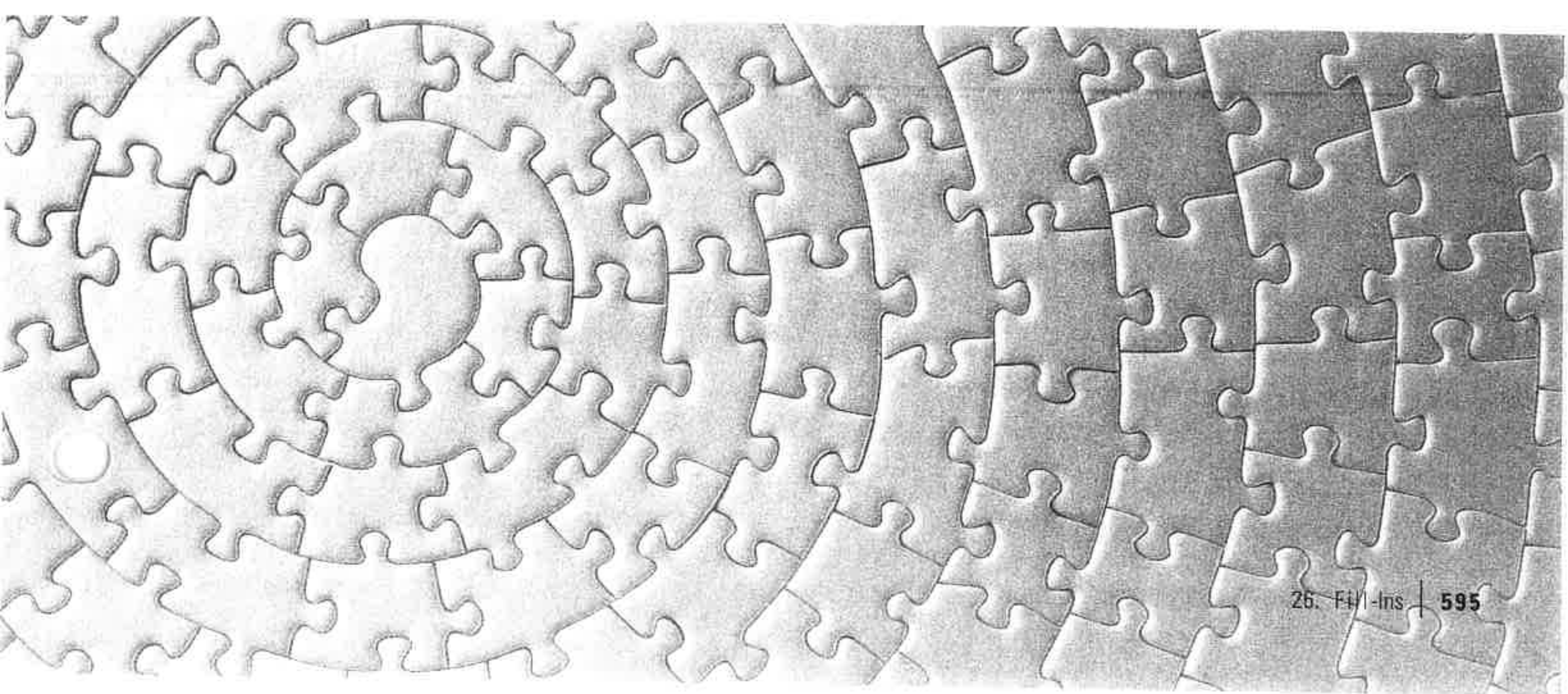
10 Mark for Review

$$x(mx + 42) + 18 = 0$$

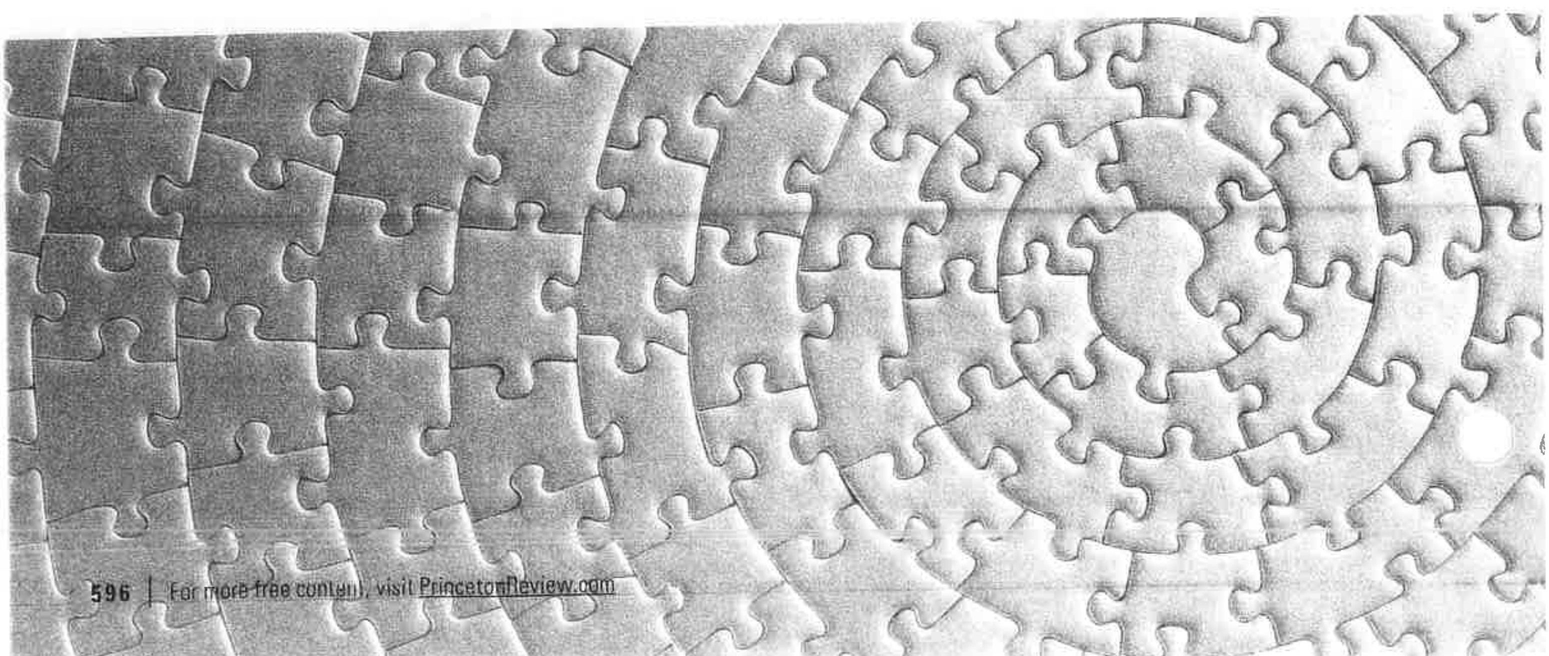
If the equation above has exactly two real solutions and m is an integer constant, what is the greatest possible value of m?

Summary

- Both of the Math modules on the Digital SAT contain several questions without answer choices. The test-writers call these questions "student-produced responses." We call them fill-ins because you have to fill in your own answer.
- Despite their format, fill-ins are really just like other Math questions on the Digital SAT, and many of the same techniques that you have learned still apply.
- The fill-in questions and multiple-choice questions are mixed together in a loose order of difficulty. Use your knowledge of your own strengths and weaknesses to decide which ones to tackle first and which ones, if any, to skip.
- The fill-in format increases the likelihood of careless errors. Know the instructions and check your work carefully.
- Just like the rest of the Digital SAT, there is no guessing penalty for fill-ins, so you should always fill in an answer, even if it's a guess.
- Enter only one answer even if the question has multiple possible answers. It doesn't matter which answer you enter, as long as it's one of the possible answers.
- Enter up to 5 characters when the answer is positive. Enter up to 6 characters, including the negative sign, when the answer is negative.
- The characters that can be entered are the digits 0–9, the negative sign, the forward slash (/) for fractions, and the decimal point. Special characters such as % or π cannot be entered.
- If the answer to a fill-in question contains a fraction or decimal, you can enter the answer in either form. Use whichever form is easier and less likely to cause mistakes.
- If your answer is a fraction that doesn't fit in the space, either reduce the fraction or convert it to a decimal.



- If a fraction fits in the space, you don't have to reduce the fraction before entering it.
- Do not enter mixed numbers. Convert mixed numbers to fractions or decimals before entering your answer.
- If your answer is a long or repeating decimal, fill up all of the space. Either keep entering digits until the space is full or round the last digit that will fit.



DESMOS CALCULATOR GUIDE

While you are allowed to bring your own calculator for the digital exam just as you would for the paper exam, the digital exam will now contain a built-in graphing calculator for you to use through the Desmos interface. This calculator is similar to the regular TI-84 calculator that you are used to in class, but there are some important differences.

Algebraic Math

Maximizing Your Performance

When tackling the algebraic challenges of the SAT, the Desmos calculator emerges as a powerful ally, mirroring the familiar functionalities of the classic TI-84 calculator. As you navigate through the exam, you'll find that inputting calculations into Desmos is a straightforward process: simply type your expression on the left side of the screen, keeping the order of operations firmly in mind. The answer will conveniently appear directly below your input.

Efficiency at Your Fingertips

One of the standout features of the Desmos calculator is its ability to enhance your workflow through the copy-paste function, allowing you to reuse previous calculations. Additionally, the expansive screen real estate means that you can view a comprehensive history of your past calculations without the need to scroll, allowing you to reference previous work.

Navigating Desmos: A Quick Guide

See below for how to execute common calculator functions, tailored specifically for the SAT algebra section, covering basic arithmetic operations to more advanced functions such as square roots, exponents, and trigonometric calculations.

$$\sqrt{9} + \sqrt[3]{27} \cdot 4^2 + \mathrm{abs}(-8)$$

$$= 59$$

$$\sin\left(\frac{\pi}{2}\right) + \cos(\pi) - \tan(0)$$

$$= 0$$

$$e^3 + \ln(10) + \log(100)$$

$$= 24.3881220162$$

$$24.3881220162 + 59$$

$$= 83.3881220162$$

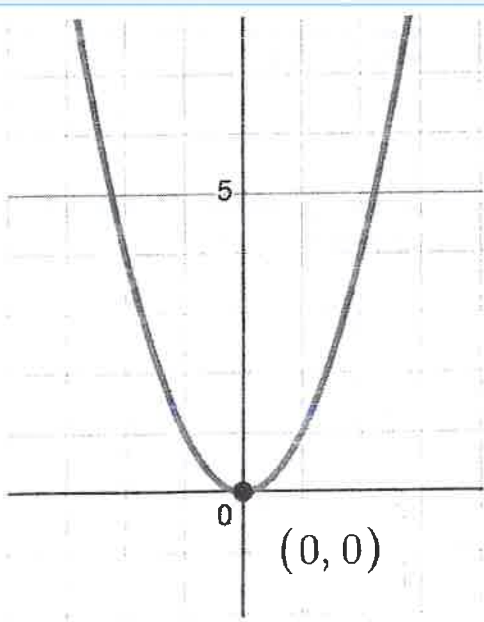
Graphing

Graphing Mastery

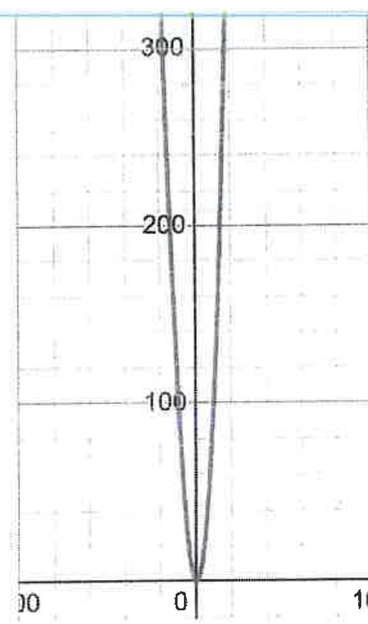
Embarking on the graphing section of the SAT, you'll find that the Desmos calculator offers an innovative approach, distinct from traditional graphing calculators. Desmos provides an expansive, infinite graphing canvas, ensuring that you are never confined by screen limitations. This feature is particularly advantageous for visualizing the behavior of functions across a wide range of values. Zooming in and out is achieved with a simple scroll of your mouse or touchpad, allowing for precise examination or a broader overview as needed.

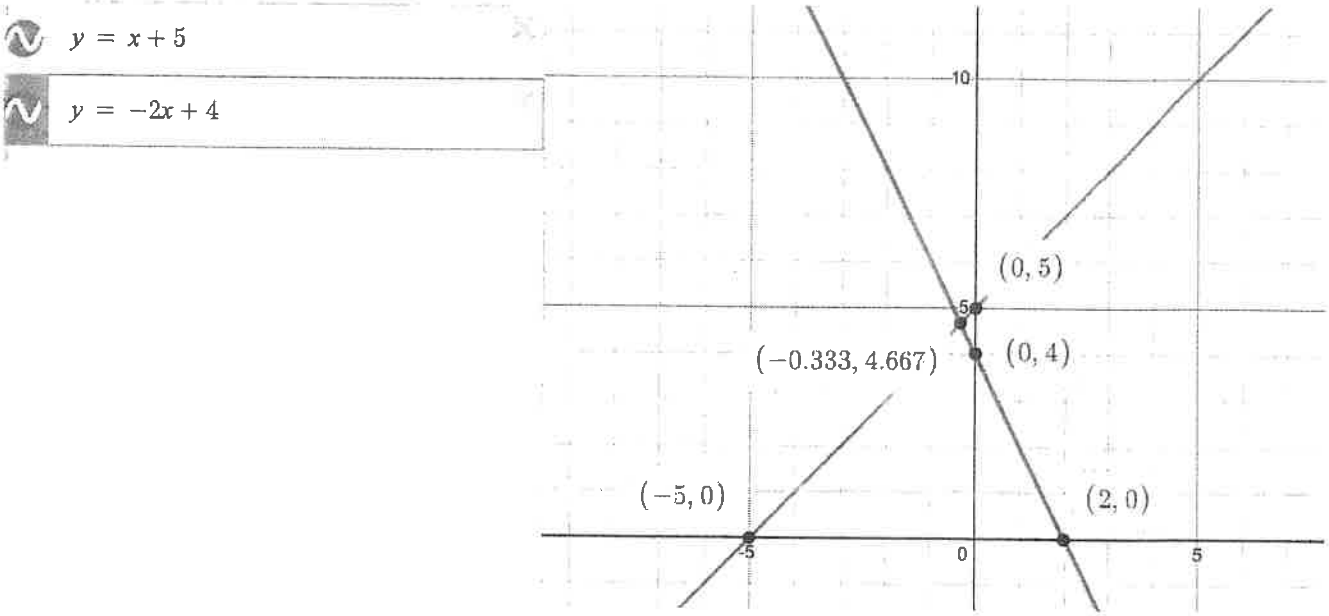
Inputting graphing equations is seamlessly integrated into the Desmos interface. You can enter equations for graphing in the same area where you perform algebraic calculations, starting with \(y = \) or another dependent variable. The Desmos calculator immediately generates a vivid, color-coded graph on the right side of the screen, providing a clear visual representation within the infinite 2-D graphing space. This color-coding is especially helpful when dealing with multiple functions, as it allows for quick and easy differentiation between graphs.

The Desmos calculator revolutionizes the way you find key features of graphs. Unlike traditional calculators, which may require a series of button presses and menu navigations, Desmos enables you to interact directly with the graph using your mouse or touchscreen. Simply click on or hover over key points on the graph to reveal their coordinates, intersection points and roots, and other important information. Additionally, Desmos assists in locating and understanding the maximum or minimum points of a graph, crucial for analyzing the behavior of various functions.



⇐ Same Graph ⇒





* Here I can see all important points on the graphs like intersections and intercepts

Function	How to Get
Square Root	Type **sqrt** then whatever you want inside it then the right arrow key to get out from under the radical
Cube Root	Type **cbrt** then whatever you want inside it then the right arrow key to get out from under the radical
Exponent	Use **Shift + 6** to get into the exponent and use the right arrow key to get out of the exponent
Sine	Type **Sin** then input what you want the sine of making sure to use parenthesis
Cosine	Type **Cos** then input what you want the cosine of making sure to use parenthesis
Tangent	Type **Tan** then input what you want the tangent of making sure to use parenthesis
e	Just type out **e** regularly and using proper order of operations, Desmos will recognize it as the constant number 2.718
Log / Ln	Just like the trig functions type **Log or Ln** followed by what you are looking for in parentheses
Pi	Just type out **pi** and it will turn into π
Absolute Value	Type **abs** and put what you want in parentheses to get the absolute value function

Some Example Question Uses

1. What is 20% of 340 ?

A) 30
B) 20
C) 34
D) 68

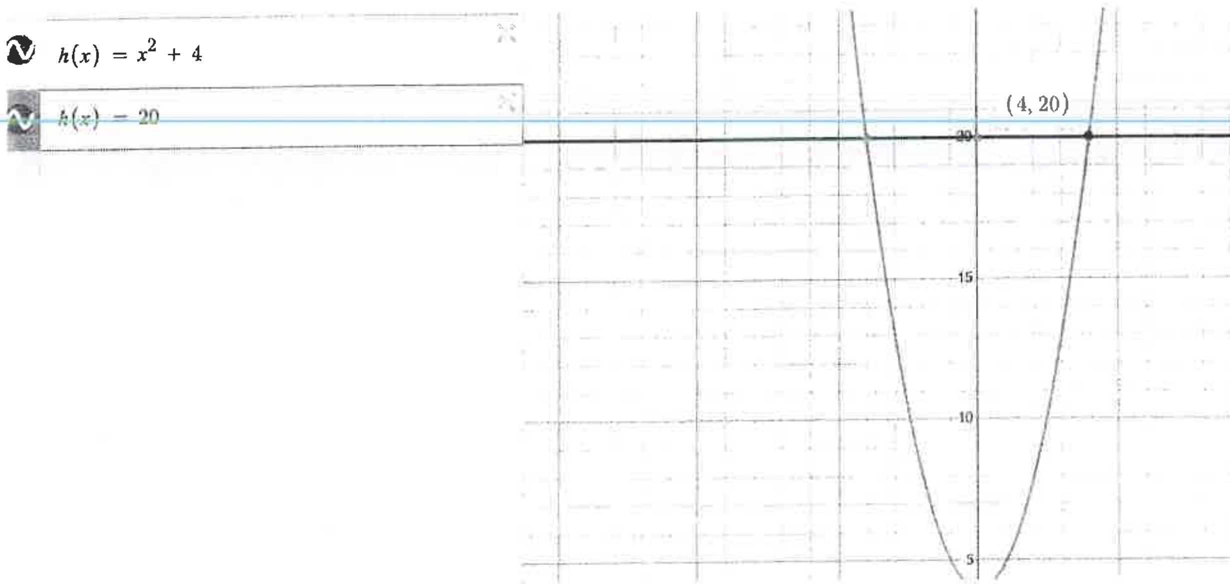
$.2 \cdot 340$

$= 68$

2. The function h is defined by $h(x) = x^2 + 4$. For which **Solution**
Type 20% as a number which means moving the decimal over 2 places to get .2. Then multiply this number by 340 and you will get answer D) 68.

h value of x is $h(x) = 20$?

A) 2
B) 3
C) 4
D) 5



Solution

First plot the graph $h(x)$ in DESMOS then plot the line we want the function to equal. Then hover over the intersection to see where the two lines intersect and see what the x value is which in this case is C) 4.

3. The function f is defined by the equation $f(x) = 5x - 6$. What is the value of $f(x)$ when $x = 5$?

A) 19
B) 21
C) 25
D) 30

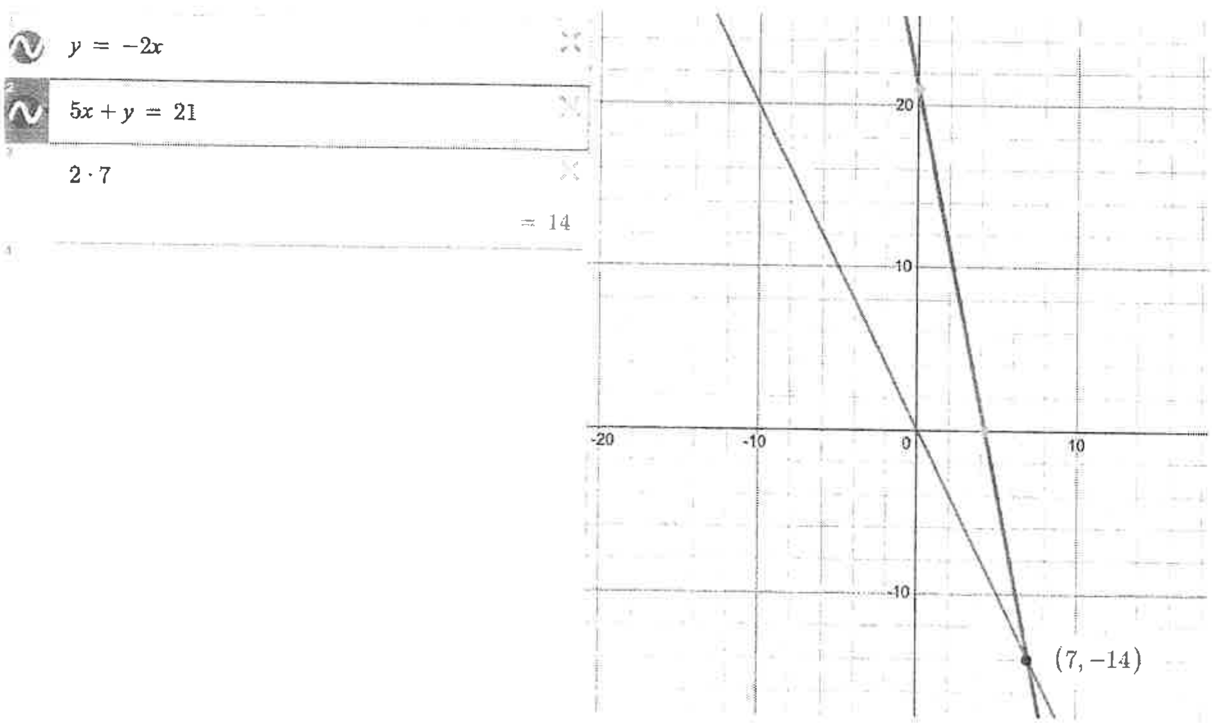
Solution

Since we know what we want x to equal and we have the function, we can just plug 5 into our function for x and figure out what the corresponding $f(x)$ value should be, which is A) 19.

$$\begin{aligned} y &= -2x \\ 5x + y &= 21 \end{aligned}$$

4. The solution to the given system for equations is (x,y). What is the value of $2x$?

A) 7
B) -14
C) 14
D) 21



Solution

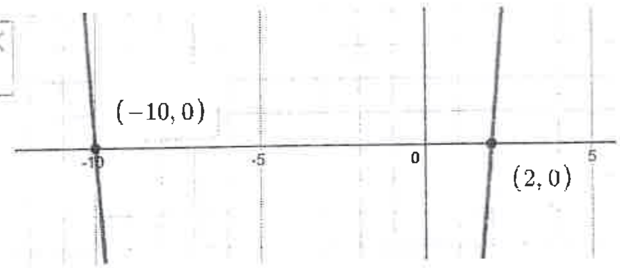
First plot the graphs in DESMOS then hover over the intersection to see where the two lines intersect and see what the x value of the intersection is which in this case is 7. Then since they ask for $2x$, we can multiply this value of 7 times 2 to get a final answer of C) 14.

$c^2 + 8c - 20$

5. What is one solution to the given equation

A) 2
B) 4
C) 8
D) 20

$x^2 + 8x - 20$



Solution

First plot the graph in DESMOS and look at the x-axis because that is where the roots/zeros/solutions will be. The variable itself does not matter so even though the question uses c, we will use x when we plug into DESMOS. Hover over these points to reveal both solutions and then look at the answer choices to see if there are any that match these points with the correct x values. The answer is A) 2.

6. Bacteria are growing in a liquid growth medium. There were 200 cells per milliliter during an initial observation. The number of cells per milliliter doubles every 5 hours. How many cells per milliliter will there be 20 hours after the initial observation?

A) 200
B) 800
C) 1000
D) 3200

$y = 200(2)^{\frac{20}{5}}$

$y = 3200$

Solution

Since this is an exponential function, we can plot this in DESMOS using our traditional equation for an exponential. We know 200 is the initial value, since it doubles the rate inside the parenthesis is 2 and we know that $t = 20$ which should go into our exponent. We then see that the answer is D) 3200.

7. A circle in the xy-plane has a diameter with endpoints (4, 2) and (4, 14). An equation of this circle is $(x-4)^2+(y-8)^2=r^2$, where r is a positive constant. What is the value of r?

A) 4
B) 5
C) 6
D) 7

$$\sqrt{(14-2)^2+(4-4)^2}$$

$$= 12$$

$$\frac{12}{2}$$

$$= 6$$

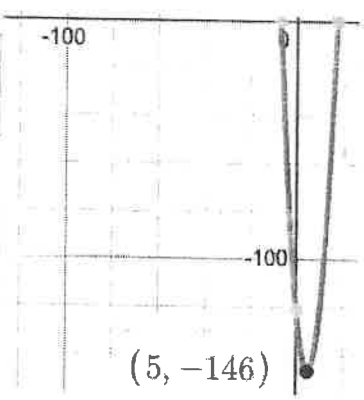
Solution

The first thing that we do is use the distance equation to find the length between the endpoints of the circle which we know will be the diameter. Once we find this, we know that the diameter is just twice the radius, so we need to divide this value (12) by 2 to get the radius C) 6.

8. In the xy-plane, the graph of the equation $y = x^2 - 10x - 121$ intersects the line $y = c$ at exactly one point. What is the value of c?

A) -100
B) -121
C) -140
D) -146

$$x^2 - 10x - 121$$

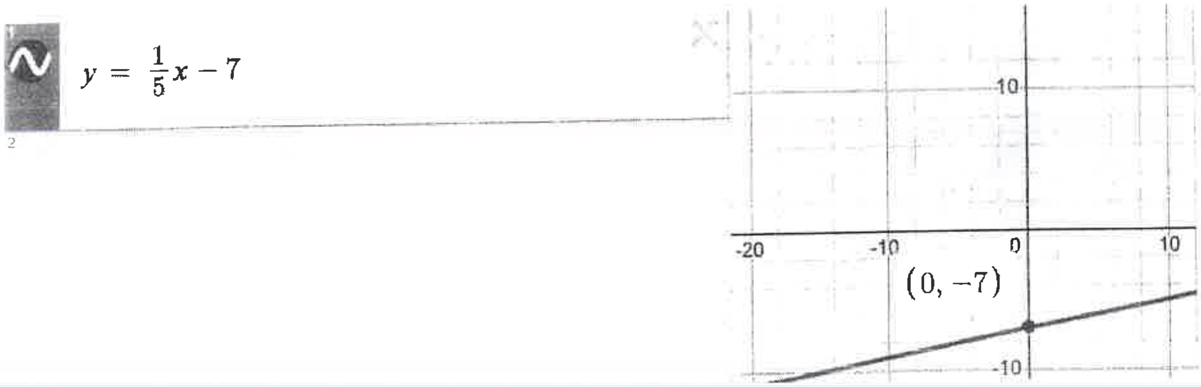


Solution

Plot this equation is DESMOS and then we know that the only time a parabola will hit a horizontal line once is at the vertex so we need to find that point. Hover over the vertex point and find the y-value associated with it and then we know that this value is the line that will only hit the parabola once, which is d) (-146).

9. The function g is defined by $g(x) = \frac{1}{5}x - 7$. What is the y-intercept of the graph $y = f(x)$ in the xy-plane?

A) -7
B) -3
C) $\frac{1}{5}$
D) 4



Solution

First plot the graph In DESMOS and since we are looking for the y-intercept we know that this is the value when $x = 0$. So we go down to the graph at this point to figure out what the corresponding y-value is. In this case it is answer C) -7.

$$x + 6 - 12$$
$$(x + 3)^2 = y$$

10. What ordered pair (x,y) is a solution to the given system of equations?

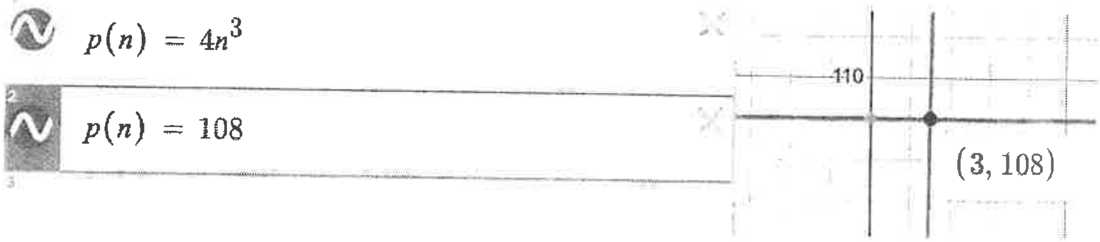
A) (3,36)
B) (4,49)
C) (0,0)
D) (6,81)

Solution

So we can plot both of these graphs just as they are in DESMOS and we are looking for the solution which is also known as the intersection point. We can highlight over this point and see the ordered pair that is the solution for this set of equations.

11. The function p is defined by $p(n) = 4n^3$. What is the value of n when $p(n) = 108$?

A) 2
B) 3
C) 4
D) 5



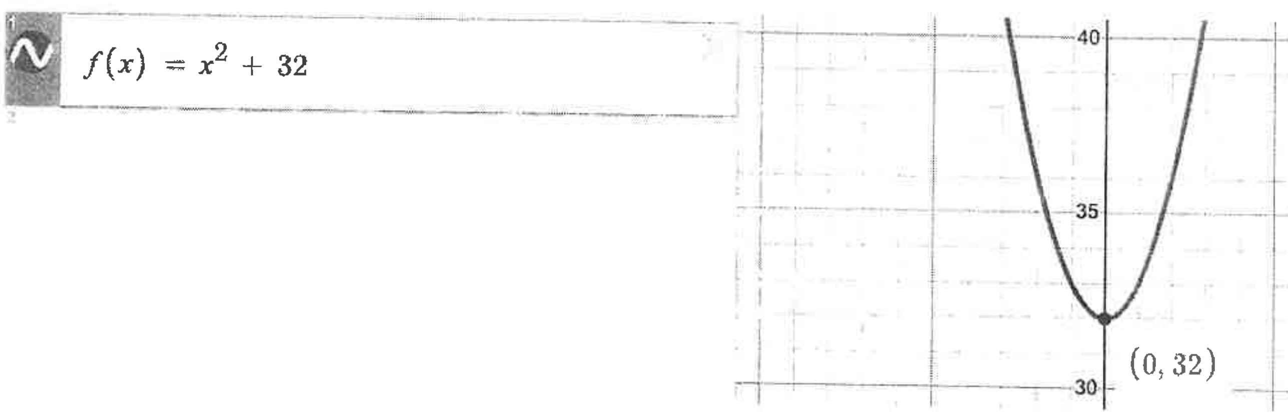
Solution

So first plot the graph $p(n)$ in DESMOS and then plot the value that we want to hit which is 108. Then we want to hover over the intersection point because this will give us the value of n where this happens, which is B) 3.

$$f(x) - x^2 + 32$$

12. What is the minimum value of the given function?

A) 0
B) 20
C) 28
D) 32



Solution

We know that the minimum value of a parabola occurs at the vertex so we are going to plot this graph to find the coordinates of the vertex. Plot this graph in DESMOS and then hover over the vertex to find the points of the vertex and see what the y-value is because that is what corresponds to the minimum value. The answer is D) 32.

3, 5, 7, 9, 3, 2, 23, 43, 7, 2

13. What is the mean of the following data?

A) 7.5
B) 8.3
C) 10.4
D) 12.2

$$\text{mean}(3,5,7,9,3,2,23,43,7,2)$$

$$= 10.4$$

Solution

So here we can use the mean function that DESMOS has. First type mean then add a parentheses. Then add in all of the values each separated by a comma and then close the parenthesis. This function will then automatically give you the mean of the numbers. The answer is C) 10.4

14. Cory is planning a party. It costs Cory a one time fee of $25 to rent the venue and $8.25 per attendee. Cory has a budget of $150. What is the greatest number of attendees possible without exceeding the budget?

A) 12
B) 13
C) 14
D) 15

$$150 - 25$$

$$= 125$$

$$\frac{125}{8.25}$$

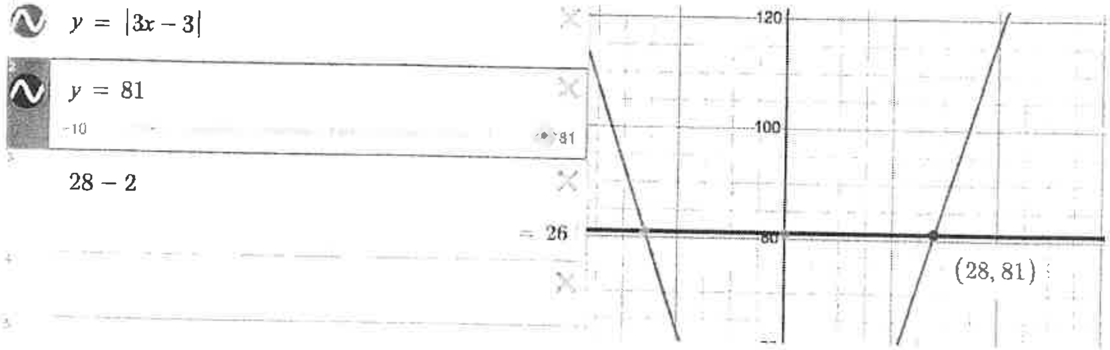
$$= 15.1515151515$$

Solution

So first subtract the 25 dollars from the original 150. Then take the answer of that (125) and divide that by the amount it costs each person to attend. In this case since we cannot have a decimal amount of a person, we must round down to get an answer of 15, which is choice D.

15. If $|3x-3| = 81$, what is the positive value of $x - 2$?

A) 20
B) 26
C) 28
D) 81



Solution

First graph the function and the value of 81 on DESMOS. Now look for the positive intersection point and hover over it to figure out the x value where this intersection happens on the positive side. Now since we want this value minus 2 we can plug that equation in below to solve for our answer, which is B) 26.

16. A cube has an edge length of 48 inches. A solid sphere with a radius of 24 inches is inside the cube, such that the sphere touches the center of each face of the cube. To the nearest cubic inch, what is the volume of the space in the cube not taken up by the sphere?

A) 516,087
B) 520,087
C) 525,243
D) 526,686

$$v = 48^3$$

$$v = 110592$$

$$v = \frac{4}{3}\pi(24)^3$$

$$v = 57905.835791$$

$$110592 - 57905.835791$$

$$= 52686.164209$$

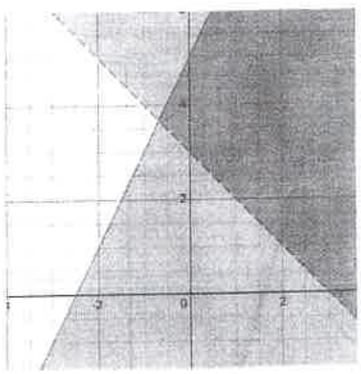
Solution

So first plug into the volume of a cube formula and then do the same thing for the volume of a sphere formula right below it. Use the numbers they give you for the components of the volumes and then subtract these two numbers to see what remaining volume is in he cube and not the sphere. The answer is choice D.

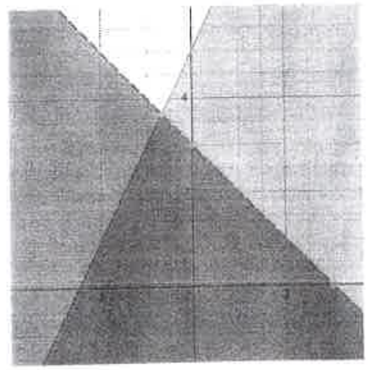
$y > -x + 3$
$y \leq 2x + 5$

17. Which graph represents the solution to the system of inequalities?

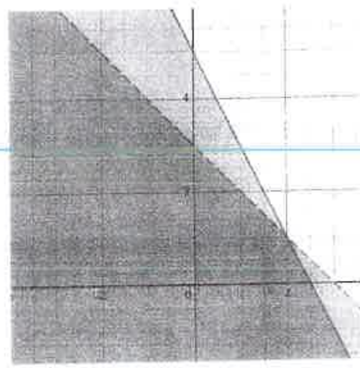
a)



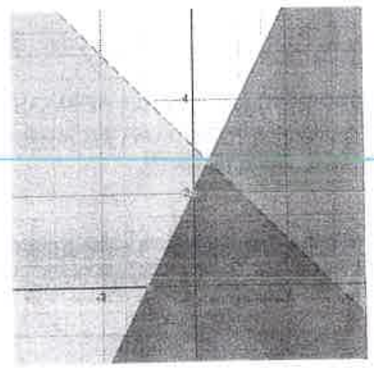
b)



c)



d)

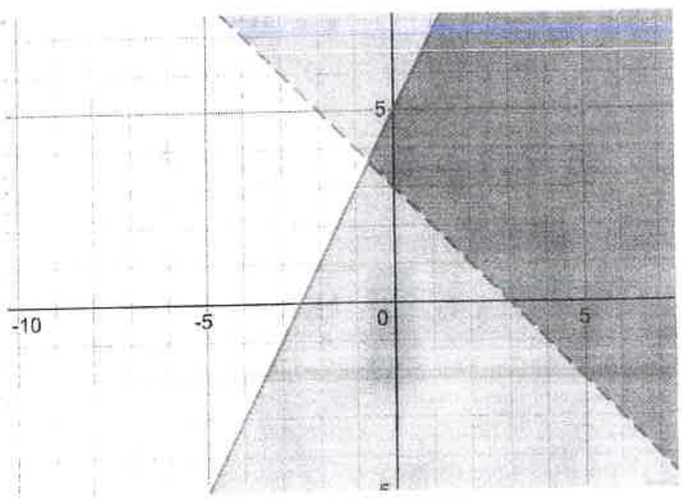


Solution

When we plug the equations into DESMOS it is important to note that the signs are different and represent dotted and solid lines respectively. When we plot the system we look at the choices and see that our graph matches choice A.

$y > -x + 3$

$y \leq 2x + 5$



CALCULATOR TIPS AND TRICKS

PERMITTED CALCULATORS ON THE SAT

Most students use some type of TI-83 or TI-84, which is allowed by the College Board on the SAT exam. Curvebreakers recommends the TI-84 Plus CE. For more information about using calculators on the SAT, read Curvebreakers' detailed blog posts online at curvebreakerstestprep.com/blog.

PROHIBITED DEVICES

Unless students have an accommodation approved by the College Board, they can't access these items during the test or breaks:

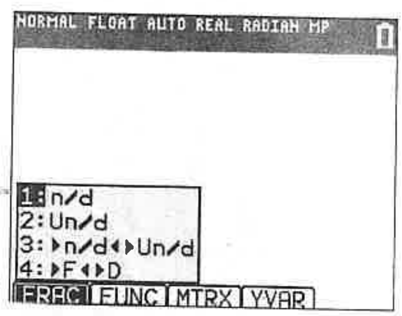
- Phones smartwatches, fitness trackers, or other wearable technology
- Audio players, Bluetooth devices (like wireless earbuds/headphones), or any other electronic devices (except your testing device)
- Detachable privacy screens
- External keyboards for use with laptops or Chromebooks (keyboards for iPads are allowed)
- Stylus for iPad
- Any cameras, recording device, or timer
- Notes, books, or any other reference materials
- Compasses, rulers, protractors, or cutting devices
- Headphones, earbuds, or earplugs
- Unacceptable calculators that have computer-style (QWERTY) keyboards, use paper tape, make noise, or use a power cord

HOW TO USE A GRAPHING CALCULATOR

Inputting A Fraction

1) Press "alpha"
2) Press "y="
3) Press "enter" to select "1: n/d"

3



2

1

NOTE: There are several other functions you'll see in this list.

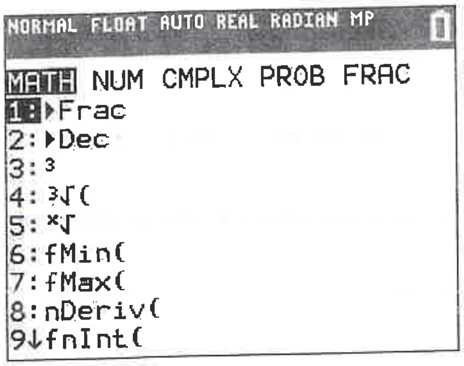
- "2: Un/d" will allow you to input a mixed number
- "3: ▸ n/d ◂ ▸Un/d" will change a fraction into a mixed number

To get a decimal in fraction form:

1) Press "math"
2) Press "enter" to select "1: ▶ Frac"
3) Press "enter"

*To get from fraction form into decimal, repeat the procedure, but select "2: ▶ Dec"

2



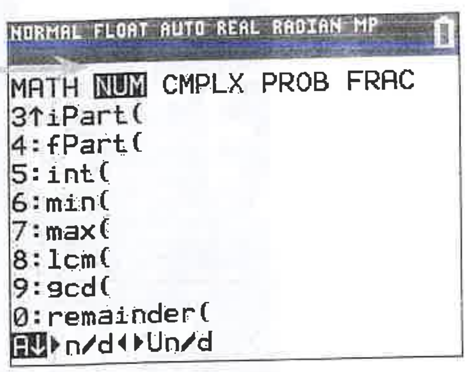
1

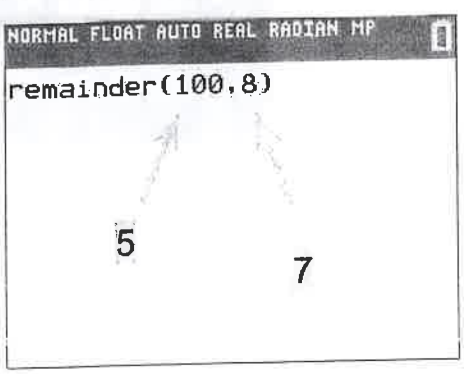
Calculating A Remainder:

1) Press "math"
2) Scroll right to "num"
3) Scroll down to "0: remainder("
4) Press "enter"
5) Input your dividend (what you're dividing)
6) Input a comma
7) Input your divisor (what you're dividing by)
8) Close parentheses and press "enter"

2

3





1

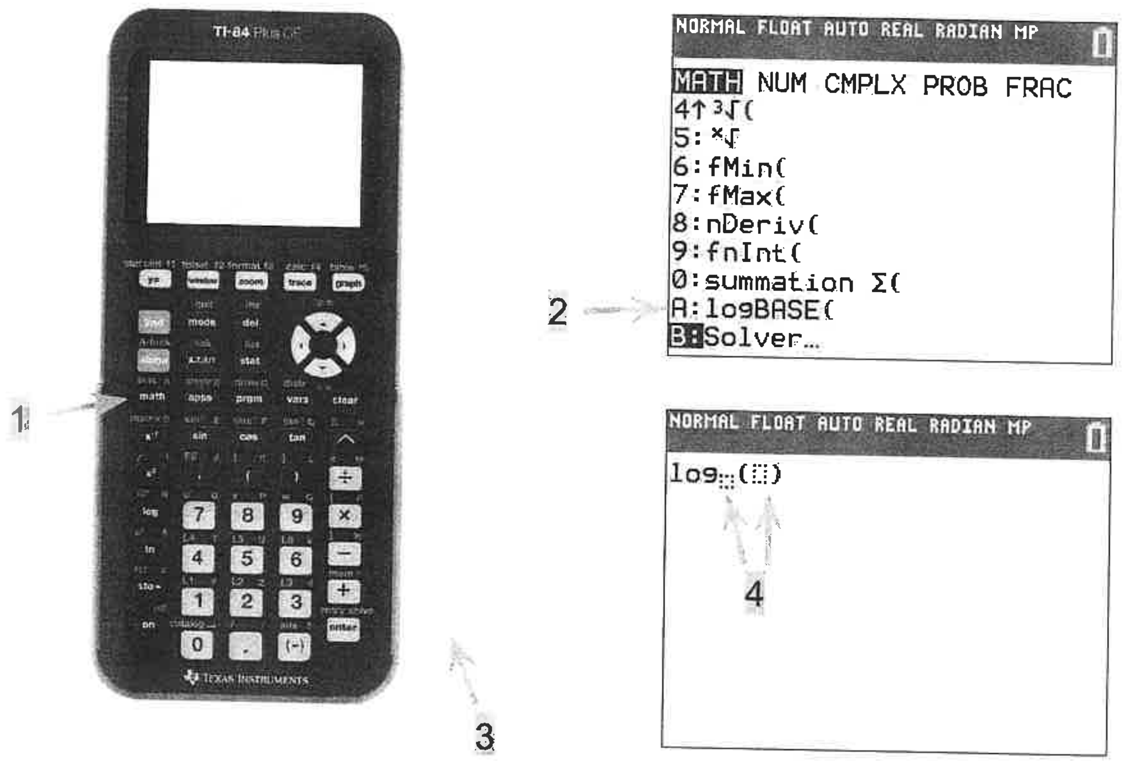
NOTE: There are several other functions you'll see in this list that can be calculated with the same steps.

- 8: lcm(" will calculate a least common multiple
- 9: gcd(" will calculate a greatest common divisor

*Place commas between your terms.

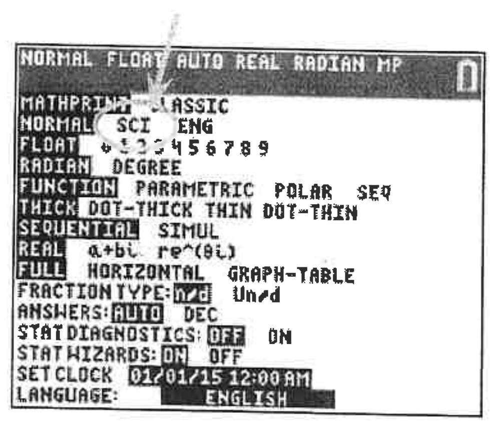
Logarithm with a Base not equal to 10:

1) Press "math"
2) Scroll down to "A: LogBASE("
3) Press "enter"
4) Input base and what you are calculating the logarithm of



Scientific Notation

1) Press "mode"
2) Scroll down to second row and select "SCI"
3) Quit (Press 2nd and then "mode" to quit)

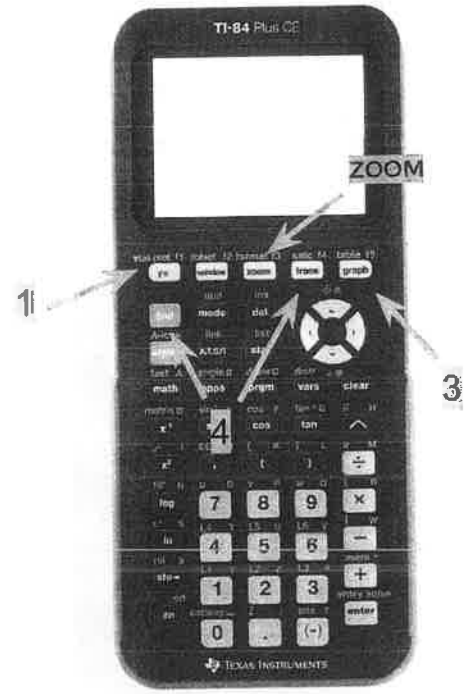


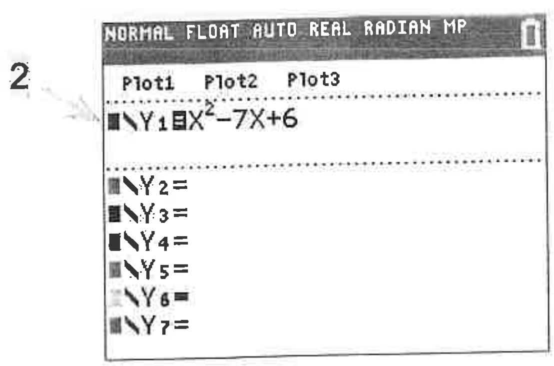
To find a zero (solution or root) by graphing:

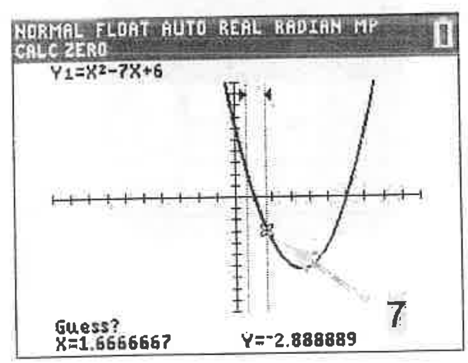
1) Press "y="
2) Input function into Y1=
3) Press "graph" (the graph will appear in your window; if it does not, try zooming out by pressing "zoom" and scrolling down to "3: zoom out")
4) Press "2ND" and then "trace"
5) Scroll down to "2: zero" and press "enter"
6) Move the cursor to a spot on the curve to the LEFT of the zero. Push "enter"
7) Move the cursor to a spot on the curve to the RIGHT of the zero. Push "enter"
8) Push "enter" a third time
9) The x and y coordinates will appear at the bottom of the screen

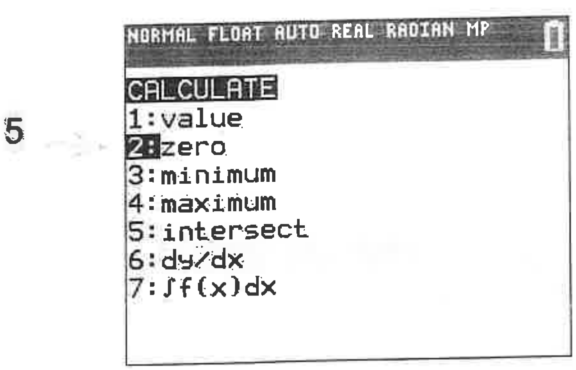
NOTE: You can find a maximum or minimum with the same procedure.
In step 5, select:

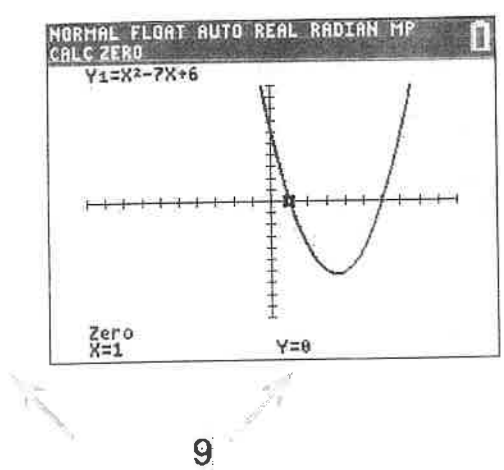
- "3: minimum" to trace a minimum
- "4: maximum" to trace a maximum

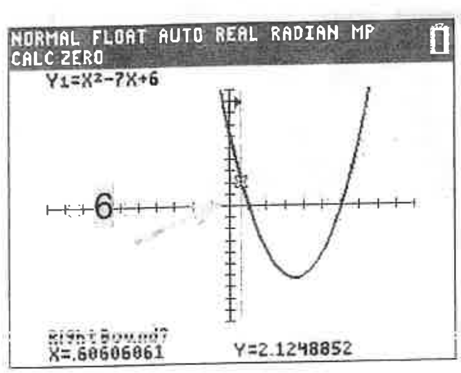






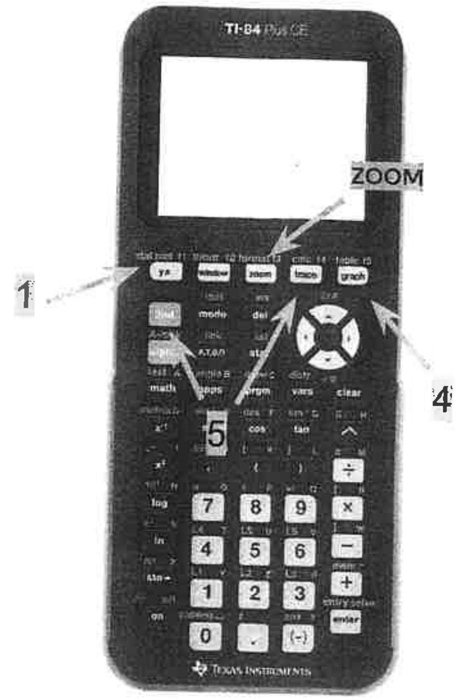






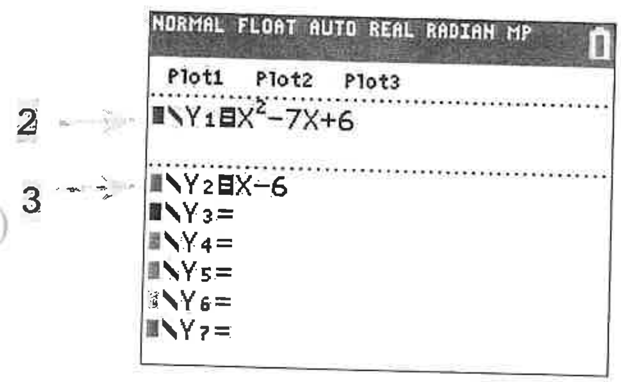
To find a point of intersection:

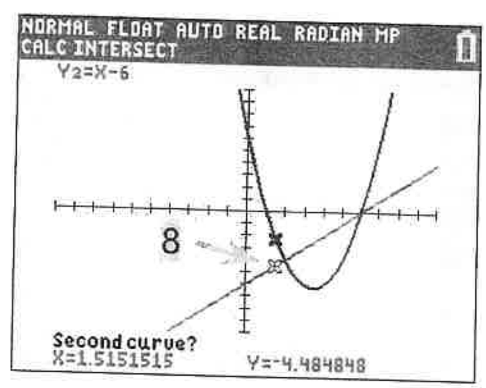
1) Press "y="
2) Input first function into Y1=
3) Input second function into Y2=
4) Press "graph" (both graphs will appear in your window; if they do not, try zooming out by pressing "zoom" and scrolling down to "3: zoom out")
5) Press "2ND" and then "trace"
6) Scroll down to "5: intersect" and press "enter"
7) Move the cursor to a spot near the intersection on the FIRST curve. Push "enter"
8) Move the cursor to a spot near the intersection on the SECOND curve. Push "enter"
9) Push "enter" a third time
10) The x and y coordinates will appear at the bottom of the screen

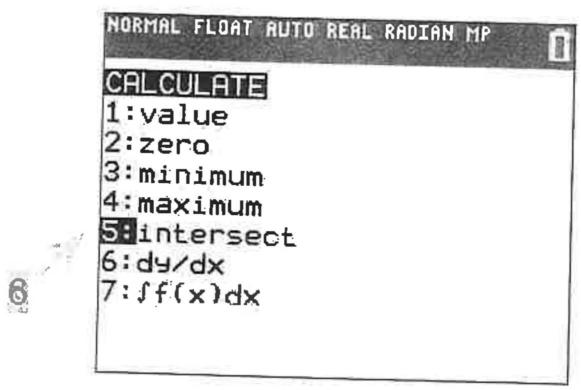


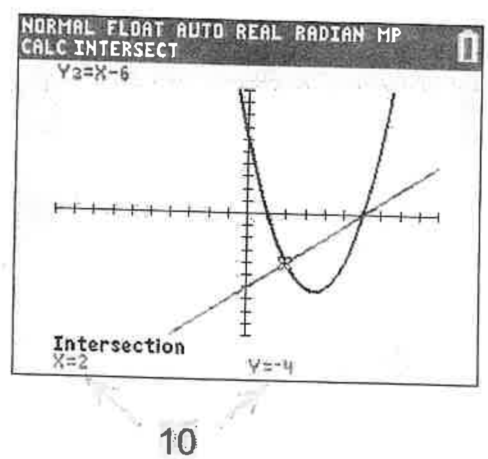
NOTE: This is a great way to solve equations that are difficult to solve by hand.

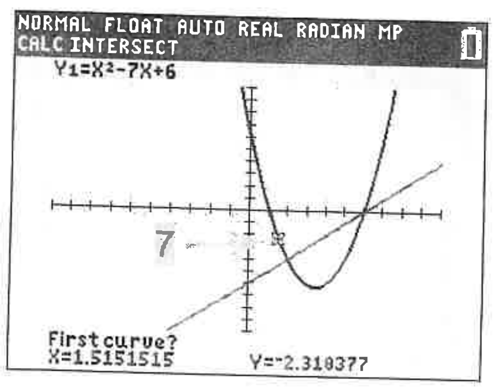
- Put one side of the equation into Y1=
- Put the other side of the equation into Y2=
- Trace the intersection











NOTE: A great way to test for **equivalency** (without needing to do the algebra) is to use the graphing component. Put one expression into Y1 and the other expression into Y2 and graph them both. If the functions are truly equivalent, they will have the same graphs.

They will also have the same table values. Viewing the table might be quicker than letting the calculator graph the entire functions. To locate a table:

1) Press "2nd"
2) Press "graph"

NORMAL FLOAT AUTO REAL RADIAN MP
PRESS + FOR △Tbl

X	Y1	Y2		
9	24	3		
10	36	4		
11	50	5		
12	66	6		
13	84	7		
14	104	8		
15	126	9		
16	150	10		
17	176	11		
18	204	12		
19	234	13		

X=9

NOTE: The physical graphs, trace functions, and table values can help you determine several features about a function without needing to perform any operations by hand. You can simply *observe* them. These features include:

1) Y- Intercepts and X- Intercepts (zeros)
2) Maximums & Minimums
3) Points of Intersection
4) Points of Discontinuity
5) Asymptotes
6) Limits
7) Equivalencies to other functions

To determine the factors of a number:

1) Press "y="
2) Input that number divided by x into Y1
3) Press "2ND" and then "graph" to bring you to the table
4) Any x-y pairs that are whole numbers are factors of that initial value

2

NORMAL FLOAT AUTO REAL RADIAN MP

Plot1 Plot2 Plot3
Y1=96/X
Y2=
Y3=
Y4=
Y5=
Y6=
Y7=
Y8=
Y9=

NORMAL FLOAT AUTO REAL RADIAN MP
PRESS + FOR △Tbl

X	Y1			
0	ERROR			
1	96			
2	48			
3	32			
4	24			
5	19.2			
6	16			
7	13.714			
8	12			
9	10.667			
10	9.6			

X=0

Week

Algebra

(13-15 questions, about 35%)

Topics: linear equations (functions) with one or two variables, System of linear equations, and Linear inequalities one or two variables. Meaning of numbers or variables in the context.

- **Linear equations**: Standard form $ax + by = c$
  Slope-intercept form $y = mx + b$,
  Point-slope form $y - y_1 = m(x - x_1)$.

(Practice problems for algebra)

1) In the XY-plane, line l is perpendicular to line $2x - 3y = 1$. If the line l passes through a point (2, 1), which of the following is an equation of line l?

A) $3x + 2y = 8$
B) $2x - 3y = -8$
C) $y = -\frac{2}{3}x + 4$
D) $y = -\frac{2}{3}x + \frac{7}{3}$

x	y
-2	-4
1/3	3
1	5
3	11

2) From the table above, four pairs of XY-coordinate points in a line are shown. If a line $ax + by = -4$, where a *and* b *are constants*, represents the relationship in the line, what is the value of a?

3) A construction contractor uses the function h defined by $h(x) = 5{,}000 + 250x$, where x is the area of floor in square feet to estimate the cost of labor, in dollars, to build a wooden floor in a certain area. If the contractor gives the estimate of labor cost to build a wooden floor of a house in that area is \$15,000, what is the area of the wooden floor, in square feet, of the house?

CONTINUE

- **The system of equations**:
  (Ways to solve system of equations)
  1) Linear combination method (match the coefficients of one variable and eliminate it)
  2) Substitution method (isolate one variable and substitute it into the other equation)

Standard form: $\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases}$ Point-slope form: $\begin{cases} y = m_1x + b_1 \\ y = m_2x + b_2 \end{cases}$

Standard form	Types of solutions	Point-slope form
$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$	No solution	$m_1 = m_2$ and $b_1 \neq b_2$
$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$	Infinitely many solutions	$m_1 = m_2$ and $b_1 = b_2$
$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$	One solution	$m_1 \neq m_2$

$$4u + 5v = a$$
$$-12u - \frac{v}{b} = 2$$

4) In the given system of equation above, a and b are constants. If the system has infinitely many solutions, what is the absolute value of $a - b$?

5) Adrian plans to work out every morning. He runs at 8 miles per hour and swims at 4 miles per hour. His goal is to practice both exercises at least a total of 15 miles in no more than 2 hours a day. If he spends k hours in running and m hours in swimming, which of the following system of inequalities represent Adrian's goal?

A) $\begin{cases} k + m \leq 2 \\ \frac{8}{k} + \frac{4}{m} \geq 15 \end{cases}$

B) $\begin{cases} k + m < 2 \\ \frac{k}{8} + \frac{m}{4} \geq 15 \end{cases}$

C) $\begin{cases} 8k + 4m \geq 15 \\ k + m \geq 2 \end{cases}$

D) $\begin{cases} 8k + 4m \geq 15 \\ k + m \leq 2 \end{cases}$

CONTINUE

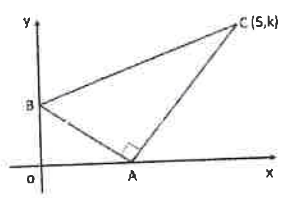
6) Janet purchased a blouse and a purse at a local store. The store offered a special discount on certain items. She spent a total of \$215.20 for both items. If the store offered no tax on the blouse and 10% sales tax on the purse she purchased and the sum of the prices before tax was \$198, what was the price, in dollars, of the purse?

- Properties of parallel and perpendicular lines (in the point-slope forms)

Parallel lines	$m_1 = m_2$ and $b_1 \neq b_2$
Perpendicular lines	$m_1 = -\frac{1}{m_2}$ or $m_1 \cdot m_2 = -1$

- Distance, midpoint, and slope formula between two points $A(x_1, y_1)$ and $B(x_2, y_2)$

Distance = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ Midpoint $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$ Slope = $\frac{y_2-y_1}{x_2-x_1}$



7) In the XY-plane above, the coordinates of points B and C are (0, 2) and (5, k), respectively. If $\overline{AB}$ is perpendicular to $\overline{AC}$ and the slope of $\overline{AB}$ is $-\frac{3}{5}$, what is the value of k?

8) The distance between two points $A(2, a)$ and $B(-1, b)$ is 5. What is the value of $\frac{1}{2}|b - a|$?

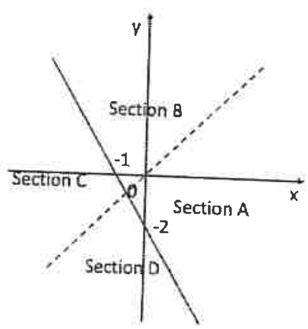
CONTINUE

9) The end points of a line segment AB are A(2, 12) and B(-4, -9), respectively. If points M is on $\overline{AB}$ such that $AM: MB = 1: 2$, what are the coordinates of point M?

A) $(0, 8)$
B) $(-1, 4)$
C) $(0, 5)$
D) $(-1, 2)$

10) Which of the following equations represent a line parallel to the graph of the equation $\frac{1}{5}x + \frac{1}{3}y = -2$?

A) $5x + 3y = 1$
B) $5x - 3y = -4$
C) $3x - 5y = 9$
D) $3x + 5y - 2 = 0$



$$\begin{cases} y - x > 0 \\ -2x - y \leq 2 \end{cases}$$

11) A system of inequalities and a graph are shown in the XY-plane above, which section of the graph could represent all of the solutions to the system?

A) Section A B) Section B C) Section C D) Section D

12) Elliott scored 85, 89, 95, and 80 on his exams before his last exam. If all exams weigh equally, which of the following inequalities could get him all the possible scores of the fifth exam, m that he would result in a mean score on all five exams at least 90?

A) $85 + 89 + 95 + 80 + m \leq 450$
B) $85 + 89 + 95 + 80 + m \geq 360$
C) $\frac{85+89+95+80+m}{5} \leq 90$
D) $85 + 89 + 95 + 80 \geq 450 - m$

CONTINUE

35:00

Section 2, Module 1: Math

Annotate

1 Mark for Review

If $6y = 12$, what is the value of y?

(A) 12

(B) 6

(C) 3

(D) 2

TESTQUBE Question 1 of 22 >

Section 2, Module 1: Math

Annotate

2 Mark for Review

Each side of a fair 2-sided coin is denoted heads and tails. If Jake flips the coin twice, what is the probability of having only heads as the result?

(A) $\frac{1}{2}$

(B) $\frac{1}{3}$

(C) $\frac{1}{4}$

(D) $\frac{1}{8}$

TESTQUBE Question 2 of 22 >

Section 2, Module 1: Math

Annotate

3 Mark for Review

What is 16% of 25?

(A) 4

(B) 14

(C) 16

(D) 400

TESTQUBE Question 3 of 22 >

Section 2, Module 1: Math

Annotate

4 Mark for Review

$F(x) = 120,000 + 4600x$

Function $F(x)$ models a nuclear power plant's total monthly operation cost, in dollars, where x represents the number of workers working at the nuclear power plant. If 27 workers work at the nuclear power plant this month, how much is the total operation cost, in dollars, of the nuclear power plant for this month?

(A) 124,200

(B) 124,600

(C) 200,600

(D) 244,200

TESTQUBE Question 4 of 22 > Back Next

Section 2, Module 1: Math

Annotate

5

Mark for Review

The area of Abraham's square-shaped cornfield equals 196 acres. The length of each side of Jennifer's square-shaped cornfield equals half of the corresponding side length of Abraham's cornfield. Which choice represents the area of Jennifer's cornfield, in acres?

(A) $196 \times \frac{1}{2}$

(B) 196×50

(C) $196 \times (\frac{1}{2})^2$

(D) $196^2 \times (\frac{1}{2})^2$

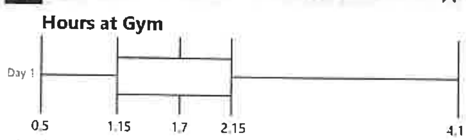
TESTQUBE Question 5 of 22 >

Section 2, Module 1: Math

Annotate

6

Mark for Review



The box plot represents the distribution of time spent at the gym on a certain day, in hours, of 20 Monster Gym members. Which of the following interpretations of the box plot is true?

(A) At least 5 gym members spent more than 4.1 hours at the gym.

(B) The mean hours spent at the gym is 2.15.

(C) The median hours spent at the gym is 2.15.

(D) At least 15 gym members spent more than 1.15 hours at the gym.

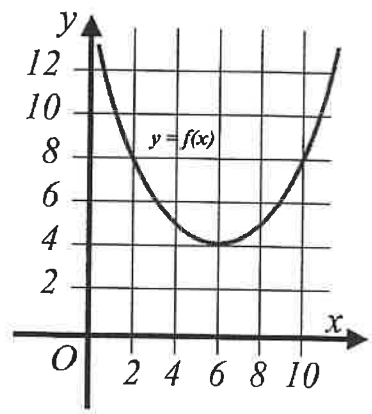
TESTQUBE Question 6 of 22 >

Section 2, Module 1: Math

Annotate

7

Mark for Review



The figure above shows the graph of the quadratic function $y = f(x)$ on the xy-plane. How many different real root(s) does the quadratic equation $f(x) = 0$ have?

(A) 0

(B) 1

(C) 2

(D) Infinitely many

TESTQUBE Question 7 of 22 > Back Next

I

II

III

IV

V

VI

VII

Section 2, Module 1: Math

Annotate

8

Mark for Review

A city government plans to spend at most $24,000$ dollars for a city environment campaign, which consists of tree planting and garbage recycling. Tree planting costs 55 dollars per tree (t), and garbage recycling costs 90 dollars per gallon of garbage (g). Which inequality represents this situation?

(A) $55t \leq 24,000$

(B) $55t + 90g \leq 24,000$

(C) $90t + 55g \leq 24,000$

(D) $55t + 90t \leq 24,000$

TESTQUBE Question 8 of 22 >

Section 2, Module 1: Math

Annotate

9

Mark for Review

$$x + 2y = 11$$
$$4xy = 20$$

(x, y) is the solution for the system of equations above. Find one value of y that satisfies the system of equations.

TESTQUBE Question 9 of 22 >

Section 2, Module 1: Math

Annotate

10

Mark for Review

Each side of a fair 6-sided dice has a different integer from 1 to 6. When the dice is rolled once, what is the probability of rolling a prime number?

(A) $\frac{1}{6}$

(B) $\frac{1}{4}$

(C) $\frac{1}{3}$

(D) $\frac{1}{2}$

TESTQUBE Question 10 of 22 >

Section 2, Module 1: Math

Annotate

11

Mark for Review

Mountain A erodes every year, causing a decrease in height by 0.1% compared to the previous year. If the current height of mountain A is M feet, which choice best models the height of mountain A, in feet, after x years?

(A) $M(0.1)^x$

(B) $M(1 - 0.9)^x$

(C) $M(1 - 0.001)^x$

(D) $M(0.001)^x$

TESTQUBE Question 11 of 22 > Back Next

Section 2, Module 1: Math

Annotate

12

Mark for Review

Student	Credits
A	11
B	14
C	13
D	18
E	12
F	10
G	13

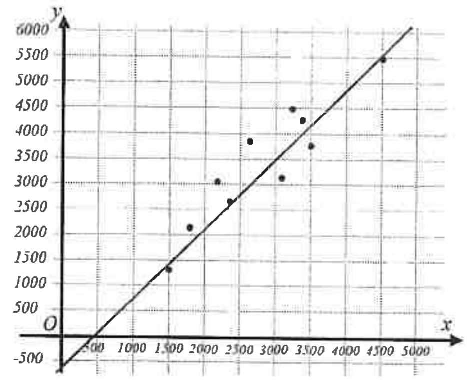
Seven students $A, B, C, D, E, F,$ and G take credit courses at Wharton High School. The total credits each student takes this semester are shown in the table above. What is the median value of the seven students' credits at Wharton High School?

Section 2, Module 1: Math

Annotate

13

Mark for Review



The scatterplot above shows the statistics of major cities in data set P, where the x-values represent the land area (in square kilometers) and the y-values represent the population (in thousands). Which choice most appropriately models the line of best fit for data set P?

(A) $y = -1.41x + 560$

(B) $y = -1.41x - 560$

(C) $y = 1.41x + 560$

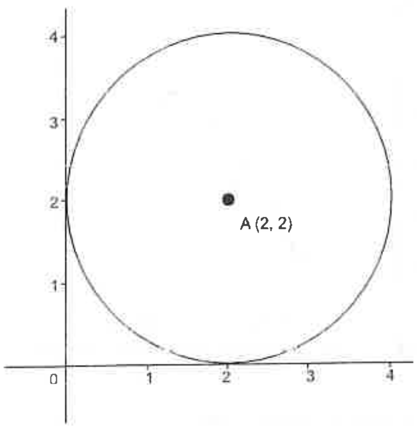
(D) $y = 1.41x - 560$

Section 2, Module 1: Math

Annotate

14

Mark for Review



What is the area of a circle whose center is $A(2, 2)$ and is tangent to both x-axis and y-axis?

Ⓐ π

Ⓑ 2π

Ⓒ 4π

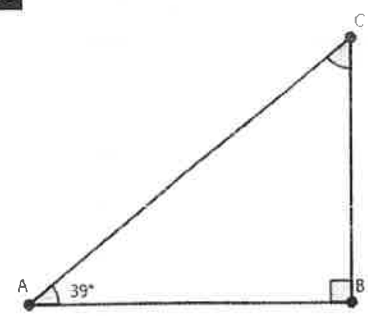
Ⓓ 8π

Section 2, Module 1: Math

Annotate

15

Mark for Review



Among the internal angles in right triangle ABC, angle B has the largest value. If angle A equals $39°$, what is the value, in degrees, of angle C?

Section 2, Module 1: Math

Annotate

16

Mark for Review

Which expression is equivalent to $x(xy)^3 \times \frac{x}{y}$ where x and y are different positive real numbers?

Ⓐ x^5y^2

Ⓑ x^4y^3

Ⓒ x^3y^{-2}

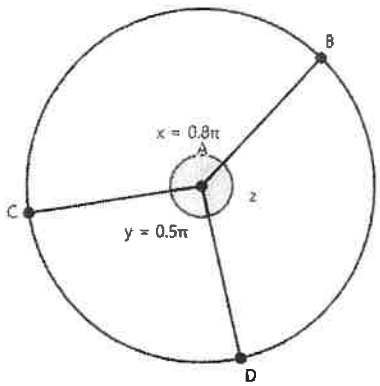
Ⓓ $(xy)^3$

Section 2, Module 1: Math

Annotate

17

Mark for Review



Points $B, C,$ and D are on a circle with center A. Angle BAC (angle x) equals 0.8π and angle CAD (angle y) equals 0.5π, each in radians. What is the measure of the smaller angle BAD, in radians? (The picture is not drawn to scale.)

(A) 2π

(B) 1.5π

(C) 0.7π

(D) 0.5π

Section 2, Module 1: Math

Annotate

18

Mark for Review

What is the correct set of solutions for equation $x^2 - 12x + 27 = 0$?

(A) $x = 3$ or $x = 9$

(B) $x = -3$ or $x = 9$

(C) $x = 3$ or $x = -9$

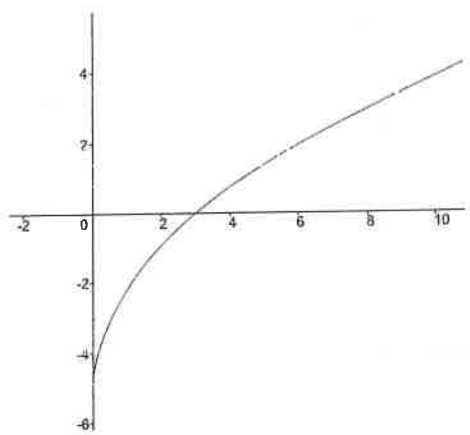
(D) $x = -3$ or $x = -9$

Section 2, Module 1: Math

Annotate

19

Mark for Review



A part of the graph of $f(x) = 2\sqrt{2x} - 5$ on the xy-plane is shown above. Each point of the graph of $f(x)$ is translated by 6 to the positive x-direction, forming a new graph identical to the graph of $g(x)$. Which equation defines $g(x)$?

Ⓐ $g(x) = 2\sqrt{12x} - 5$

Ⓑ $g(x) = 2\sqrt{x-6} - 5$

Ⓒ $g(x) = 2\sqrt{2(x-6)} - 5$

Ⓓ $g(x) = 2\sqrt{2x} - 11$

TESTQUBE Question 19 of 22 >

Section 2, Module 1: Math

Annotate

20

Mark for Review

$f(x) = \frac{1}{x-11}$

What is the value of x, if $f(x) = \frac{1}{35}$?

TESTQUBE Question 20 of 22 >

Section 2, Module 1: Math

Annotate

21

Mark for Review

In a city, 60% of the population owns a car, and 40% of those car owners also own a motorcycle. If the city has a population of 50,000, how many people own both a car and a motorcycle?

TESTQUBE Question 21 of 22 > Back Next

Section 2, Module 1: Math

Annotate

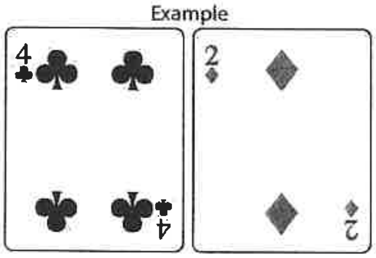
22

Mark for Review

Table

Suit	Numerals	Faces
Spades	1, 2, 3, 4, 5, 6, 7, 8, 9, and 10	Jack, Queen, and King
Hearts	1, 2, 3, 4, 5, 6, 7, 8, 9, and 10	Jack, Queen, and King
Clubs	1, 2, 3, 4, 5, 6, 7, 8, 9, and 10	Jack, Queen, and King
Diamonds	1, 2, 3, 4, 5, 6, 7, 8, 9, and 10	Jack, Queen, and King

Example



A standard card deck contains 52 unique cards. On each card, either a numeral or a face is denoted as shown in the table. (Aces are considered as 1.) Julia randomly picked two different numeral cards from a deck and placed one on the left and one on the right. What is the chance of the numeral denoted on the card Julia placed on the left is exactly two times bigger than the one on the right?

(A) $\frac{80}{40\times39}$

(B) $\frac{16}{40\times39}$

(C) $\frac{80}{52\times52}$

(D) $\frac{16}{52\times52}$

Move on to the Next Section

35:00

Section 2, Module 2: Math

Annotate

1

Mark for Review

140 grams of white rice contains 42 grams of carbohydrates. What is the percentage of carbohydrates in white rice?

(A) 10%

(B) 20%

(C) 30%

(D) 42%

TESTQUBE Question 1 of 22 >

Section 2, Module 2: Math

Annotate

2

Mark for Review

Kepler school library charges a 1.50 dollars daily fee for books returned after 14 days of borrowing. Steven borrowed a book at Kepler school library and returned it late, and paid 4.50 dollars as a late return fee. How long is the period, in days, between Steven borrowing and returning the book?

(A) 1

(B) 4

(C) 14

(D) 17

TESTQUBE Question 2 of 22 >

Section 2, Module 2: Math

Annotate

3

Mark for Review

$2(x+y)=14$

$x-by=-13$

$x=2$ is a part of the solution for the system of equations above, where b is a constant real number. What is the value of b?

(A) 1

(B) 2

(C) 3

(D) 4

TESTQUBE Question 3 of 22 >

Section 2, Module 2: Math

Annotate

4

Mark for Review

$f(x)=380,000-75x$

$f(x)$ models the distance, in million miles, between the Earth and an outer Galaxy A, x years past 1970 for $x<50$. Which choice best describes this context?

(A) Galaxy A was traveling away from the Earth in 1970.

(B) Galaxy A approaches earth 75 million miles every year.

(C) Galaxy A was 75 million miles away from the Earth in 1970.

(D) Galaxy A would be 384, 500 miles away from the Earth in the year 2030.

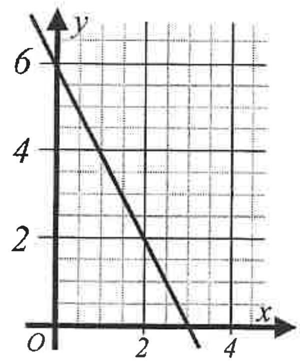
TESTQUBE Question 4 of 22 > Back Next

Section 2, Module 2: Math

Annotate

5

Mark for Review



What is the y-intercept of the graph shown?

Ⓐ 6

Ⓑ 4

Ⓒ 3

Ⓓ 0

TESTQUBE Question 5 of 22 >

Section 2, Module 2: Math

Annotate

6

Mark for Review

Vigo's school band consists of guitar players and flute players, where a band member plays exactly one kind of instrument between the two. If there are twice as many guitar players as there are flute players, and there are 18 members in Vigo's band, how many guitar players are present in Vigo's band?

Ⓐ 12

Ⓑ 8

Ⓒ 6

Ⓓ 4

TESTQUBE Question 6 of 22 >

Section 2, Module 2: Math

Annotate

7

Mark for Review

Factory F has a 2% chance of producing defective products. Out of $5,000$ items produced from factory F, which of the following values is closest to the expected quantity of defective products?

Ⓐ 2

Ⓑ 10

Ⓒ 100

Ⓓ $1,000$

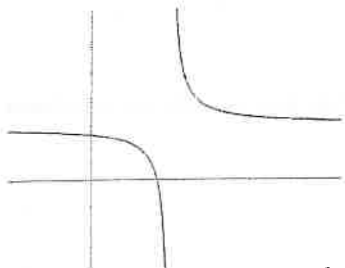
TESTQUBE Question 7 of 22 > Back Next

Section 2, Module 2: Math

Annotate

8

Mark for Review



An overview of the graph of $y = \frac{1}{x-3} + 2$ is shown. What is the x-intercept of the graph?

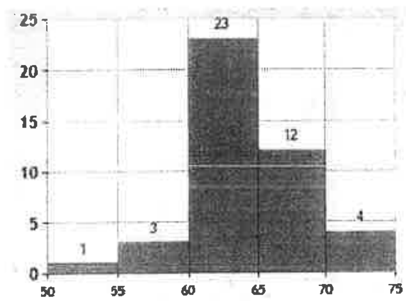
TESTQUBE Question 8 of 22 >

Section 2, Module 2: Math

Annotate

9

Mark for Review



The histogram shows the distribution of the ages of all 43 students in a certain senior literature course. Which of the following is true?

(A) The median age of the students is greater than or equal to 60, and less than 65.

(B) A student at the age of 57 is above the 90th percentile of age.

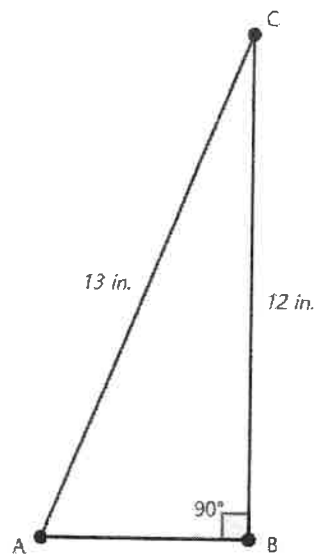
(C) 60 students are 23 years old.

(D) 12 students are older than or at the age of 70.

TESTQUBE Question 9 of 22 > Back Next

10

Mark for Review



In a right triangle ABC where angle $B = 90^{\circ}$, the hypotenuse A is 13 inches long, and the opposite B is 12 inches long. What is $\cos(A)$?

(A) $\frac{1}{13}$

(B) $\frac{5}{13}$

(C) $\frac{13}{5}$

(D) $\frac{13}{12}$

11

Mark for Review

Approve Relocation	44
Against Relocation	56

This winter, City A is planning a vote for or against the relocation of the city hall. Kristin visited 100 restaurants near the city hall to interview restaurant owners and collected the data above. Based on the data, she concluded that among 200, 000 voters in City A, approximately 88, 000 will vote for the relocation. Which of the following is the best strategy Kristin can apply to improve the accuracy of her research?

(A) Interviewing 10 restaurant owners instead of 100.

(B) Revising the conclusion such that 112, 000 voters will vote for the relocation.

(C) Calling random voters in City A on the phone to conduct the interview rather than visiting the restaurants.

(D) Interviewing additional 100 restaurant owners in another city to enlarge the sample size.

Section 2, Module 2: Math

Annotate

12

Mark for Review

$f(x) = \frac{1}{3}\sqrt{x}$

What is the x-value when $f(x) = 3$?

Section 2, Module 2: Math

Annotate

13

Mark for Review

$f(x) = 2,260(1.05)^{\frac{x}{6}}$

$f(x)$ models the population of certain fungi per square centimeter of an experimental medium x hours after the initial observation. Which of the following functions best models the population of the fungi per square centimeter of the medium y days after the initial observation?

(A) $24 \times 2,260(1.05)^{\frac{y}{6}}$

(B) $2,260(1.05)^{4y}$

(C) $2,260(1.05)^{\frac{y}{6}}$

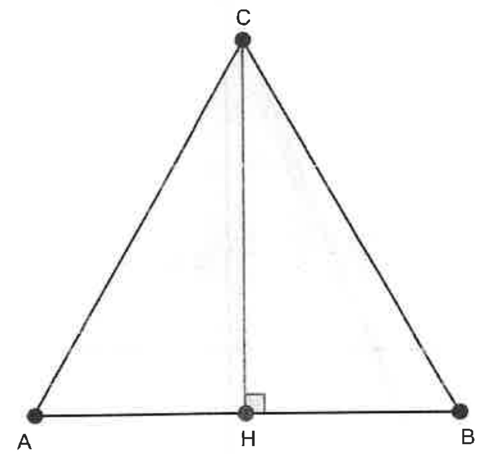
(D) $2,260(1.05 \times 24)^{\frac{y}{6}}$

Section 2, Module 2: Math

Annotate

14

Mark for Review



Straight line CH bisects the area of an equilateral triangle ABC. What is the value of angle ACH in radians?

(A) $\frac{\pi}{6}$

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{3}$

(D) $\frac{\pi}{2}$

Section 2, Module 2: Math

Annotate

15

Mark for Review

$x^2 + 16x + k = 0$

The given equation has two distinct real solutions for x, where k is a constant. Which of the following values can k be to satisfy the condition above?

(A) 0

(B) 64

(C) 128

(D) $1,024$

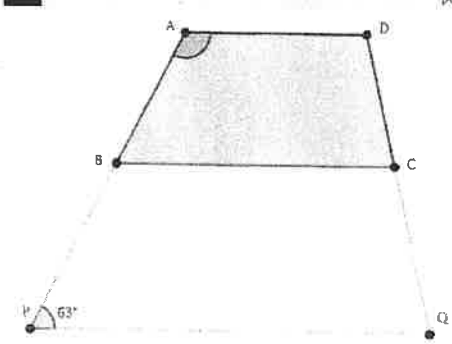
TESTQUBE Question 15 of 22 >

Section 2, Module 2: Math

Annotate

16

Mark for Review



Trapezoids $ABCD$ and $BPQC$ are similar, where sides CD and DA of $ABCD$ correspond to QC and CB of $BPQC$, respectively. If angle P measures $63°$, what is the value, in degrees, of angle A?

TESTQUBE Question 16 of 22 >

Section 2, Module 2: Math

Annotate

17

Mark for Review

$$x = \frac{1}{3}(y+1)$$
$$xz = 27$$
$$c = x + y + z$$

$x = 3$ is a part of the solution to the given system of equations where c is a constant real number. What is the value of c?

TESTQUBE Question 17 of 22 > Back Next

I II III IV V VI VII

I

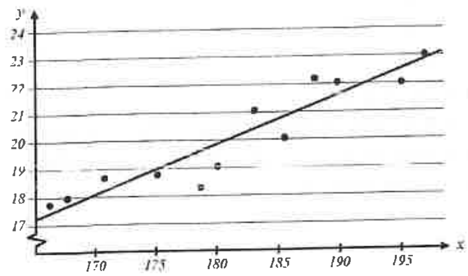
Section 2, Module 2: Math

Annotate

18

Mark for Review

II



III

A group of geologists analyzed fossils from the Jurassic Era excavated In a certain region to measure the average atmospheric temperature at the time when the fossil was formed. The x values of the scatterplot represent the ages of the fossil records (in million years ago), and the y values represent the calculated average atmospheric temperature of the region based on each record (in degrees Celsius). The slope of the line of best fit is 0.19. Which of the following interpretations for the number 0.19 is most appropriate in this context?

IV

A) The average atmospheric temperature increased approximately 0.19 degrees Celsius per million years during the Jurassic Era.

V

B) The average atmospheric temperature in the Jurassic Era was 0.19 degrees Celsius.

C) The average atmospheric temperature in the Jurassic Era was 0.19 degrees Celsius higher than now.

VI

D) The average atmospheric temperature increased 0.19 degrees Celcius every year during the Jurassic Era.

VII

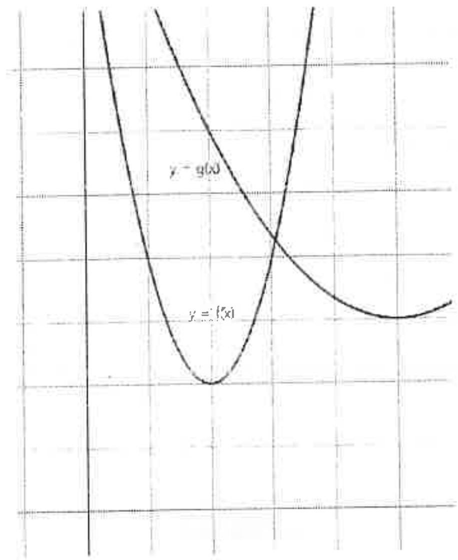
TEST QUBE Question 18 of 22 >

Section 2, Module 2: Math

Annotate

19

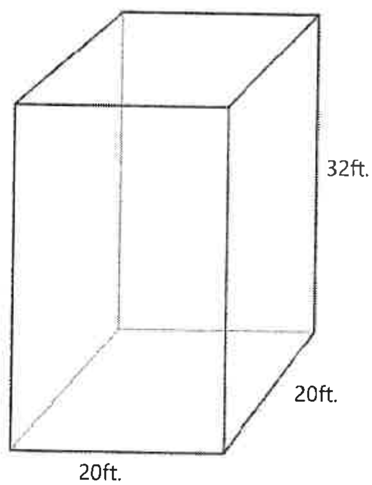
Mark for Review



$f(x) = 2(x-2)^2 + 2$
$g(x) = \frac{1}{3}(x-5)^2 + 3.$
The distance between the vertex of $y = f(x)$ and the vertex of $y = g(x)$ plotted on xy-plane is $\sqrt{d}$. What is the value of d?

TEST QUBE Question 19 of 22 > Back Next

20 Mark for Review



The base of a 32-foot-deep cuboid diving pool is a square with 20 feet of each side. The pool is full of water with which the density equals 62.4 pounds per cubic foot. Which of the following is closest to the total mass of the water in pounds?

(A) $800,000$ pounds

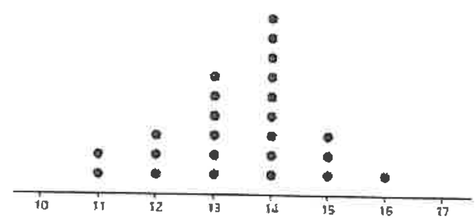
(B) $40,000$ pounds

(C) $13,000$ pounds

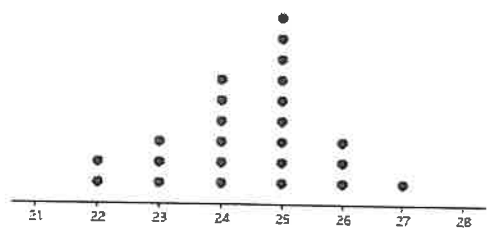
(D) 200 pounds

21 Mark for Review

Dot Plot A: Data Set X



Dot Plot B: Data Set Y



Dot plots A and B represent the distribution of 25 integer values each from data sets X and Y, respectively. Each value in data set Y is greater than the corresponding value in data set X by 11. Which of the following descriptions about the relationship between data sets X and Y is true?

(A) The mean value of data set X equals 15, and the mean value of Y equals 26.

(B) The median of data set Y equals the median of data set X.

(C) The data set Y contains 11 more values than data set X.

(D) The range of data set Y equals the range of data set X.

Section 2, Module 2: Math

Annotate

22

Mark for Review

$f(x) = 39.1(0.98)^{\frac{x}{12}}$ models the air pressure of a car tire, in psi, after x hours of filling the tire with air. What does the number 0.98 mean in this context?

(A) An average rate of decrease in the tire pressure per 12 hours in percent.

(B) The ratio of the tire pressure at a certain time to the tire pressure 12 hours after the certain time.

(C) An average decrease in the tire pressure per 12 hours in psi/hour.

(D) The ratio of the tire pressure 12 hours after a certain time to the tire pressure at the certain time.

Advanced Math

(13-15 questions, about 35%)

Topics: Equivalent equations, non-linear equations in one variable and systems of equations in two variables, non-linear functions. Solving rational equations and graphs of rational and polynomial functions.

- **Equivalent equations**: The expressions on both sides of the given equation are always identical for all values of the given variables. It means that the equation is true for all values of the variable.

(Practice problems for Advanced math)

1) $x^3 + 2x^2 - 4x + 1 = (x + b)(x^2 + cx - 1)$
In the given equation above, b and c are constants. If the equation is true for all values of x, what is the value of $b + c$?

A) 2
B) 3
C) 1
D) -2

2) Which of the following is equivalent to $3b^2 + 18a^4 - 5b^2$?

A) $2(3a - b)(3a + b)$
B) $2(3a - b^2)(3a + b^2)$
C) $3(2a^2 - b)(2a^2 + b)$
D) $2(3a^2 - b)(3a^2 + b)$

- **Quadratic Equation:**

i. Standard form $\boldsymbol{y = ax^2 + bx + c}$.
Vertex (h, k) : $h = -\frac{b}{2a}$. And substitute h into the equation to find the value of k.
Axis of symmetry: $x = h$. (In SAT math, very useful information to find the unknown Y values or unknown x-intercept.)
Shape: $a > 0$: opens upwards. $a < 0$: opens downwards.
This form shows the y-intercept: the value of c.

ii. Vertex form $\boldsymbol{y = a(x - h)^2 + k}$.
Vertex (h, k). when $a > 0$, the graph has a minimum value of k at $x = h$.
When $a < 0$, the graph has a maximum value of k at $x = h$.
This form shows the maximum/minimum values of the function.

iii. Zeros form $\boldsymbol{y = a(x - x_1)(x - x_2)}$.
Vertex (h, k): $h = \frac{x_1+x_2}{2}$ (the mid-point of x_1 *and* x_2) And substitute h into the equation to find the value of k.
This form shows two x-intercepts (zeros) in the equation.

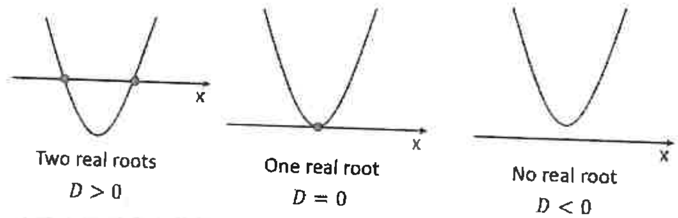
CONTINUE

- How to find solutions of quadratic equations (x-intercepts or zeros)

  i. Factor if possible.
  $a(x-x_1)(x-x_2)=0$. Then, $x=x_1$ or $x=x_2$.

  ii. Use the quadratic formula.
  $x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$.

Note: **D(Discriminant)**$= b^2-4ac$.
The value of D will tell you how many real solutions the equation will have.



  iii. Complete the square.
  $x^2+bx+c=0$. Add $\left(\frac{b}{2}\right)^2$ on both sides of the equation. And factor it.

- Sum or Product of two roots of a quadratic equation in standard form (known as Fundamental theorem of algebra)

$$y=ax^2+bx+c$$
$$\text{Sum} = -\frac{b}{a} \quad \text{Product} = \frac{c}{a}$$

(Quadratic Equations/Functions Practice Problems)

3) What is the value of m if $20x^2+mx-21=(5x+a)(bx+7)$, where $a, b, and\ m\ are\ constant$?

A) -13
B) 13
C) 23
D) -23

4) $f(x)=kx^2+2$, where k is a constant. what is the same value of $f(3)$?

A) $f(1)$
B) $f(4)$
C) $f(-4)$
D) $f(-3)$

CONTINUE

5) A ball is launched straight upward from a cliff (10m above the ground). The motion of a ball could be described as $h(t) = -4.9t^2 + 24t + 10$, where t is the time, in second, the ball is in the air and h is the height, in meters, of the ball after it was launched. How long will it take for a ball to reach its peak?

A) 2.45 sec
B) 24 sec
C) 10 sec
D) 4.9 sec

6) In the system of equations $\begin{cases} y = ax^2 + b \\ y = 1 \end{cases}$, for which of the following values of a and b does the system have two solutions?

A) $a = 2, b = 1$
B) $a = -1, b = 5$
C) $a = 2, b = 5$
D) $a = -2, b = 0$

7) In the system of equations $\begin{cases} f(x) = -x^2 + 2x + 6 \\ h(x) = k \end{cases}$, $where\ k\ is\ a\ constant.$ $f(x) \leq h(x)$ for all real values of x. What is the minimum value of k?

8) In the system of equations $\begin{cases} y = -(x+2)^2 \\ y = m \end{cases}$, where m is a constant. If the system has two points of intersections in the XY-plane and the distance between two points is 12, what is the value of m?

A) -36
B) 36
C) -64
D) 64

CONTINUE

9) In a quadratic equation $2x^2 + 6x - 8 = 0$, Let $a\ and\ b$ are two different roots. What is the value of $(a-1)(b-1)$?

10) The graph of $y = x^2 - 4x + k$ intersects the line $y = -2$ at one point. What is the value of k?

11) In a quadratic equation $h(t) = -16t^2 + 128t$. The function $h(t)$ represents the height, in feet, of an object t seconds after it is thrown upwards from the ground with an initial speed 128 feet per second. How long will the object stay above 192 ft from the ground?

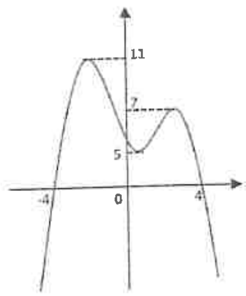
A) 2 sec
B) 4 sec
C) 6 sec
D) 8 sec

12) In the equation $\frac{x^2}{4} - 3x + k^2 = 0$, where k is a constant. If the equation has exactly one solution, what could be the value of k?

A) 3
B) 2
C) 1
D) 0

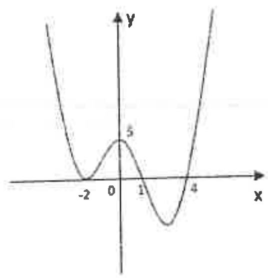
CONTINUE

- Graphs of Polynomial and rational functions



13) The graph of a polynomial function $y = f(x)$ is shown above. If a horizontal line $g(x) = k$(not shown), where k is a constant, meets four times with $y = f(x)$, which of the following could be the value of k?

A) 7
B) 8
C) 6
D) 4



14) Which of the following could be the equation of the graph shown above?

A) $y = \frac{5}{16}\cancel{(x-2)^2(x+1)(x+4)}$
B) $y = -\frac{5}{16}(x+2)^2(x-1)(x-4)$
C) $y = \frac{5}{16}(x+2)^2(x-1)(x-4)$
D) $y = 5(x+2)^2(x-1)(x-4)$

$$\frac{k^2}{\sqrt{k^2 - x^2}} = \frac{x^2}{\sqrt{k^2 - x^2}} + 29$$

15) In the equation shown above, k is a constant. Which of the following is one of the solutions to the equation?

A) $-k$
B) $k^2 - 29^2$
C) $-\sqrt{k^2 - 29^2}$
D) $\sqrt{29^2 - k^2}$

CONTINUE

35:00

Section 2, Module 1: Math

Annotate

1 Mark for Review

$f(x)$ is defined by $f(x) = 2x - 12$. What is the value of x when $f(x) = 2$?

(A) -12

(B) -8

(C) 2

(D) 7

TEST QUBE Question 1 of 22 >

Section 2, Module 1: Math

Annotate

2 Mark for Review

Stephanie is planning to hold a party. She intends to spend no more than 140 dollars on decorating the party room and preparing the food. The room decoration costs 38 dollars regardless of the number of guests, and food preparation costs 9 dollars per guest. Which of the following expressions is most appropriate to find the maximum number of guests, g?

(A) $38 - 9g \leq 140$

(B) $38 + 9g \geq 140$

(C) $9 + 39g \leq 140$

(D) $38 + 9g \leq 140$

TEST QUBE Question 2 of 22 >

Section 2, Module 1: Math

Annotate

3 Mark for Review

Which of the following is a common factor of $x(x+2)$ and $x^2 + 4x + 4$?

(A) x

(B) x^2

(C) $x + 2$

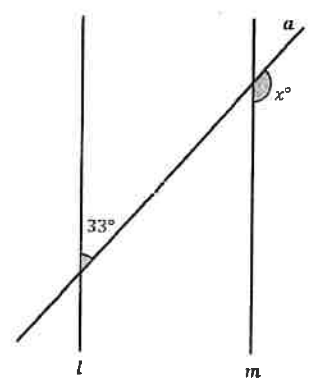
(D) $x + 4x + 4$

TEST QUBE Question 3 of 22 >

Section 2, Module 1: Math

Annotate

4 Mark for Review



Note: Figure Not Drawn to Scale

Line a intersects with parallel lines l and m in the diagram shown above. What is the value of x?

TEST QUBE Question 4 of 22 > Back Next

I II III IV V VI VII

Section 2, Module 1: Math

Annotate

5

Mark for Review

What is the sum of two distinct real solutions for $x^2 - 6x - 16$?

TEST QUBE Question 5 of 22 >

Section 2, Module 1: Math

Annotate

6

Mark for Review

Each side of a right hexagon H whose area is $24\sqrt{3}$ is 4 units long. What is the area of a right hexagon J that has a side length of 2?

(A) $3\sqrt{3}$

(B) $6\sqrt{3}$

(C) $12\sqrt{3}$

(D) $48\sqrt{3}$

TEST QUBE Question 6 of 22 >

Section 2, Module 1: Math

Annotate

7

Mark for Review

Light travels at a constant speed of $300,000$ kilometers per second in a vacuum. However, the speed of light is halved in water. How long, in kilometers, would light travel in 10 seconds underwater?

(A) $3,000,000km$

(B) $1,500,000km$

(C) $750,000km$

(D) $300,000km$

TEST QUBE Question 7 of 22 >

Section 2, Module 1: Math

Annotate

8

Mark for Review

$y = -4x + 4$
$x - 1 = 16$
(x, y) is a solution for the given system of equations. What is the value of y?

(A) -1

(B) -4

(C) -16

(D) -64

TEST QUBE Question 8 of 22 > Back Next

9 Mark for Review

$11, 9, 17, 15, 3$

Data set S consists of five values as shown above. What is the mean value of data set S?

10 Mark for Review

The graphs of $y = \sqrt{x} - 2$ and $x = 4$ intersect at exactly one point $P(x, y)$. What is the value of y?

(A) -3

(B) 0

(C) 3

(D) 6

11 Mark for Review

Which of the following expressions is equivalent to $2ab$?

(A) $\frac{a^2b^2}{2ab}$

(B) $\frac{2a^{-1}b}{b}$

(C) $\frac{2a}{b}$

(D) $\frac{2a^2}{ab^{-1}}$

12 Mark for Review

Nutrient	Calories per gram
Carbohydrate	4 kcal/g
Fat	9 kcal/g
Protein	4 kcal/g

The table above shows the calories for each nutrient: carbohydrate, fat, and protein. For example, 1 gram of protein contains 4 kcal. A serving of certain food consists of 30 grams of fat and carbohydrates together and contains no protein. If the food contains 210 kcal per serving, how many calories, in kcal, in a serving of the food are from fat?

Section 2, Module 1: Math

Annotate

13 Mark for Review

1 foot equals 12 inches. How much is 1 cubic foot (ft^3) in cubic inches?

Ⓐ 12

Ⓑ 144

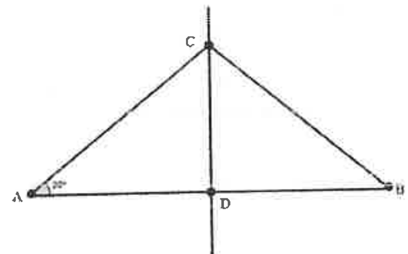
Ⓒ 1,728

Ⓓ 20,736

Section 2, Module 1: Math

Annotate

14 Mark for Review



Straight line CD bisects the edge AB of an isosceles triangle ABC as shown. If the angle CAB is 30 degrees, what is the value, in radians, of angle ACD?

Ⓐ $\frac{\pi}{3}$

Ⓑ $\frac{\pi}{6}$

Ⓒ $\frac{\pi}{2}$

Ⓓ $\frac{2\pi}{3}$

Section 2, Module 1: Math

Annotate

15

Mark for Review

$S(t) = 30 + 2t$

The formula above models the speed of a car t seconds after passing the speed enforcement camera in $\frac{miles}{hour}$. Find the speed of the car, in $\frac{miles}{hour}$, 3 seconds after the event.

TESTQUBE Question 15 of 22 >

Section 2, Module 1: Math

Annotate

16

Mark for Review

Function g is defined by $g(x) = 1.5^x$. What is the value of x if $g(x) = 1.5$?

A) 0

B) 1

C) 1.5

D) 2

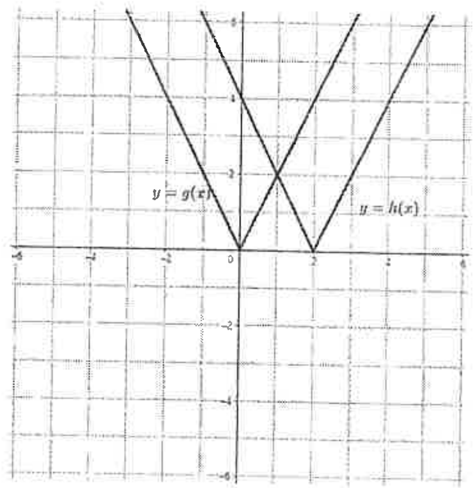
TESTQUBE Question 16 of 22 >

Section 2, Module 1: Math

Annotate

17

Mark for Review



The graph of $g(x) = |2x|$ on the xy-plane is given. The graph of $h(x)$ is generated by pushing the graph of $g(x)$ by 2 units to the right. Which of the following correctly defines $h(x)$?

A) $h(x) = |2x| + 2$

B) $h(x) = |2(x + 2)|$

C) $h(x) = |2(x - 2)|$

D) $h(x) = |2x| - 2$

TESTQUBE Question 17 of 22 > Back Next

Section 2, Module 1: Math

Annotate

18 Mark for Review

Function f is defined by $f(x) = x^2 - 7$. What is the minimum value of $f(x)$?

Section 2, Module 1: Math

Annotate

19 Mark for Review

Data set X
$12, 9, 5, 5, 1, 1, 1, 9, 1, 8$

Restaurant B replaces the knife when the durability of the knife reaches below 95%. Durability is defined as the proportion of the knife's original strength or effectiveness that remains after a certain period of use. The formula $D(w) = 100(0.99)^w$ models the durability of a knife in Restaurant B, w weeks after the purchase. Data set X represents the period of use of all 10 knives in Restaurant B in weeks. How many knives in Restaurant B are subject to replacement?

Section 2, Module 1: Math

Annotate

20 Mark for Review

What is a solution for an equation $x^3 - 27 = 0$?

Ⓐ -3

Ⓑ 0

Ⓒ 3

Ⓓ 9

21 Mark for Review

The median payment of 21 employees of Company A is $49,000$ dollars per year. Which of the following changes in Company A cannot possibly change the median payment?

(A) The company hires 2 more interns each of who receives $32,000$ dollars per year.

(B) The company decides to cut down every employee's annual payment by $1,000$ dollars.

(C) The company pays an extra $2,000$ dollars for an employee who receives the top payment.

(D) The company doubles all employee payments.

22 Mark for Review

The density of a certain steel is 0.25 pounds per cubic inch. Which of the following answer choices most accurately shows the mass, in pounds, of a metal sphere with a diameter of 6 inches?

(A) 13 pounds

(B) 28 pounds

(C) 113 pounds

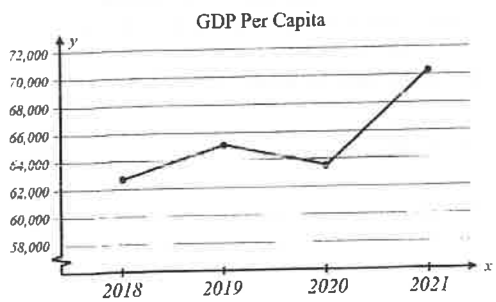
(D) 339 pounds

Move on to the Next Section

Section 2, Module 2: Math

Annotate

1 Mark for Review



The given graph represents the four-year change in a particular country's Gross Domestic Product per capita, in dollars per capita. In which year in the four years did the country have the highest GDP per capita?

A) 2018

B) 2019

C) 2020

D) 2021

Section 2, Module 2: Math

Annotate

2 Mark for Review

Various dating systems call a year differently. For example, the year 2023 in Gregorian Calendar equals the year 2567 in Buddhist Calendar. Assuming the starting date and length of a year are the same, the year in Gregorian Calendar G can be modeled in a linear function $f(B)$ where B is the year in Buddhist Calendar, such that $f(B) = B + c$ where c is constant. What is the value of c?

A) 2567

B) 544

C) -544

D) -2567

Section 2, Module 2: Math

Annotate

3 Mark for Review

Which of the following expressions is equivalent to $xy + 2x^2y^2 - xy^3$?

A) $xy(xy - y^2)$

B) $2xy(x + y^2)$

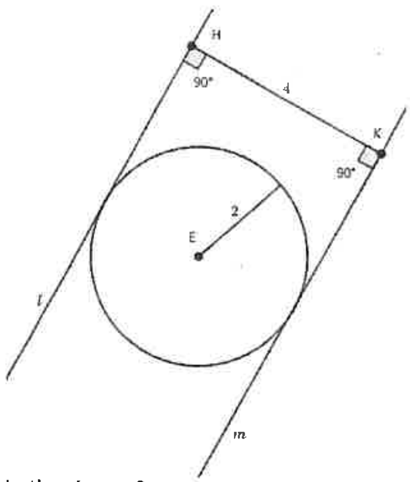
C) $xy(1 + 2xy - y^2)$

D) $y(xy + 2x^2 - y)$

Section 2, Module 2: Math

Annotate

4 Mark for Review



In the shown figure, straight lines l and m are tangent to a circle with a radius of 2 units. HK is perpendicular to both lines l and m and is 4 units long. At how many points do lines l and m intersect?

(A) 0

(B) 1

(C) 2

(D) Infinitely many

TESTQUBE Question 4 of 22 >

Section 2, Module 2: Math

Annotate

5 Mark for Review

Emily has 21 chairs in her office. Some of the chairs have four legs each, while the others have three legs each. If there are a total of 72 legs, how many three-legged chairs does Emily have in her office?

TESTQUBE Question 5 of 22 >

Section 2, Module 2: Math

Annotate

6 Mark for Review

Function f is defined by $f(x) = \frac{1}{x} + 2$. What is the value of $f(\frac{1}{2})$?

(A) $\frac{5}{2}$

(B) 3

(C) $\frac{7}{2}$

(D) 4

TESTQUBE Question 6 of 22 > Back Next

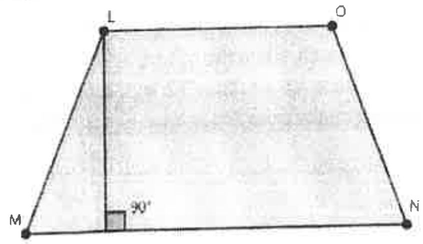
I

Section 2, Module 2: Math

Annotate

7

Mark for Review



In a given figure, LO and MN are parallel sides of trapezoid $LMNO$. The mean of the lengths LO and MN is 8 units. If the area of trapezoid $LMNO$ is 32 square units, what is the height of the trapezoid $LMNO$?

TESTQUBE Question 7 of 22 >

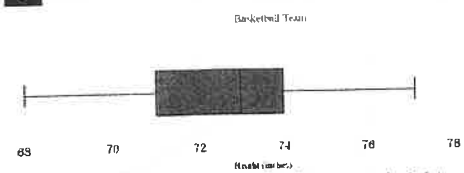
II

Section 2, Module 2: Math

Annotate

9

Mark for Review



The box plot shows the distribution of the heights, in inches, of 19 players in a high school basketball team. What is the height, in inches, of the player at 25th percentile?

(A) 68

(B) 71

(C) 73

(D) 74

TESTQUBE Question 9 of 22 >

III

IV

Section 2, Module 2: Math

Annotate

8

Mark for Review

Which of the following statements is true for a circle that is defined by $(x+2)^2+(y-1)^2=4$?

(A) It has a radius of 4 units.

(B) It is tangent to the y-axis.

(C) Its center is $(1, 2)$.

(D) It has a diameter of 2 units.

TESTQUBE Question 8 of 22 >

V

Section 2, Module 2: Math

Annotate

10

Mark for Review

Factory F produces ballpoint pens. 7% of the produced pens contain more than 1.1mL of ink. Which of the following quantities is most reasonable for the number of pens that contain more than 1.1mL of ink out of $20,000$ randomly chosen ballpoint pens produced at factory F?

(A) 1.1

(B) 7

(C) 140

(D) $1,400$

TESTQUBE Question 10 of 22 > Back Next

VI

VII

Section 2, Module 2: Math

Annotate

11

Mark for Review

$x^2 - 6x = -8$

What is one of the values of x that satisfies the given equation?

TEST QUBE Question 11 of 22 >

Section 2, Module 2: Math

Annotate

12

Mark for Review

A power plant absorbs solar energy and converts it. During the conversion, 60% of the energy absorbed is lost and the rest is converted to electric energy. How much energy, in Joules, would the power plant convert out of 1, 000 Joules of solar energy?

(A) 200

(B) 400

(C) 600

(D) 800

TEST QUBE Question 12 of 22 >

Section 2, Module 2: Math

Annotate

13

Mark for Review

Which of the following expressions has the same value as $2\sqrt{2}$?

(A) 2^2

(B) $\frac{8}{2}$

(C) $\sqrt{8}$

(D) $\sqrt{4}$

TEST QUBE Question 13 of 22 >

Section 2, Module 2: Math

Annotate

14

Mark for Review

When a printer starts printing, it loads document data from a computer and prints out the document. Amma's printer always takes 12 seconds to load data and prints 10 pages per minute. Which of the following functions correctly models the time T, in seconds, it takes for Amma's printer to start loading a document and complete printing p pages?

(A) $T(p) = 12 + 6p$

(B) $T(p) = 6 + \frac{1}{12}p$

(C) $T(p) = 12 + 10p$

(D) $T(p) = 12 + \frac{1}{10}p$

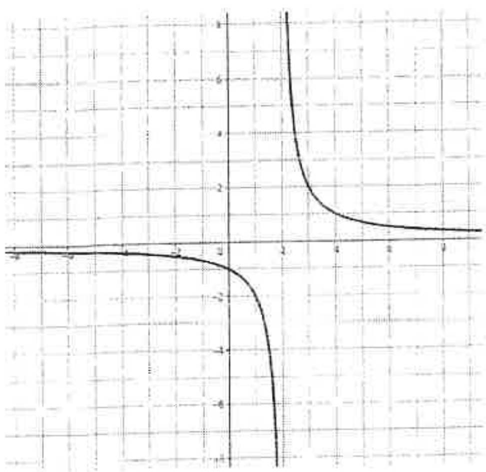
TEST QUBE Question 14 of 22 > Back Next

Section 2, Module 2: Math

Annotate

15

Mark for Review



The graph of $y = \frac{2}{x-2}$ is shown. Which of the following values of x does not have a corresponding y value?

(A) 1

(B) 2

(C) 3

(D) 4

TESTQUBE Question 15 of 22 >

Section 2, Module 2: Math

Annotate

16

Mark for Review

Which of the equations represents a line that has a slope of -1 and a y-intercept of 6?

(A) $y = x + 6$

(B) $y = x - 6$

(C) $y = -x + 6$

(D) $y = -x - 6$

TESTQUBE Question 16 of 22 >

Section 2, Module 2: Math

Annotate

17

Mark for Review

$x = y(y^2 - 11y - 20) + 1$

$(a, 0)$ is one of the solution sets for the given equation. What is the value of a?

TESTQUBE Question 17 of 22 > Back Next

18

Mark for Review

What is the area of a square with a perimeter of 12?

(A) 16

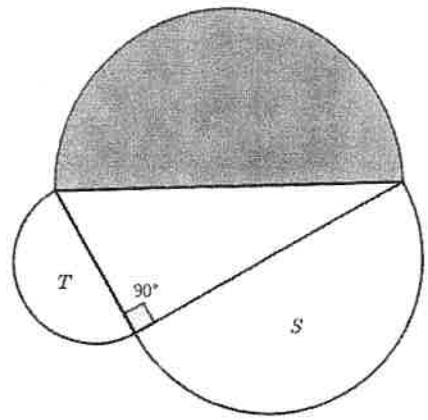
(B) 9

(C) 8

(D) 4

19

Mark for Review



In the given figure, each of the three semicircles has a diameter that equals a corresponding side length of a right triangle. S and T are the area of two smaller semicircles. What is the area of the largest semicircle in terms of S and T?

(A) $S+T$

(B) S^2+T^2

(C) ST

(D) $\sqrt{ST}$

Section 2, Module 2: Math

Annotate

20

Mark for Review

A bicycle lock consists of a three digit password. Each digit of the password can have an integer value from 0 to 6, where the same number may appear more than once. A different order of the same combination sets a different password. For example, 656 and 566 are two different valid passwords. What is the number of different passwords that can be set for the lock?

(A) 3^6

(B) 6^3

(C) 3^7

(D) 7^3

Section 2, Module 2: Math

Annotate

21

Mark for Review

Factory A produces 12-ounce packs of breakfast cereal in four parallel production lanes. The owner conducted research to find if the production lanes are operating as intended. She randomly selected a production lane and measured the masses of 5 consecutively produced packs from the lane. She found the mean value of 13.5 ounces per pack of breakfast cereal. She concluded that Factory A has a defect in all production lanes. Which of the following strategy is most appropriate for the owner to apply in order to improve her research?

(A) Measuring the mass of just one pack instead of 5.

(B) Using pounds as a unit instead of ounces.

(C) Examining 50 randomly selected packs from Factory A.

(D) Comparing the result of the same research from another factory.

Section 2, Module 2: Math

Annotate

22

Mark for Review

The clock marks the time $2:00$. The angle between the hour hand and the minute hand is $\frac{\pi}{d}$ radians. What is the value of d?

TESTQUBE Question 22 of 22 >

Problem-Solving and Data analysis

(5-7 questions, about 15%)

Topics: Ratios, rates, proportional relationship, units. Percentages, one or two variable data: distributions and measures of center and spread, Scatterplots. Probability and conditional probability, Inference from sample statistics and margin of error.

- **Exponential growth/decay**

$p(t) = A(1+r)^t$, where r is % in decimal and t is the time unit. (Exponential growth)
$p(t) = A(1-r)^t$, where r is % in decimal and t is the time unit (Exponential decay)

(Exponential growth/decay Practice Problems)

1) A certain animal population in South Africa is about 2,000 currently. The scientists expect the population will continue to decay 5% every year due to the environmental issues.
$P(t) = 2{,}000\ a^t$, where t in in years. If this exponential model, P(t) represents the population t years from now, what is the value of a?

A) 1.05
B) 0.95
C) 5
D) 0.05

2) Elliott measured the temperature of a tea placed in his room with a constant temperature of 75 degrees Fahrenheit. The temperature of tea was 180°F at 7:00 a.m. and 120°F at 7:10 a.m. Assume that the temperature of tea continues to decrease close to the room temperature. Which of the following best models the temperature T(m), in degrees Fahrenheit, of the tea m minutes after it was placed in his room at 7:00 a.m.?

A) $T(m) = 180(0.67)^{\frac{m}{10}}$
B) $T(m) = 180(1.67)^{\frac{m}{10}}$
C) $T(m) = 75 + 105(0.43)^m$
D) $T(m) = 75 + 105(0.43)^{\frac{m}{10}}$

3) A spacecraft launches from a launching pad from the ground. The spacecraft ascends from the ground to an altitude of 100,000 ft at a constant rate of 1,000 feet per minute. What type of function best represents the relationship between the altitude of the spacecraft and time?

A) Decreasing exponential
B) Decreasing linear
C) Increasing exponential
D) Increasing linear

CONTINUE

- **Percent increase/decrease**

Percent increase = $\frac{Increase}{Original\ amount} \times 100\ (\%)$

Percent decrease = $\frac{decrease}{Original\ amount} \times 100\ (\%)$

- **Ratio / Proportions**

A ratio is a comparison of two numbers in the same unit.
A proportion is an equation with a ratio on each side. It is a statement that two ratios are equal.

(Practice problems with related topics)

4) If $5a + 3b$ is equal to 300 percent of 6b, what is the value of $\frac{a}{b}$?

5) The price of a certain product was first decreased by 20% and then the new price was increased by 20%. Which of the following is true about the price change from the initial price?

A) The price stayed the same.
B) The price went up by 4 percent.
C) The price went down by 4 percent.
D) Not enough information to compute.

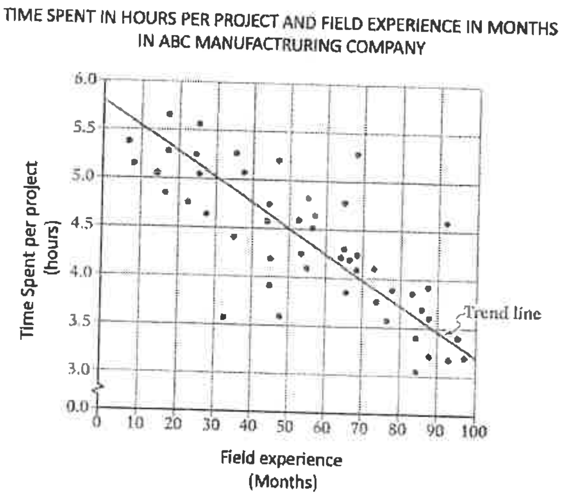
6) On an engineer's blueprint, 1 inch represents 2 feet in real dimensions. If a heating oven is represented on the blueprint by a rectangle that has sides of lengths 4 inches and 5 inches, what is the actual area of the oven, in square feet?

A) 40
B) 80
C) 120
D) 160

7) County A has two school districts (1st grade to 12th grade). The first school district has an area of 110 square miles and a population density of 90 students per square miles. And the second district has an area of 70 square miles and a population density of 120 students per square miles. What is the student population density, the number of students per square miles, for all of County A? (Answer into the nearest integer.)

CONTINUE

- Interpreting data / relationships in scatterplot, table, and equations in Statistics



(Types of questions could be asked in this context from the scatterplot above)

8) According to the scatterplot, a line of best fit is drawn in the graph above. Which of the following equations best represents the line of best fit?

A) $y = \frac{1}{40}x - 5.75$
B) $y = -\frac{1}{40}x + 4.5$
C) $y = -40x + 5.75$
D) $y = -\frac{1}{40}x + 5.75$

9) For how many data shown was the number of hours spent per project less than the number of hours predicted by the line of best fit if the data were chosen over 80-month field experience?

A) 5
B) 6
C) 7
D) 8

10) Which of the following best interpret the slope of the line of best fit in this context?

A) An engineer in the company would spend an hour less on each project if the person experienced every 40 projects.
B) An engineer in the company would spend an hour less on each project if the person had every 40 months field experience.
C) An engineer in the company could finish one more project if the person had more than 40-month field experience.
D) An engineer in the company would spend an hour less on each project if the person had every 40 hours field experience.

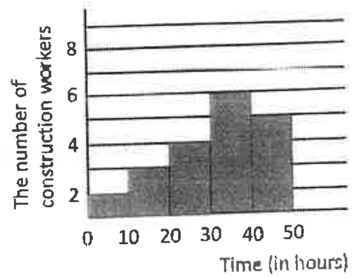
CONTINUE

Pet distribution in ABC High school students' household

	Less than 10 kg	10-15 kg	Greater than 15 kg	Total
DOG	14	20	48	82
CAT	9	7	22	38
Total	23	27	70	120

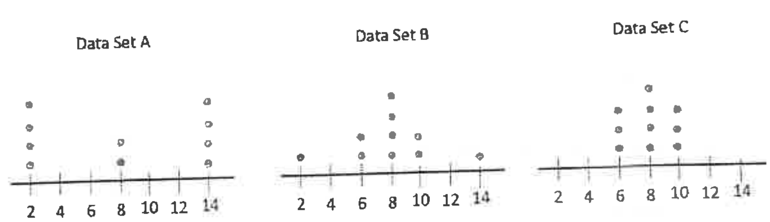
11) The table above summarizes the distribution of Pets weight, in kilograms, in ABC High school students' household. What is the probability of selecting a cat, given that the pet's weight is greater than 15kg?

A) $\frac{24}{35}$
B) $\frac{11}{35}$
C) $\frac{11}{60}$
D) $\frac{2}{5}$



12) In the histogram shown summarizes the distribution of time, in hours, worked by 20 construction workers last week. The first bar represents the number of workers who worked at least 0 hours and less than 10 hours and the second bar represents the number of workers who worked at least 10 hours and less than 20 hours and so on. Which of the following could be the median and mean amount of time worked, in hours, for 20 construction workers assuming only whole numbers of working hour considered?

A) Median = 25, Mean = 25
B) Median = 32, Mean = 27
C) Median = 33, Mean = 35
D) Median = 27, Mean = 33



13) The dot plots show the distribution of math quiz scores in 3 classes of 10 students each class. Which of the following is the correct order about the standard deviations?

A) $A < B < C$ B) $A < C < B$ C) $C < B < A$ D) $B < C < A$

CONTINUE

- **Strategy in survey problems**

1) If the subjects in the sample of a study were selected at random from the entire population, the result can be generalized to the entire population.
2) If the subjects in the sample were randomly assigned to the treatments, it may be appropriate to make conclusions about cause and effect.

	Subjects selected at random	Subjects NOT selected at random
Subjects randomly assigned to treatments	• Results can be generalized to the entire population • Conclusions about cause and effect can be appropriately be drawn	• Results CANNOT be generalized to the entire population • Conclusions about cause and effect can be appropriately be drawn
Subjects NOT randomly assigned to treatments	• Results can be generalized to the entire population • Conclusions about cause and effect should NOT be drawn	• Results CANNOT be generalized to the entire population • Conclusions about cause and effect should NOT be drawn

(Practice problems in data analysis and statistics)

14) A sample of 50 sixth-grade students were randomly selected from a certain elementary school. The 50 students completed a survey regarding to morning meditation before the first class starts, and 45 students replied that the meditation in the morning was helpful to focus in class. Which of the following is the largest population to which the results of the survey can be applied?

A) All students at the same school
B) The 50 students who were surveyed
C) All sixth-grade students in the county in which the school is located
D) All sixth-grade students at the same school

15) A community college offered a Japan tour over the summer to students who would take Japanese course in the fall. The students who visited Japan over the summer through the program did better in the course than students who didn't visit Japan over the summer. Based on the results, which of the following is the most appropriate conclusion?

A) Visiting foreign countries over the summer will cause an improvement for any student who takes the same foreign language course.
B) Visiting Japan over the summer will cause an improvement for any student who takes Japanese language course.
C) Visiting Japan over the summer was the cause of the improvement for the students at the specific community college.
D) No conclusion about the cause can be made regarding students who visited Japan over the summer and their performance in the Japanese course because students who visited Japan were volunteered.

CONTINUE

16) A data set of 20 different numbers has a mean of 55 and a median of 55 as well. A new data set was created by adding 10 to each number in the original set of data that is greater than the median and subtracting 10 to each number in the original set of data that is less than the median. Which of the following does not have the same value in both the original and the new data set?

A) Median
B) Mean
C) Standard Deviation
D) Sum of the data

Sample	Percent in favor of new menu	Margin of error
A	78%	1.5%
B	68%	4.3%

17) The results of two random samples of survey for a new item in a local ice-cream shop are shown above. The samples were selected from the same population and the margins of error were calculated using the same method. Which of the following is the most proper reason that the margin of error for sample A is less than that of sample B?

A) Sample A had a larger sample size.
B) Sample B had a larger sample size.
C) Sample A had a higher percentage of favorable response.
D) Sample A could be more knowledgeable in new item.

18) In which of the following tables is the relationship between the values of x and their corresponding y-values non-linear?

A)

x	1	2	3	4
y	4	6	8	10

B)

x	1	2	3	4
y	-12	-6	0	6

C)

x	1	2	3	4
y	2	4	8	16

A)

x	1	2	3	4
y	4.5	6.5	8.5	10.5

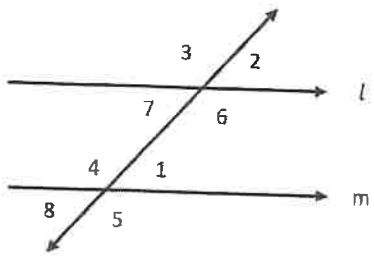
CONTINUE

Geometry and Trigonometry

(5-7 questions, about 15%)

Topics: Area and volume. Lines, angles, and triangles, right triangles, trigonometry, Circles.

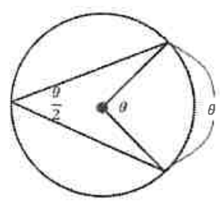
- Angles associated with parallel lines



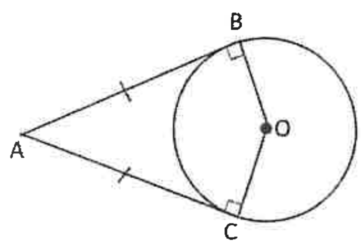
when $l \parallel m$,

- Corresponding angles are congruent: $\angle 1 \cong \angle 2$, $\angle 3 \cong \angle 4$, $\angle 5 \cong \angle 6$, $\angle 7 \cong \angle 8$
- Alternate interior angles are congruent: $\angle 1 \cong \angle 7, \angle 4 \cong \angle 6$
- Same side interior angles are supplementary: $\angle 1 + \angle 6 = 180°$, $\angle 4 + \angle 7 = 180°$
- Alternate exterior angles are congruent: $\angle 3 \cong \angle 5, \angle 2 \cong \angle 8$
- Vertical angles are congruent: $\angle 1 \cong \angle 8, \angle 4 \cong \angle 5$

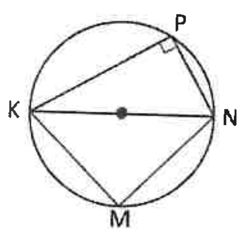
- Angles associated with circles



- Measure of arc is the same as the central angle
- The measure of inscribed angle is half the measure of the same intercepted arc



- The lengths of two tangent segments from a point outside of a circle are congruent ($AB = AC$)
- the segments form a right angle with the radius drawn to the point of tangency ($OB \perp AB, OC \perp AC$)

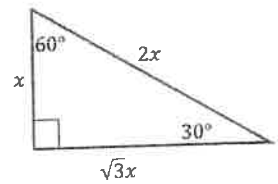


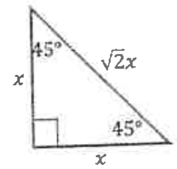
- Any inscribed angle drawn from the endpoints of diameter ($\angle KPN$ *or* $\angle$KMN are right angles)
- If a quadrilateral is inscribed in a circle, its opposite angles are supplementary. ($\angle KPN + \angle KMN = 180°$)

- Angles in polygons

  i. Sum of interior angles in any polygon = $(n-2) \cdot 180°$, where n is the number of sides.
  ii. Each interior angle of a regular polygon = $\frac{(n-2) \cdot 180°}{n}$, where n is the number of sides.
  iii. Sum of exterior angles of any polygon = $360°$
  iv. Each exterior angle of a regular polygon = $\frac{360°}{n}$, where n is the number of sides.

- Angles associated with triangles



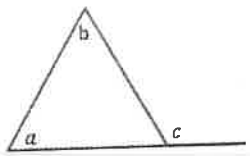


$3-4-5$

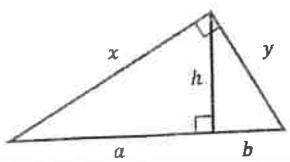
$5-12-13$

$7-24-25$

$8-15-17$



- One exterior angle is same as the sum of two remote interior angles
  $\angle a + \angle b = \angle c$



$x = \sqrt{a \cdot (a+b)}$

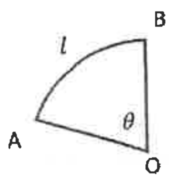
$y = \sqrt{b \cdot (a+b)}$

$h = \sqrt{a \cdot b}$

- Circle equation

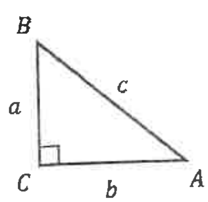
$(x-h)^2 + (y-k)^2 = r^2$, where (h, k) is the coordinates of the center of the circle and r is the radius.

- Area of a sector and Length of an arc



- Area of a sector = $\frac{\theta}{360°} \cdot \pi r^2$, where θ is the central angle in degrees
- Length of an arc = $\frac{\theta}{360°} \cdot 2\pi r$, where θ is the central angle in degrees

- Trigonometry



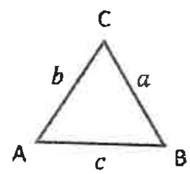
- sin A = $\frac{a}{c}$, cos A = $\frac{b}{c}$, tan A = $\frac{a}{b}$ (Soh-Cah-Toa)
- sin A = cos B (co-functions)

$Note: A = 90° - B$ or A + B = 90° (complementary angles)

- How to convert angles in different mode (radians to degrees or degrees to radians)
  - Radians to degrees : multiply by $\frac{180°}{\pi}$
  - Degrees to radians : multiply by $\frac{\pi}{180°}$

CONTINUE

- Isosceles triangle property and triangle inequalities



<u>Isosceles Triangle Property</u>

- If $\angle A \cong \angle B$, then $a = b$
- If $a = b$, then $\angle A \cong \angle B$

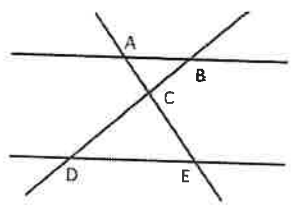
<u>Triangle inequalities</u>

- $|a - b| < c < a + b$

(Practice Problems in Geometry and Trigonometry)

1) In a right triangle, the tangent of one of the two acute angle is $\frac{1}{3}$. Which of the following is the sine of the other angle?

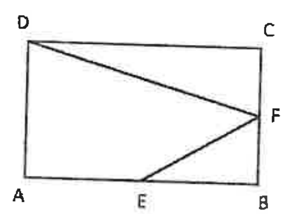
A) $\frac{1}{\sqrt{10}}$ B) $\sqrt{10}$ C) $\frac{\sqrt{10}}{3}$ D) $\frac{3}{\sqrt{10}}$



Note: Figure not drawn to scale

2) In the figure above, $\Delta\ ABC$ is similar to ΔEDC. Which of the following must be true?

A) $\overline{AB} \perp \overline{AE}$ B) $\overline{BD} \perp \overline{DE}$ C) $\overline{AB} \parallel \overline{DE}$ D) $\overline{BC} \perp \overline{CE}$

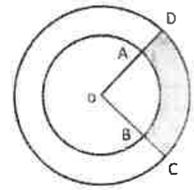


Note: Figure not drawn to scale

3) In the rectangle ABCD above, Points E and F are midpoints of the sides AB and BC, respectively. If $\tan \angle FDC = \frac{1}{2}$, what is the value of $sin\angle BEF$?

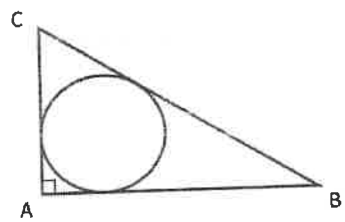
A) $\frac{1}{2}$
B) $\frac{1}{\sqrt{2}}$
C) $\frac{\sqrt{3}}{2}$
D) $\sqrt{2}$

CONTINUE



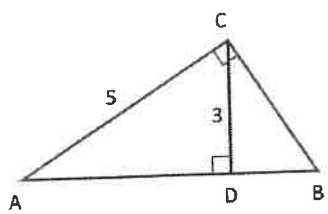
4) Two concentric circles with a center O are drawn above. The area of the shaded region is 16π. If the measure of angle AOB is $\frac{2\pi}{9}$ and $OA : AD = 3 : 2$, what is the length of a minor arc CD?

A) $\frac{10\pi}{9}$ B) $\frac{10\pi}{3}$ C) $\frac{9\pi}{5}$ D) 3π



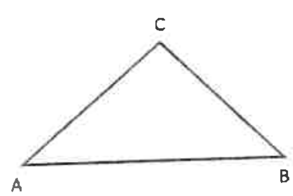
5) In the right triangle ABC above, $AC = 7, BC = 25$. What is the length of the radius of the inscribed circle?

A) 2 B) 3 C) 4 D) 5



6) In the right triangle ABC above, what is the value of $cos\angle CBD$?

A) $\frac{3}{4}$ B) $\frac{3}{5}$ C) $\frac{3}{5}$ D) $\frac{5}{3}$



Note: Figure not drawn to scale

7) In the triangle ABC above, $\sin\angle A = cos\angle B$. Which of the following is correct for triangle ABC?

A) Acute triangle
B) Obtuse triangle
C) Right triangle
D) Isosceles triangle

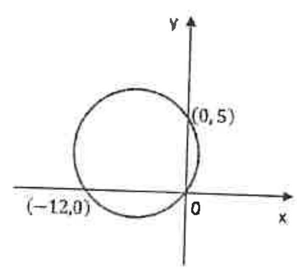
CONTINUE

8) The number of radians in a 540-degree angle can be written as $k\pi$, where k is a constant. What is the value of k?

9) A circle in the XY-plane has equation $(x-1)^2+(y+2)^2=9$. Which of the following points lie in the interior of the circle?

A) $(2,-1)$
B) $(-3,2)$
C) $(2,4)$
D) $(-2,0)$

10) The graph of $2x^2+x+2y^2+y=\frac{1}{4}$ in the XY-plane is a circle. What is the length of the radius of the circle?



11) The graph of a circle is drawn in the XY-plane above. If the circle intersects at three points as shown, what is the length of the radius of the circle?

CONTINUE

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SAT Prep Test 1—Math

Module 1

Turn to Section 2 of your answer sheet (p. 76) to answer the questions in this section.

DIRECTIONS

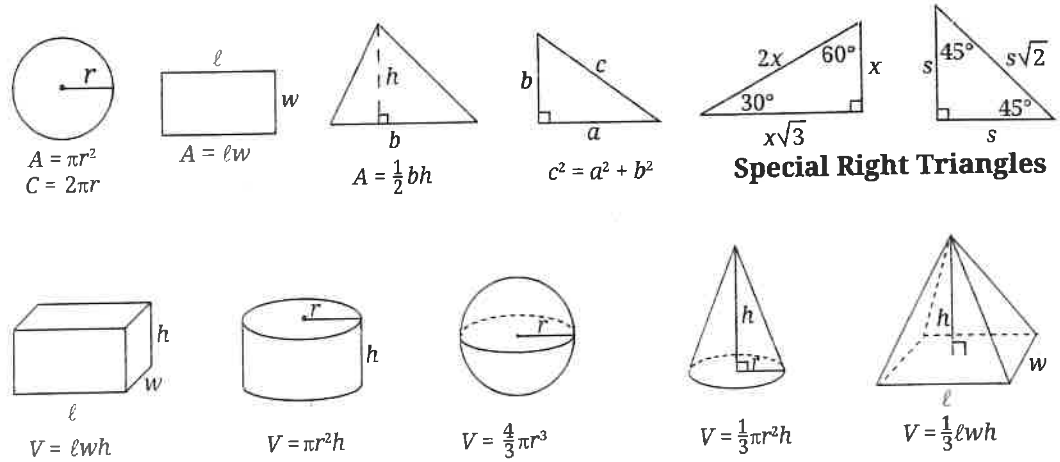
The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

Unless otherwise indicated:

- All variables and expressions represent real numbers.
- Figures provided are drawn to scale.
- All figures lie in a plane.
- The domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE



The number of degrees of arc in a circle is 360.
The number of radians of arc in a circle is 2π.
The sum of the measures in degrees of the angles of a triangle is 180.

CONTINUE

For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled or for questions with no answers circled.

For student-produced response questions, solve each problem and write your answer next to or under the question in the test book as described below.

- Once you've written your answer, circle it clearly. You will not receive credit for anything written outside the circle or for any questions with more than one circled answer.
- If you find **more than one correct answer**, write and circle only one answer.
- Your answer can be up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer, but no more.
- If your answer is a **fraction** that is too long (over 5 characters for positive, 6 characters for negative), write the decimal equivalent.
- If your answer is a **decimal** that is too long (over 5 characters for positive, 6 characters for negative), truncate it or round at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), write it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign in your circled answer.

CONTINUE

1 Mark for Review

Data set S: 2, 3, 8, 8, 11, 24

Data set T: 3, 8, 8, 11, 24

The values in data sets S and T are given. Which of the following is a true statement comparing the means of the two data sets?

(A) There is not enough information to compare the means.

(B) The means of data set S and data set T are equal.

(C) The mean of data set S is less than the mean of data set T.

(D) The mean of data set S is greater than the mean of data set T.

2 Mark for Review

If $x = 8$, what is the value of $30 - x$?

(A) 14

(B) 22

(C) 30

(D) 38

3 Mark for Review

The valve of a water tank remained open for 44 seconds. The water tank contained 7,854 liters of water before the valve was opened and 1,192 liters of water once the valve was closed. Approximately how many liters of water, on average, drained from the tank each second while the valve was open?

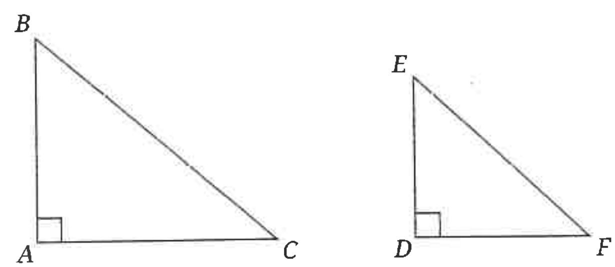
(A) 27

(B) 151

(C) 206

(D) 6,662

4 Mark for Review



Note: Figures not drawn to scale.

Similar right triangles ABC and DEF are shown, where B corresponds to E. If the measure of angle E is $52°$, what is the measure of angle C?

(A) 38°

(B) 52°

(C) 128°

(D) 142°

CONTINUE

5 Mark for Review

Last week, a softball equipment company made \$26,000 from the sale of equally-priced bats and equally-priced gloves. When b is the number of bats sold and g is the number of gloves sold, the equation $500b + 300g = 26{,}000$ represents this situation. The sales price of each glove is how many dollars less than the sales price of each bat?

6 Mark for Review

$$8x - y = -17$$
$$-7x = 14$$

The given system of equations has one solution at (x, y). What is the value of $x - y$?

Ⓐ −31

Ⓑ −3

Ⓒ 3

Ⓓ 31

7 Mark for Review

The function $f(d) = 750d + 12{,}000$ models the amount of money, in dollars, that an arts organization has in its account d days after starting a fundraising campaign. Based on this model, how much money, in dollars, did the organization have in its account before starting the fundraising campaign?

Ⓐ 16

Ⓑ 750

Ⓒ 11,250

Ⓓ 12,000

8 Mark for Review

Equilateral triangle T has a perimeter that is one-third the perimeter of equilateral triangle S. If one side of triangle S is 9 inches long, what is the length, in inches, of one side of triangle T?

Ⓐ 3

Ⓑ 9

Ⓒ 12

Ⓓ 27

CONTINUE

9 Mark for Review

If $|7x + 14| = 49$, what is one possible value of $x + 2$?

______

10 Mark for Review

x	$g(x)$
−30	−116
−24	−86
−18	−56
12	26
−6	4

Five values of x and their corresponding values of $g(x)$ are shown in the table. The relationship between x and $g(x)$ is linear. If the function g is defined by $g(x) = kx + 34$, what is the value of the constant k?

______

11 Mark for Review

The expression $(-3x^4 + 13) + (-8x^4 - 9)$ is equivalent to $cx^4 + 4$, where c is a constant. What is the value of c?

______

12 Mark for Review

$$\frac{22m}{n} = \frac{2}{3s}$$

The given equation relates the numbers m, n, and s, where n is not equal to 0 and $s > 1$. Which equation correctly expresses m in terms of n and s?

(A) $m = 2n - 66s$

(B) $m = 66ns$

(C) $m = \frac{3ns}{44}$

(D) $m = \frac{2n}{66s}$

13 Mark for Review

A certain ant colony contains 96,000 ants. A disease infects the colony, causing the number of ants to decrease by one-half every 4 days. How many ants remained in the colony 20 days after the infection started?

(A) 3,000

(B) 4,800

(C) 6,000

(D) 24,000

CONTINUE

14 Mark for Review

$$x < 53$$
$$x - 7y < 16$$

When the given system of inequalities is graphed in the xy-plane, one of the solutions is $(39, y)$. Which of the following could be the value of y?

Ⓐ -4

Ⓑ -3

Ⓒ 3

Ⓓ 4

15 Mark for Review

$$\sqrt[9]{s^5t^5}$$

If s and t are positive, which of the following expressions is equivalent to the given expression?

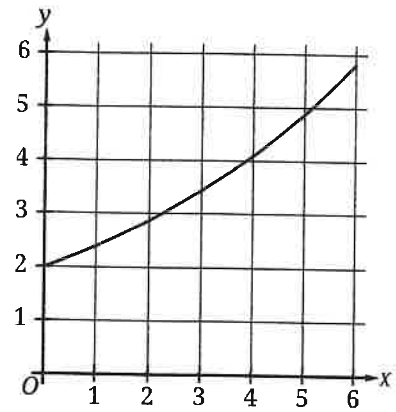
Ⓐ $(st)^{\frac{5}{9}}$

Ⓑ $(st)^{\frac{9}{5}}$

Ⓒ $(st)^{14}$

Ⓓ $(st)^{45}$

16 Mark for Review



A certain collector's item increased in value every month for the first six months after it was purchased. The graph shows the value, y, in hundreds of dollars, of the item x months after it was purchased, where $0 \leq x \leq 6$. Which of the following is the best interpretation of the y-intercept of the graph in this context?

Ⓐ The value of the item when it was purchased was \$2.

Ⓑ The value of the item increased by \$200 over the first six months after it was purchased.

Ⓒ The value of the item when it was purchased was \$200.

Ⓓ The value of the item six months after it was purchased was \$2,000.

CONTINUE

17 Mark for Review

Which of the following systems of linear equations has exactly one real solution?

Ⓐ $y = 2$
$y = 4$

Ⓑ $y = 2x$
$y = 2x - 4$

Ⓒ $y = 4x - 4$
$y = 4x + 4$

Ⓓ $y = 4x - 2$
$y = 8x - 4$

18 Mark for Review

The nth term of a sequence is represented by s, and each term after the first term is one-half of the preceding term. If the first term of the sequence is 56, which of the following equations expresses s in terms of n?

Ⓐ $s = \frac{1}{2}(56^{n-1})$

Ⓑ $s = \frac{1}{2}(56^{n})$

Ⓒ $s = 56\left(\frac{1}{2}\right)^{n-1}$

Ⓓ $s = 56\left(\frac{1}{2}\right)^{n}$

19 Mark for Review

In 2023, a certain streaming service decreased its number of movies available by 9% from the number of movies available in 2022. If the number of movies available in 2023 is m times the number of movies available in 2022, what is the value of m?

Ⓐ 0.09

Ⓑ 0.91

Ⓒ 1.09

Ⓓ 1.91

20 Mark for Review

The maximum value of q is 14 more than 7 times the value of r. Which inequality represents the relationship between q and r?

Ⓐ $q \leq 7r + 14$

Ⓑ $q \leq 14r + 7$

Ⓒ $q \geq 7r + 14$

Ⓓ $q \geq 14r + 7$

CONTINUE

21 Mark for Review

The equation $5x + 3y = -8$ represents line l. Line m is obtained by shifting line l up 2 units in the xy-plane. The x-intercept of the graph of line m is (a, b). What is the value of a?

22 Mark for Review

$$AB = 42$$
$$BC = 56$$
$$CA = 70$$

Right triangle ABC is similar to triangle DEF, where A corresponds to D and B corresponds to E. The lengths of the sides of triangle ABC are given. What is the value of $\cos D$?

Ⓐ $\frac{3}{5}$

Ⓑ $\frac{3}{4}$

Ⓒ $\frac{4}{5}$

Ⓓ $\frac{4}{3}$

YIELD

Once you've finished (or run out of time for) this section, use the answer key to determine how many questions you got right. If you got fewer than 14 questions right, move on to Module 2—Easier, otherwise move on to Module 2—Harder.

SAT Prep Test 1—Math

Module 2—Easier

Turn to Section 2 of your answer sheet (p. 76) to answer the questions in this section.

DIRECTIONS

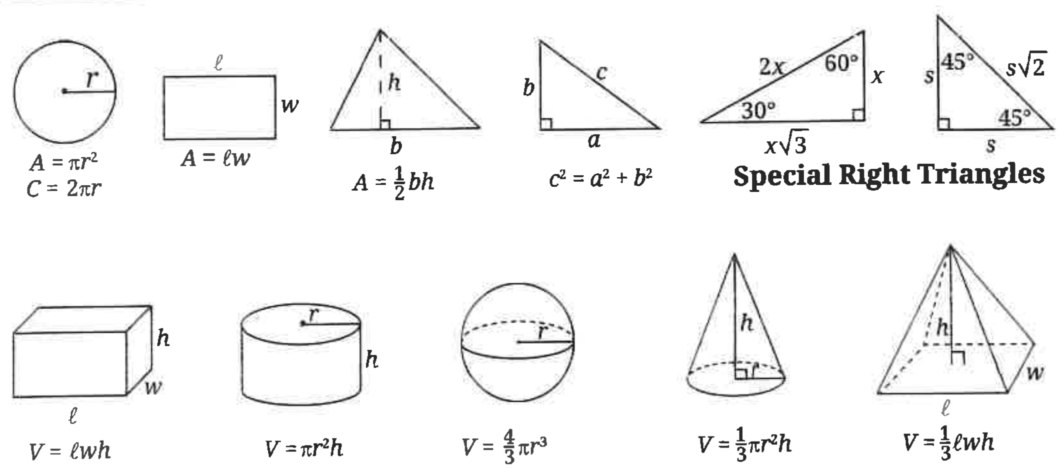
The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

Unless otherwise indicated:

- All variables and expressions represent real numbers.
- Figures provided are drawn to scale.
- All figures lie in a plane.
- The domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE



The number of degrees of arc in a circle is 360.
The number of radians of arc in a circle is 2π.
The sum of the measures in degrees of the angles of a triangle is 180.

CONTINUE

For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled or for questions with no answers circled.

For student-produced response questions, solve each problem and write your answer next to or under the question in the test book as described below.

- Once you've written your answer, circle it clearly. You will not receive credit for anything written outside the circle or for any questions with more than one circled answer.
- If you find **more than one correct answer**, write and circle only one answer.
- Your answer can be up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer, but no more.
- If your answer is a **fraction** that is too long (over 5 characters for positive, 6 characters for negative), write the decimal equivalent.
- If your answer is a **decimal** that is too long (over 5 characters for positive, 6 characters for negative), truncate it or round at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), write it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign in your circled answer.

CONTINUE

1 Mark for Review

An automobile factory makes 1,200 vehicles in 1 day. If the factory operates continuously for 1 week at this rate, how many vehicles will it make?

2 Mark for Review

If $10a = 30$, what is the value of $9a$?

Ⓐ 27

Ⓑ 40

Ⓒ 270

Ⓓ 360

3 Mark for Review

Talula baked 340 cookies and gave 20% of them to her neighbors. How many of the cookies did Talula give to her neighbors?

Ⓐ 14

Ⓑ 34

Ⓒ 54

Ⓓ 68

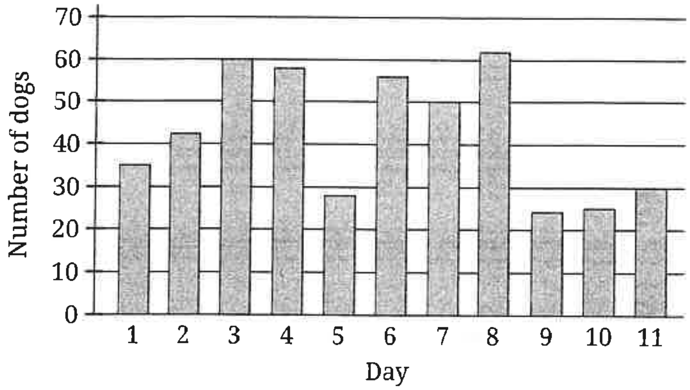
4 Mark for Review

$$h(x) = 11x - 5$$

The function h is defined by the given equation. When $x = 6$, what is the value of $h(x)$?

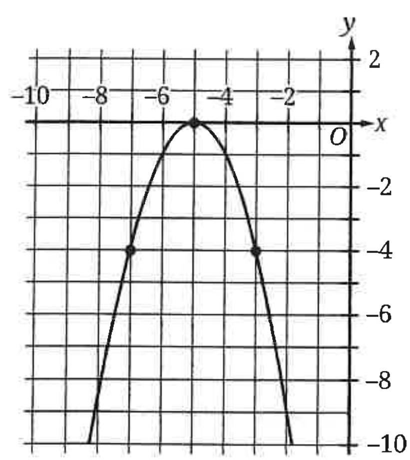
CONTINUE

Mark for Review



The distribution of 469 dogs that visited a dog park over an 11-day period is shown in the bar graph. How many dogs visited this dog park on day 7?

6 Mark for Review



The graph shown intercepts the x-axis at $(x, 0)$. What is the value of x?

7 Mark for Review

Which of the following expressions is equivalent to $14a + 7ab^2$?

Ⓐ $2a(7 + 7b^2)$

Ⓑ $7a(2 + b^2)$

Ⓒ $7a(a + 14b)$

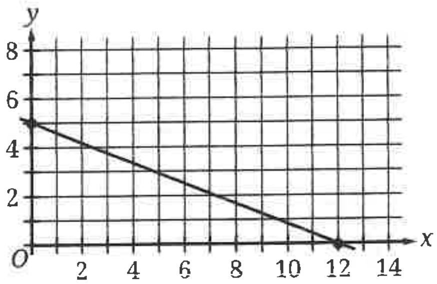
Ⓓ $7b(2ab)$

8 Mark for Review

Rectangle A has a length of 80 and a width of 24. What is the perimeter of rectangle A?

CONTINUE

9 Mark for Review



The point with coordinates $(3, n)$ lies on the line shown. What is the value of n?

Ⓐ $\frac{13}{4}$

Ⓑ $\frac{15}{4}$

Ⓒ $\frac{19}{5}$

Ⓓ $\frac{24}{7}$

10 Mark for Review

A magician has a hat with 18 cards inside. The face of each card has a number from 1 to 18 written on it, with a different number on each card. If the magician takes out a single card, what is the probability that the number written on it is <u>not</u> 6?

Ⓐ $\frac{1}{18}$

Ⓑ $\frac{6}{18}$

Ⓒ $\frac{12}{18}$

Ⓓ $\frac{17}{18}$

11 Mark for Review

The function g is defined by $g(x) = -2x^2$. What is the value of $g(3)$?

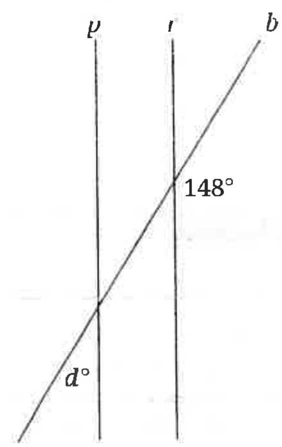
Ⓐ -18

Ⓑ -12

Ⓒ -10

Ⓓ -6

12 Mark for Review



Note: Figure not drawn to scale.

In the figure shown, line b intersects parallel lines p and r. What is the value of d?

Ⓐ 16

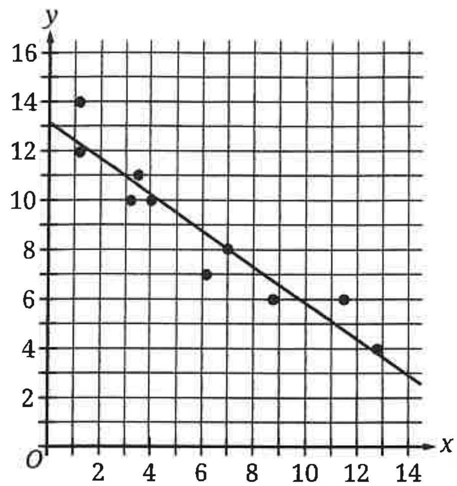
Ⓑ 32

Ⓒ 74

Ⓓ 148

CONTINUE

13 Mark for Review



Which of the following equations could define the line of best fit for the scatterplot shown?

Ⓐ $y = -13 - 0.7x$

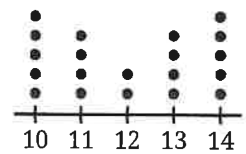
Ⓑ $y = -13 + 0.7x$

Ⓒ $y = 13 - 0.7x$

Ⓓ $y = 13 + 0.7x$

14 Mark for Review

Data Set R



There are 20 values in data set R, represented by the dot plot shown. Data set S is created by subtracting 8 from each of the values in data set R. Which of the following correctly compares the ranges and the means of data sets R and S?

Ⓐ The range of data set S is less than the range of data set R, and the mean of data set S is equal to the mean of data set R.

Ⓑ The range of data set S is less than the range of data set R, and the mean of data set S is less than the mean of data set R.

Ⓒ The range of data set S is equal to the range of data set R, and the mean of data set S is equal to the mean of data set R.

Ⓓ The range of data set S is equal to the range of data set R, and the mean of data set S is less than the mean of data set R.

CONTINUE

15 Mark for Review

During a video game session, a player scored a total of 1,000 points for c cooperative missions and s solo missions. The equation $20c + 25s = 1{,}000$ represents this situation. Which of the following is the best interpretation of the number 25 in this context?

Ⓐ The player completed 25 cooperative missions during this session.

Ⓑ The player scored 25 points for each cooperative mission during this session.

Ⓒ The player completed 25 solo missions during this session.

Ⓓ The player scored 25 points for each solo mission during this session.

16 Mark for Review

$$g(x) = \frac{\sqrt{x}}{2}$$

The function g is defined by the given equation. If $g(x) = 5$, what is the value of x?

Ⓐ 10

Ⓑ 25

Ⓒ 50

Ⓓ 100

17 Mark for Review

$$y = x^2 - 7$$
$$x = -7$$

When graphed in the xy-plane, the given equations intersect at the point (x, y). What is the value of y?

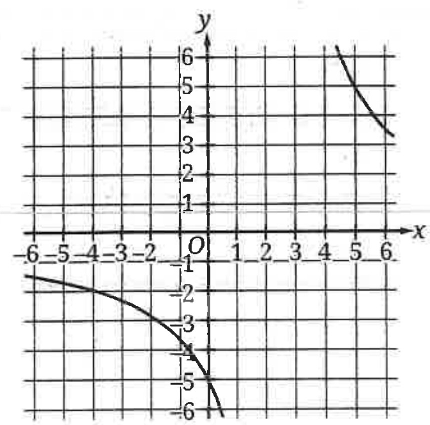
Ⓐ -21

Ⓑ -7

Ⓒ 42

Ⓓ 49

18 Mark for Review



A partial graph of $y = f(x)$ in the xy-plane is shown. Which of the following is the value of $f(0)$?

Ⓐ -5

Ⓑ $-\frac{1}{4}$

Ⓒ 0

Ⓓ 5

CONTINUE

19 Mark for Review

Which of the following equations defines a line in the xy-plane that has a slope of $\frac{1}{6}$ and passes through the point $(12, -7)$?

Ⓐ $y = \frac{x}{6} - 9$

Ⓑ $y = \frac{x}{6} - 7$

Ⓒ $y = -9x + \frac{1}{6}$

Ⓓ $y = 12x - 7$

20 Mark for Review

How many distinct real solutions does the equation $4x^2 - 8x - 5 = 0$ have?

Ⓐ Exactly one

Ⓑ Exactly two

Ⓒ Infinitely many

Ⓓ Zero

21 Mark for Review

A jar contains a total of 37 red and blue tokens used to play a game. The mass of one red token is 90 grams, and the mass of one blue token is 120 grams. If the combined mass of the tokens is 3,810 grams, how many of the tokens in the jar are blue?

Ⓐ 5

Ⓑ 16

Ⓒ 21

Ⓓ 32

22 Mark for Review

Circle R is defined by the equation $(x + 3)^2 + y^2 = 64$. If circle S is the result of shifting the graph of circle R to the right 7 units in the xy-plane, what is the equation of circle S?

Ⓐ $(x + 3)^2 + (y - 7)^2 = 64$

Ⓑ $(x + 3)^2 + (y + 7)^2 = 64$

Ⓒ $(x - 4)^2 + y^2 = 64$

Ⓓ $(x + 10)^2 + y^2 = 64$

STOP

If you finish before time is called, you may check your work on this module only. Do not turn to any other module in the test.

SAT Prep Test 1—Math

Module 2—Harder

Turn to Section 2 of your answer sheet (p. 76) to answer the questions in this section.

DIRECTIONS

The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

Unless otherwise indicated:

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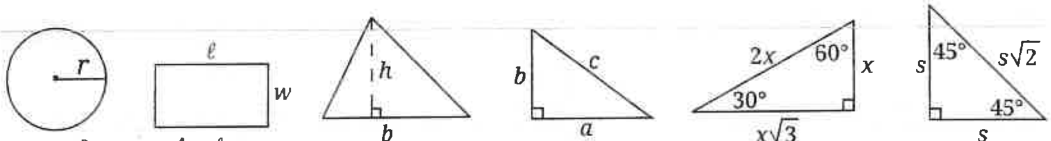
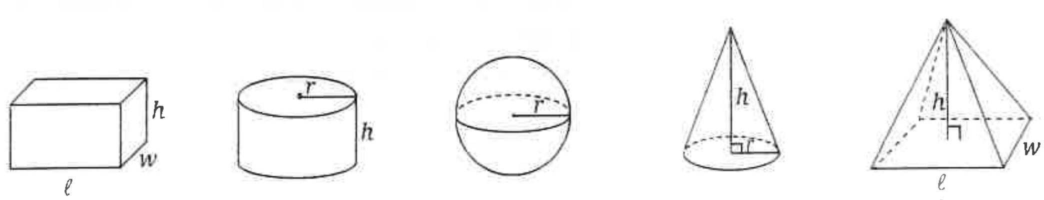
REFERENCE



$A = \pi r^2$, $C = 2\pi r$ | $A = \ell w$ | $A = \frac{1}{2}bh$ | $c^2 = a^2 + b^2$

Special Right Triangles



$V = \ell wh$ | $V = \pi r^2 h$ | $V = \frac{4}{3}\pi r^3$ | $V = \frac{1}{3}\pi r^2 h$ | $V = \frac{1}{3}\ell wh$

The number of degrees of arc in a circle is 360.
The number of radians of arc in a circle is 2π.
The sum of the measures in degrees of the angles of a triangle is 180.

CONTINUE

multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled or for questions with no answers circled.

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- If your answer is a **mixed number** (such as $3\frac{1}{2}$), write it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign in your circled answer.

CONTINUE

1 Mark for Review

A fruit stand sells a total of 200 apples and bananas. The apples are sold in bags of 4 apples per bag, and the bananas are sold in bunches of 6 bananas each. Which of the following equations best represents the number of bags of apples, a, and bunches of bananas, b, that could be sold at the fruit stand?

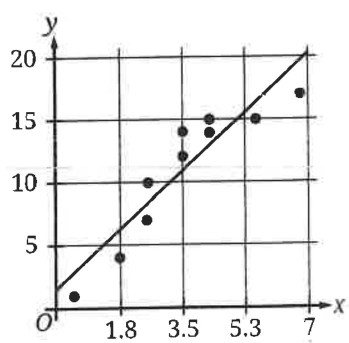
Ⓐ $(a + b)(4 + 6) = 200$

Ⓑ $(4 + a)(6 + b) = 200$

Ⓒ $4a + 6b = 200$

Ⓓ $6a + 4b = 200$

2 Mark for Review



The relationship between two variables, x and y, is shown on the scatterplot, and a line of best fit is also shown. Which of the following equations best represents the line of best fit?

Ⓐ $y = -1.5 - 2.7x$

Ⓑ $y = -1.5 + 2.7x$

Ⓒ $y = 1.5 - 2.7x$

Ⓓ $y = 1.5 + 2.7x$

3 Mark for Review

A random sample of the 120 members of a cycling club was given a survey. The survey asked the cycling club members whether they plan to compete in an upcoming race. Of those surveyed, 45% responded that they do not plan to compete in the upcoming race. Which of the following is the best estimate of the total number of cycling club members who do not plan to compete in the upcoming race, based on the survey?

Ⓐ 45

Ⓑ 54

Ⓒ 66

Ⓓ 120

4 Mark for Review

x	$g(x)$
–1	–11
0	–5
1	–7
2	–17

The table shows four values of x and their corresponding values of $g(x)$ for the quadratic function g. Which of the following equations defines function g?

Ⓐ $g(x) = -x^2 - 4x - 5$

Ⓑ $g(x) = -5x^2 + 3x - 5$

Ⓒ $g(x) = -4x^2 + 2x - 5$

Ⓓ $g(x) = -2x^2 + 4x - 5$

CONTINUE

Mark for Review

$$f(x) = \frac{x-12}{8}$$

The equation given defines function f. If c is a constant, for which value of c does $f(c) = 20$?

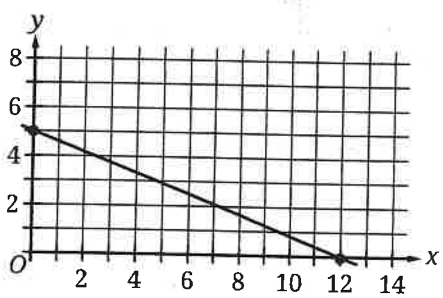
Ⓐ 1

Ⓑ 20

Ⓒ 148

Ⓓ 172

6 Mark for Review



The point with coordinates $(3, n)$ lies on the line shown. What is the value of n?

Ⓐ $\frac{13}{4}$

Ⓑ $\frac{15}{4}$

Ⓒ $\frac{19}{5}$

Ⓓ $\frac{24}{7}$

7 Mark for Review

Line l is graphed in the xy-plane and is defined by $7 + 4y = -16x$. If line m is parallel to line l, what is the slope of line m?

Ⓐ -4

Ⓑ $-\frac{1}{4}$

Ⓒ $\frac{1}{4}$

Ⓓ 4

8 Mark for Review

A concrete block in the shape of a rectangular solid has a mass of 1,690 kilograms. The block has a length of 1.1 meters, a width of 0.8 meters, and a height of 0.8 meters. To the nearest whole number, what is the density, in kilograms per cubic meter, of the concrete block?

Ⓐ 1,082

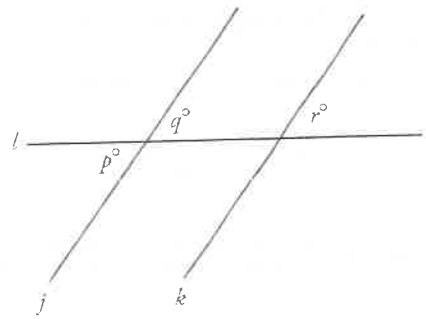
Ⓑ 1,190

Ⓒ 2,401

Ⓓ 2,641

CONTINUE

9 Mark for Review



Note: Figure not drawn to scale.

In the figure shown, line l intersects parallel lines j and k. If $q = 10c - 11$ and $r = 15c + 41$, what is the value of p?

Ⓐ 6

Ⓑ 49

Ⓒ 115

Ⓓ 131

10 Mark for Review

$$-7(px + q) = \frac{42}{23}x + \frac{35}{9}$$

The given equation, where p and q are constants, has infinitely many solutions. If $p < 0$, what is the value of q?

11 Mark for Review

During the last business quarter, the number of unique visitors to a small e-commerce website decreased by 25% from its previous average of 620 unique visitors each day. At the start of the upcoming quarter, the website will launch a promotion, and the resulting number of unique visitors per day is projected to be 180% of the number of visitors last quarter. What is the projected average number of unique visitors per day the website will receive during its promotion?

12 Mark for Review

The amount, in micrograms, of a certain radioactive isotope h hours after its initial creation is modeled by the function $M(h) = 302(0.87)^{\left(\frac{4}{5}\right)h}$. According to the model, the amount of the isotope is predicted to decrease by $d\%$ every 75 minutes. What is the value of d?

Ⓐ 0.87

Ⓑ 13

Ⓒ 16.25

Ⓓ 87

CONTINUE

13 Mark for Review

Packages in a warehouse are split into two groups. Group X contains 40 packages, and group Y contains 110 packages. If the mean mass of the packages in group X is 24 kilograms (kg), and the mean mass of the packages in group Y is 9 kg, what is the mean mass, in kg, of all 150 packages?

14 Mark for Review

$$11x^2 - kx + 63$$

The given expression, where k is a constant, can be rewritten as $(px - q)(x - r)$, where p, q, and r are integer constants. Which of the following must be an integer?

(A) $\frac{63}{r}$

(B) $\frac{63}{p}$

(C) $\frac{k}{r}$

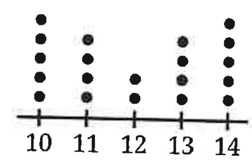
(D) $\frac{k}{p}$

15 Mark for Review

The equation $3x^2 - 36x + k = 0$ has exactly two real solutions. If k is a constant and $k < m$, what is the least possible value of m?

16 Mark for Review

Data Set R



There are 20 values in data set R, represented by the dot plot shown. Data set S is created by subtracting 8 from each of the values in data set R. Which of the following correctly compares the ranges and the means of data sets R and S?

(A) The range of data set S is less than the range of data set R, and the mean of data set S is equal to the mean of data set R.

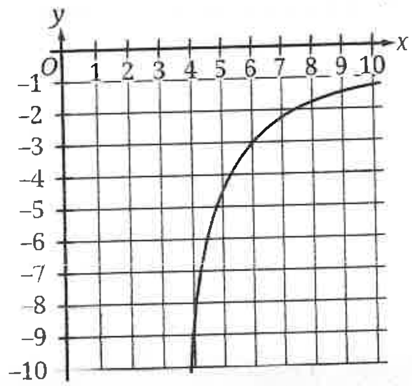
(B) The range of data set S is less than the range of data set R, and the mean of data set S is less than the mean of data set R.

(C) The range of data set S is equal to the range of data set R, and the mean of data set S is equal to the mean of data set R.

(D) The range of data set S is equal to the range of data set R, and the mean of data set S is less than the mean of data set R.

CONTINUE

17 Mark for Review



The function g is defined by the equation $g(x) = \frac{c}{x-d}$, where c and d are constants. The partial graph of $y = g(x)$ is shown. Which equation could define function h if $h(x) = g(x - 3)$?

Ⓐ $h(x) = -\frac{9}{x-6}$

Ⓑ $h(x) = -\frac{9}{x-3}$

Ⓒ $h(x) = \frac{-9}{x}$

Ⓓ $h(x) = \frac{-9(x-3)}{x-3}$

18 Mark for Review

$$y = x^2 - 6x - c$$
$$y = 3.5$$

If the given system of equations has exactly one real solution, and c is a negative constant, what is the value of c?

19 Mark for Review

A circle with the equation $x^2 - \frac{1}{2}x + y^2 - \frac{1}{2}y = \frac{7}{8}$ is graphed in the xy-plane. What is the length of the radius of this circle?

20 Mark for Review

In the equation $18x^2 - (18n - m)x - mn = 0$, m and n are positive constants. If the product of the solutions to the given equation is kmn, where k is a constant, what is the value of k?

Ⓐ -18

Ⓑ $-\frac{1}{18}$

Ⓒ $\frac{1}{9}$

Ⓓ 1

CONTINUE

21 Mark for Review

The equation of a parabola is written in the form $y = ax^2 + bx + c$, where a, b, and c are constants. When graphed in the xy-plane, the parabola has vertex $(-1, 4)$ and does not intersect the x-axis. Which of the following could be the value of $a - b - c$?

Ⓐ -5

Ⓑ -4

Ⓒ 0

Ⓓ 4

22 Mark for Review

A rectangular prism has a height of 50 centimeters (cm). The base of the prism is a square and the surface area of the prism is S cm^2. If the prism is divided into two identical rectangular prisms by making a cut parallel to the square base, each resulting prism has a surface area of $\frac{31}{56}S$ cm^2. What is the side length, in cm, of each square base?

Ⓐ 5

Ⓑ 6

Ⓒ 12

Ⓓ 24

STOP

If you finish before time is called, you may check your work on this module only. Do not turn to any other module in the test.